Adopted Levels, Gammas

$Q(\beta^-)=-5966.2$ *7*; S(n)=12644.76 *5*; S(p)=6370.81 *4*; $Q(\alpha)=-6997.90$ *4* 2021Wa16
S(2n)=24152.8 *4*, S(2p)=17254.1 *11* (2021Wa16).
Isotope discovery (2012Th10): positive-ray mass spectrograph at Cambridge, UK (1919As01).

^{35}Cl Levels

Cross Reference (XREF) Flags

A	^{35}S β^- decay (87.61 d)	K	^{31}P(^{6}Li,d)	U	^{35}Cl(p,p′γ)
B	^{35}Ar ε decay (1.7755 s)	L	^{32}S(α,p)	V	^{36}Ar(n,d)
C	^{36}K εp decay (341 ms)	M	^{32}S(α,pγ)	W	^{36}Ar(d,^{3}He)
D	^{1}H(^{34}S,^{35}Clγ)	N	^{33}S(α,d)	X	^{36}Ar(pol d,^{3}He)
E	^{2}H(^{34}S,nγ)	O	^{34}S(p,γ)	Y	^{37}Cl(p,t)
F	^{12}C(^{28}Si,αpγ)	P	^{34}S(p,p),(p,p′γ):resonances	Z	^{39}K(γ,α)
G	^{16}O(^{24}Mg,αpγ)	Q	^{34}S(d,n)	Others:	
H	^{24}Mg(^{16}O,αpγ)	R	^{34}S(^{3}He,d)	AA	^{40}Ca(μ^-,$\nu\alpha$nγ)
I	^{27}Al(^{14}N,αpnγ)	S	^{35}Cl(γ,γ′)	AB	Coulomb excitation
J	^{31}P(α,p),(α,n):resonances	T	^{35}Cl(n,n′γ)		

E(level)[†]	J$^\pi$[‡]	T$_{1/2}$ or Γ[&]	XREF	Comments
0.0	3/2$^+$	stable	ABC EFGHI KLM O QRSTUVWXYZ	XREF: Others: AA, AB μ=+0.821870 *2* (2013Ja17,2019StZV) Q=−0.0817 *8* (2008Py02,2021StZZ) J$^\pi$: L=2 from 0$^+$ in (d,n), (n,d), (^{3}He,d), (d,^{3}He), and (pol d,^{3}He) and L-1/2 transfer from analyzing power. Spin=3/2 from microwave spectroscopy and atomic beam methods (1948To10,1949Da14). μ: Nuclear Magnetic Resonance (2013Ja17). Q: Atomic beam magnetic resonance (2008Py02). Others: −0.08249 *2* (1972St38), −0.076 *5* (1986El09), −0.817 *8* (1993Su36), 0.0819 *11* (2000Ha64), 0.0850 *11* (2004Al08). Evaluated rms nuclear charge radius R=3.365 fm *19* (2013An02).
1219.38 *7*	1/2$^+$	0.119 ps *14*	B E H KLM O QRSTUVWXY	XREF: Others: AA, AB B(E2)↑=0.00076 *7* B(E2)↑: from Coulomb excitation. E(level): weighted average of 1219.2 *2* from ^{35}Ar ε decay, 1219.39 *7* from (p,γ), and 1219.42 *10* from (p,p′γ), 1219.0 *10* from (^{16}O,αpγ), and 1219.4 *6* from (α,pγ). J$^\pi$: L=0 from 0$^+$ in (d,n), (n,d), (^{3}He,d), (d,^{3}He), and (pol d,^{3}He). T$_{1/2}$: from lifetime τ=0.172 ps *20*: weighted average of 0.29 ps *4* from (^{34}S,nγ) (1973Wa10, DSAM), 0.21 ps *+10−8* from (α,pγ) (1969In04, DSAM), 80 fs *40* (1971Wi13, DSAM), 175 fs *20* (1972Hu11, DSAM), 150 fs *20* (1973Fa07, DSAM), 145 fs *30* (1976Me12, DSAM) from (p,γ), 0.13 ps *+3−2* from (γ,γ′) (1966Ho02, resonance fluorescence), 200 fs *80* from (n,n′γ) (1989Ge09, DSAM), 0.215 ps *75* (1972Va06, DSAM), 0.21 ps *+6−4* (1972Br45, DSAM), 0.27 ps *+19−10* (1969Du08, DSAM) from (p,p′γ), 0.16 ps *5* (1969Ha17, DSAM), and 0.230 ps *25* (1977Sc36, DSAM) from Coulomb excitation.
1763.15 *4*	5/2$^+$	0.36 ps *3*	B EFGHI KLM O QRSTUVWXYZ	XREF: Others: AA, AB

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	J^{π}‡	$T_{1/2}$ or Γ&	XREF	Comments
				T=1/2 B(E2)↑=0.0106 *7* XREF: X(1785)Y(1750) B(E2)↑: from Coulomb excitation. E(level): weighted average of 1763.1 *2* from ^{35}Ar ε decay, 1763.0 *6* from (^{28}Si,αpγ), 1762.95 *17* from (^{24}Mg,αpγ), 1763.3 *2* from (^{16}O,αpγ), 1763.19 *10* from (^{14}N,αpnγ), 1763.14 *4* from (p,γ), 1763.26 *10* from (p,p′γ). J^{π}: L=2 from 0^+ in (d,n); spin=5/2 from pγ(θ) or γ(θ) in (α,pγ), (p,γ), and (p,p′γ). $T_{1/2}$: lifetime τ=0.52 ps *5*: weighted average of 0.54 ps *7* from (^{34}S,nγ) (1973Wa10, DSAM), 0.55 ps *15* from (α,pγ) (1969In04, DSAM), 0.50 ps *10* (1972Hu11, DSAM), 0.40 ps *14* (1973Fa07, DSAM), 0.53 ps *9* (1976Me12, DSAM) from (p,γ), 0.60 ps *21* from (γ,γ′) (1962Bo17, resonance fluorescence), 0.205 ps *75* (1972Va06, DSAM), 0.49 ps *+9−6* (1972Br45, DSAM), 1.15 ps *+45−63* (1969Du08, DSAM) from (p,p′γ), 0.57 ps *7* (1969Ha17, DSAM), and 0.63 ps *5* (1977Sc36, DSAM) from Coulomb excitation.
2645.67 *8*	$7/2^+$	0.159 ps *21*	B FGHI lM O Qr TU y	XREF: Others: AA, AB XREF: r(2645)y(2650) E(level): weighted average of 2645.81 *18* from (^{24}Mg,αpγ), 2645.9 *3* from (^{16}O,αpγ), 2645.45 *12* from (^{14}N,αpnγ), 2645.70 *8* from (p,γ), and 2645.74 *17* from (p,p′γ). Others: 2645.1 *10* from ^{35}Ar ε decay, 2645.8 *6* from (^{28}Si,αpγ), and 2645.8 *8* from (α,pγ). J^{π}: spin=7/2 from pγ(θ) in (α,pγ) and γ(θ) in (p,γ); parity from 882γ, M1+E2 to $5/2^+$, 1763 level in (α,pγ). $T_{1/2}$: lifetime τ=0.229 ps *30*: weighted average of 0.30 ps *9* from (α,pγ) (1969In04, DSAM), 0.255 ps *65* (1972Hu11, DSAM), 0.200 ps *30* (1976Me12, DSAM) in (p,γ), 0.27 ps *9* in (n,n′γ) (1989Ge09, DSAM), 0.185 ps *50* (1972Va06, DSAM), 0.26 ps *6* (1972Br45, DSAM), 0.35 ps *+10−7* (1969Du08, DSAM) in (p,p′γ), and 0.26 ps *9* in Coulomb excitation (1977Sc36, DSAM). Other: τ<1.3 ps in (^{28}Si,αpγ) (2007Ks01, DSAM).
2694.00 *20*	$3/2^+$	32 fs *6*	B LM O QRSTU WXy	XREF: Others: AA XREF: L(2693?)X(2676)y(2650) E(level): weighted average of 2693.8 *2* from ^{35}Ar ε decay, 2693.6 *13* from (α,pγ), 2693.9 *3* from (p,γ), 2694.50 *28* from (p,p′γ). J^{π}: spin=3/2 from γ(θ) in (p,p′γ); L=2 from 0^+ in (^{3}He,d) and (d,^{3}He). $T_{1/2}$: lifetime τ=46 fs *9*: unweighted average of 70 fs *30* in (α,pγ) from 1972Br33 with DSAM, 21 fs *3* in (α,pγ) from 1972Hu11 with DSAM, 19 fs *4* in (α,pγ) from 1973Fa07 with DSAM, 20 fs *4* in (p,γ) from 1976Me12 with DSAM, 65 fs *20* in (p,γ) from 1972Va06 with DSAM, 62 fs *16* in (p,p′γ) from 1969Du08 with DSAM, and 62 fs *8* in (n,n′γ) from 1989Ge09 with DSAM. Others: 115 fs *+95−59* in

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	$T_{1/2}$ or Γ&	XREF	Comments
				(p,p′γ) from 1969Du08 (using 931γ) is considered less reliable by the authors, and 11 fs *7* in (p,p′γ) from 1971Ca40 with DSAM is considered an outlier.
3002.64 *11*	5/2$^+$	22 fs *6*	B F HI LM O QRSTU WXY	E(level): weighted average of 3002.3 *3* from ^{35}Ar ε decay, 3003.1 *8* from (^{28}Si,αpγ), 3002.0 *10* from (^{16}O,αpγ), 3001.98 *22* from (^{14}N,αpnγ), 3002.6 *15* from (α,pγ), 3002.75 *8* from (p,γ), 3002.6 *5* from (p,p′γ). Jπ: L=2 from 0$^+$ in (^{3}He,d), (d,^{3}He), and (pol d,^{3}He) with L+1/2 transfer from analyzing power. Discrepancy: L=3 from 0$^+$ in (d,n) and the angular distribution fit seems good. $T_{1/2}$: lifetime τ=32 fs *8*: unweighted average of 48 fs *9* in (α,pγ) from 1972Br33 with DSAM, 22 fs *3* in (α,pγ) from 1972Hu11 with DSAM, 14 fs *2* in (α,pγ) from 1973Fa07 with DSAM, 16 fs *4* in (p,γ) from 1976Me12 with DSAM, 72 fs *12* in (n,n′γ) from 1989Ge09 with DSAM, 18 fs *+26−15* in (p,p′γ) from 1972Va06 with DSAM, and 31 fs *13* in (p,p′γ) from 1969Du08 with DSAM.
3162.87 *9*	7/2$^-$	29.2 ps *12*	FGHI KLM O QR TU WX	T=1/2 E(level): weighted average of 3163.1 *6* from (^{28}Si,αpγ), 3163.43 *19* from (^{24}Mg,αpγ), 3162.9 *3* from (^{16}O,αpγ), 3162.64 *9* from (^{14}N,αpnγ), 3163.1 *11* from (α,pγ), 3162.99 *7* from (p,γ), 3162.75 *10* from (p,p′γ). Jπ: spin=7/2 from pγ(θ) in (α,pγ) and γ(θ) in (p,γ) and (p,p′γ); L=3 from 0$^+$ in (d,n), (^{3}He,d), (d,^{3}He), and (pol d,^{3}He). $T_{1/2}$: lifetime τ=42.1 ps *17*: weighted average of 42 ps *2* (1974Va13, RDM), 41.7 ps *17* (1976Ke02, RDM), 45 ps *6* (1991Ja11, RDM) from ^{24}Mg(^{16}O, αpγ), 41.8 ps *18* from ^{27}Al(^{14}N, αpnγ) (1976Wa11, RDM), 60 ps *7* from (α,pγ) (1969In04, DSAM), 37 ps *4* from (p,γ), (1971Ba23, delayed coincidence), and 46 ps *6* from (p,p′γ) (1974Hu09, RDM). Others: τ=140 ps *40* (1968Az02, delayed coincidence; obtained using NaI(Tl) detectors with broad time resolution) from (p,γ).
3918.47 *12*	3/2$^+$	4.9 fs *14*	B M O RS u x	XREF: R(3908)u(3930)x(3948) E(level): weighted average of 3918.4 *5* from ^{35}Ar ε decay, 3915 *2* from (α,pγ), 3918.48 *10* from (p,γ). Jπ: spin=3/2 determined by the feeding 5162γ(θ) from 5/2$^+$, 9081 level in (p,γ); parity=+ determined by 3918γ, M1+E2 to 3/2$^+$ ground state. L=2 from 0$^+$ in (pol d,^{3}He). $T_{1/2}$: lifetime τ=7 fs *2* from (p,γ), (1972Hu11, DSAM). Others: <22 fs from (α,pγ) (1972Br33, DSAM).
3942.9 *2*	9/2$^+$	0.18 ps *3*	F HI LM O	E(level): weighted average of 3942.0 *7* from (^{28}Si,αpγ), 3943.4 *10* from (^{16}O,αpγ), 3942.3 *10* from (^{14}N,αpnγ), 3943.4 *12* from (α,pγ), and 3942.9 *2* from (p,γ). Jπ: spin=9/2 from pγ(θ) in (α,pγ); 2180.1γ E2 to 5/2$^+$. $T_{1/2}$: lifetime τ=0.26 ps *5*: weighted average of 0.33 ps *5* in (α,pγ) (1972Br33, DSAM), 0.39 ps *11* (1972Hu11,

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				DSAM), 0.145 ps *50* (1973Fa07, DSAM), and 0.29 ps *5* (1976Me12, DSAM) from (p,γ). Other: τ<0.5 ps in (^{28}Si,αpγ) (2007Ks01, DSAM).
3967.6 *4*	1/2$^+$	11.8 fs *28*	B M O R	E(level): weighted average of 3967.7 *4* from ^{35}Ar ε decay, 3968 *2* from (α,pγ), and 3967.3 *6* from (p,γ). J^{π}: L=0 from 0$^+$ in (^{3}He,d). $T_{1/2}$: lifetime τ=17 fs *4*: weighted average of 13 fs *4* (1972Hu11, DSAM), 20 fs *5* (1973Fa07, DSAM), and 20 fs *4* (1976Me12, DSAM) in (p,γ). Other: τ<54 fs from (α,pγ) (1972Br33, DSAM).
4059.1 *4*	(3/2)$^-$	13.9 fs *21*	LM O QR	E(level): weighted average of 4056.9 *27* from (α,p), 4058 *3* from (α,pγ), 4059.2 *4* from (p,γ). J^{π}: L=1 from 0$^+$ in (^{3}He,d); (3/2$^-$) from FRESCO-DWBA analysis of measured $\sigma(\theta)$ of (α,p) (2019Se09). $T_{1/2}$: lifetime τ=20 fs *3*: weighted average of 31 fs *13* from (α,pγ) (1972Br33, DSAM), 22 fs *3* (1972Hu11, DSAM), 20 fs *4* (1973Fa07, DSAM), and 18 fs *3* (1976Me12, DSAM) from (p,γ).
4112.4 *16*	7/2$^+$	49 fs *11*	LM O Q	E(level): weighted average of 4111.1 *29* from (α,p), 4108 *2* from (α,pγ), and 4113.7 *10* from (p,γ). J^{π}: 4113.4γ E2(+M3) to 3/2$^+$. $T_{1/2}$: lifetime τ=70 fs *16* from (α,pγ) (1972Br33, DSAM).
4173.43 *23*	(5/2$^+$)	35 fs *6*	M O q	T=1/2 XREF: q(4170) E(level): weighted average of 4171 *2* from (α,pγ) and 4173.45 *19* from (p,γ). J^{π}: based on primary transitions from proton resonances in (p,γ): 3827.3γ from 8001.0, (7/2$^+$), 4456.4γ from 8630.1, 7/2$^-$ level, 4243.0γ from 8416.7, 1/2$^+$ level. Note that primary 3099.1γ from 7272.7, 1/2$^-$ resonance disfavors 5/2$^+$ assignment, but this γ branch is relatively weak. $T_{1/2}$: lifetime τ=51 fs *8*: weighted average of 68 fs *24* in (α,pγ) (1972Br33, DSAM), 105 fs *+35−30* (1971Wi13, DSAM), 55 fs *6* (1972Hu11, DSAM), and 34 fs *9* (1976Me12, DSAM) in (p,γ).
4177.9 *2*	3/2$^-$	27.0 fs *28*	L O qR	XREF: q(4170)R(4167) J^{π}: spin=3/2 from $\gamma(\theta)$ in (p,γ); L=1 from 0$^+$ in (^{3}He,d). $T_{1/2}$: lifetime τ=39 fs *4*: weighted average of 32 fs *5* (1972Hu11, DSAM), 42 fs *4* (1973Fa07, DSAM), and 47 fs *10* (1976Me12, DSAM) from (p,γ).
4347.76 *7*	9/2$^-$	1.0 ps *6*	FGHI LM O	E(level): weighted average of 4347.8 *6* from (^{28}Si,αpγ), 4348.29 *24* from (^{24}Mg,αpγ), 4347.8 *3* from (^{16}O,αpγ), 4347.43 *17* from (^{14}N,αpnγ), 4347.8 *12* from (α,pγ), and 4347.76 *6* from (p,γ). J^{π}: spin from $\gamma(\theta)$ in (p,γ); parity from 1184.6γ, M1+E2 to 7/2$^-$, 3163 level in (p,γ) (1974Lo17). $T_{1/2}$: lifetime τ=1.5 ps *8*: unweighted average of 1.31 ps *+30−23* from (^{28}Si,αpγ) (2007Ks01, DSAM), 2.9 ps *+14−7* (1972Br33, DSAM), and 0.33 ps *6* (1976Me12, DSAM) from (p,γ).
4624.3 *3*	3/2,5/2	40 fs *17*	B LM O	XREF: B(?) E(level): other: 4625.3 *28* from (α,p); 4618 *2* from

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				(α,pγ) is discrepant. J^{π}: from p$\gamma(\theta)$ in (α,pγ). $T_{1/2}$: lifetime τ=58 fs *24* from (α,pγ) in 1972Br33 with DSAM.
4769.86 *23*	($5/2^-$)	76 fs *20*	LM O	E(level): others: 4770.4 *26* from (α,p), 4770 *15* from (α,pγ). J^{π}: from primary transitions in (p,γ): 4059.2γ from 8829.3, $1/2^-$ level, 2778.9γ from 7548.9, $7/2^-$ level, 3231.0γ from 8001.0, ($7/2^+$) level. Other: 7/2,9/2 from p$\gamma(\theta)$ for a group at E=4770 *15* in (α,pγ) is inconsistent and disfavored by γ rays to $1/2^+$ and $3/2^+$. $T_{1/2}$: lifetime τ=110 fs *29*: weighted average of 270 fs *95* (1972Hu11, DSAM) and 105 fs *17* (1976Me12, DSAM) in (p,γ).
4839.10 *11*	($1/2^+$,3/2)	9.7 fs *35*	O x	XREF: x(4839) J^{π}: from primary γ transitions in (p,γ): 2339.4γ from 7178.6, $1/2^{(+)}$ level, 2355.4γ from 7194.6, $1/2^-$ level, 2867.2γ from 7706.4, $5/2^+$ level. $T_{1/2}$: lifetime τ=14 fs *5* from (p,γ) (1976Me12, DSAM).
4854.4 *19*	(1/2,3/2)	4.9 fs *14*	L O x	XREF: x(4839) E(level): other: 4859.2 *28* from (α,p). J^{π}: from primary γ transitions in (p,γ): 2340.1γ from 7194.6, $1/2^-$ level, 3183.9γ from 8038.4, $1/2^+$,$3/2^+$ level; decay 3634.8 γ to 1219.38, $1/2^+$. $T_{1/2}$: lifetime τ=7 fs *2* from (p,γ) (1976Me12, DSAM).
4880.96 *15*	$7/2^{(+)}$	5.6 fs *14*	LM O	E(level): others: 4881.8 *29* from (α,p), 4880.8 *21* from (α,pγ). J^{π}: spin=7/2 from p$\gamma(\theta)$ in (α,pγ); 3417.1 primary γ from 8298.2, $3/2^+$ resonance in (p,γ). $T_{1/2}$: lifetime τ=8 fs *2*: weighted average of 7 fs *2* (1972Hu11, DSAM), 7 fs *3* (1973Fa07, DSAM), and 10 fs *3* (1976Me12, DSAM) from (p,γ). Other: τ=0.28 ps *6* from (α,pγ) in 1972Br33 with DSAM.
5009.1 *18*	(1/2,3/2)	7.6 fs *21*	LM O R	E(level): weighted average of 5006.9 *27* from (α,p) and 5010.1 *18* from (p,γ). Others: 5015 *20* from (α,pγ) and 5010 *15* from (^{3}He,d). J^{π}: primary 2185.4γ from 7194.6, $1/2^-$ level and 3407.4γ from 8416.7, $1/2^+$ level. $T_{1/2}$: lifetime τ=11 fs *3* from (p,γ) (1976Me12, DSAM).
5158.6 *29*	$3/2^+$,$5/2^+$		Lm R WX	XREF: m(5175) E(level): weighted average of 5158.9 *29* from (α,p), 5163 *15* from (^{3}He,d), 5150 *20* from ^{36}Ar(d,^{3}He), and 5155 *11* from ^{36}Ar(pol d,^{3}He). J^{π}: L=2 from 0^+ in (pol d,^{3}He) and (d,^{3}He).
5163.34 *22*	$7/2^-$		m O	XREF: m(5175) J^{π}: spin=7/2 from the feeding 3466$\gamma(\theta)$ from 8630, $7/2^-$ level and RUL in (p,γ); parity=− from 3466γ, M1+E2 from 8630, $7/2^-$ level.
5214.5 *20*	($5/2^+$)[#]	<4.85 fs	LM O	XREF: M(5230) E(level): weighted average of 5209.5 *28* from (α,p),

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				5230 *20* from (α,pγ), and 5215.8 *15* from (p,γ). J^{π}: other: 3/2,5/2 from pγ(θ) in (α,pγ). $T_{1/2}$: lifetime τ<7 fs from (p,γ) (1976Me12, DSAM).
5403.3 *10*	3/2$^-$	11.8 fs *28*	L O R	E(level): weighted average of 5402.0 *29* from (α,p) and 5403.5 *10* from (p,γ). Other: 5409 *12* from (^{3}He,d). J^{π}: L=1 from 0$^+$ in (^{3}He,d); 2400.6γ to 5/2$^+$ rules out 1/2$^-$ based on RUL, since it would require a large B(M2)(W.u.) exceeding RUL. $T_{1/2}$: lifetime τ=17 fs *4* from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 36% of Iγ(4183)=100 *18* from (p,γ).
5407.07[a] *21*	11/2$^-$	0.28 ps *7*	FGHI LM	E(level): weighted average of 5407.4 *8* from (^{28}Si,αpγ), 5407.64 *27* from (^{24}Mg,αpγ), 5407.3 *3* from (^{16}O,αpγ), 5406.68 *19* from (^{14}N,αpnγ), and 5407.1 *15* from (α,pγ). J^{π}: spin=11/2 from pγ(θ) in (p,γ); parity from 1059γ, M1+E2 to 9/2$^-$, 4347 level. $T_{1/2}$: lifetime τ=0.4 ps *1* from (α,pγ) (1974Lo17, DSAM). Other: τ<1.1 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
5531 *5*			LM O	E(level): weighted average of 5531 *5* from (α,p) and 5535 *20* from (α,pγ).
5586.0 *20*	5/2$^+$		Lm O wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5586.0 *26* from (α,p) and 5586 *2* from (p,γ). J^{π}: L=2 from 0$^+$ in (pol d,^{3}He) and L+1/2 transfer from analyzing power.
5598.4 *22*	3/2$^+$,5/2$^+$	2.1 fs *7*	Lm O wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5594.6 *26* from (α,p) and 5599.7 *15* from (p,γ). J^{π}: L(d,^{3}He)=2 from 0$^+$. $T_{1/2}$: lifetime τ=3 fs *1* from (p,γ) (1976Me12, DSAM).
5633.6 *30*			Lm r w y	XREF: m(5650)r(5660)w(5600)y(5650) E(level): from (α,p).
5646 *2*	(5/2,7/2,9/2$^+$)	2.8 fs *7*	Lm O r y	XREF: m(5650)r(5660)y(5650) E(level): weighted average of 5645 *4* from (α,p) and 5646 *2* from (p,γ). J^{π}: from primary transitions in (p,γ): 1903γ from 7548.9, 7/2$^-$ level, 2355γ from 8001.0, (7/2$^+$) level, 3261γ from 8906.8, 5/2$^+$ level. $T_{1/2}$: lifetime τ=4 fs *1* from (p,γ) in 1976Me12 with DSAM.
5655 *2*	(3/2)$^+$	13.9 fs *35*	Lm O r w y	T=3/2 XREF: m(5650)r(5660)w(5600)y(5650) E(level): weighted average of 5653 *4* from (α,p) and 5655 *2* from (p,γ). J^{π}: L=2 from 0$^+$ in (^{3}He,d); primary 2142γ from 7797.2, 1/2$^-$ level in (p,γ). Note that a weak primary 1903γ from 7548.9, 7/2$^-$ level in

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				(p,γ) favors $5/2^+$. $T_{1/2}$: lifetime τ=20 fs *5* from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 18% of Iγ(2652)=100 *6* from (p,γ).
5682.2 *20*	$1/2^-,3/2^-$		Lm O R	XREF: m(5650) E(level): weighted average of 5679.7 *34* from (α,p), 5683 *2* from (p,γ), and 5684 *12* from (^{3}He,d). J^{π}: L=1 from 0^+ in (^{3}He,d). The contribution of Iγ to unknown levels adds up to 18% of Iγ(1504)=100 from (p,γ).
5723.52 *18*	$5/2^+$		L O WX	E(level): others: 5731.2 *25* from (α,p), 5720 *20* from ^{36}Ar(d,^{3}He), and 5720 *11* from ^{36}Ar(pol d,^{3}He). J^{π}: L=2 from 0^+ in (pol d,^{3}He) and (d,^{3}He) and L+1/2 transfer from analyzing power. The contribution of Iγ to unknown levels adds up to 100% of Iγ(3960)=100 *11* from (p,γ).
5758 *3*	$(1/2,3/2^-)$		L O R	E(level): weighted average of 5757 *4* from (α,p), 5758 *3* from (p,γ), and 5760 *12* from (^{3}He,d). J^{π}: L(^{3}He,d)=(0,1) from 0^+. The contribution of Iγ to unknown levels adds up to 56% of Iγ(5758)=100 *22* from (p,γ).
5806.6 *20*	(1/2,3/2,5/2)	3.5 fs *7*	L O	E(level): weighted average of 5808.1 *31* from (α,p) and 5806 *2* from (p,γ). J^{π}: primary 2890.2γ from 8696.9, $3/2^-$ level; 5806.1γ to $3/2^+$ g.s. $T_{1/2}$: lifetime τ=5 fs *1* from (p,γ) (1976Me12, DSAM).
5830 *4*	$(5/2^-)$[#]		LM	XREF: M(5850) E(level): weighted average of 5829 *4* from (α,p) and 5850 *25* from (α,pγ). J^{π}: other: spin=5/2,9/2 from pγ(θ) in (α,pγ).
5926.65 *29*	$(11/2)^-$	<0.28 ps	F HI LM	E(level): weighted average of 5927.1 *7* from (^{28}Si,αpγ), 5926.9 *5* from (^{16}O,αpγ), 5926.47 *29* from (^{14}N,αpnγ), 5928 *3* from (α,p), and 5927.1 *19* from (α,pγ). J^{π}: 1579γ M1+E2, ΔJ=1 to 4348, $9/2^-$ in ^{12}C(^{28}Si,αpγ), also assuming spins increase with excitation energies in fusion-evaporation reactions. Spin=7/2,11/2 from pγ(θ) in (α,pγ). $T_{1/2}$: lifetime τ<0.4 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
6087.31 *24*	$13/2^-$	6.2 ps *6*	FGHI LM	E(level): weighted average of 6087.6 *8* from (^{28}Si,αpγ), 6088.05 *29* from (^{24}Mg,αpγ), 6087.2 *4* from (^{16}O,αpγ), 6086.89 *23* from (^{14}N,αpnγ), 6084.2 *29* from (α,p), and 6087.0 *18* from (α,pγ). J^{π}: spin=13/2 from pγ(θ) in (α,pγ); parity from 680γ, M1+E2 to $11/2^-$, 5406 level in (α,pγ). $T_{1/2}$: lifetime τ=8.9 ps *8*: weighted average of 9.3 ps *8* from (α,pγ) in 1974Lo17 with RDM and 7.7 ps *14* from (^{14}N,αpnγ) in 1976Wa11.
6106.2 *21*	$(3/2,5/2^+)$	8.3 fs *21*	l O wx	XREF: l(6104)w(6140)x(6136) E(level): weighted average of 6104 *4* from (α,p)

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	XREF	Comments
			and 6106.2 *21* from (p,γ). J^{π}: 4886.5γ to 1/2$^+$, 4342.8γ to 5/2$^+$; primary 1675.5γ from 7781.7, (5/2$^-$) level. $T_{1/2}$: lifetime τ=12 fs *3* from (p,γ) in 1976Me12 with DSAM.
6141 *4*	5/2$^+$	L wx	XREF: w(6140)x(6136) J^{π}: L=2 from 0$^+$ in (d,^{3}He) and (pol d,^{3}He), and L+1/2 transfer from analyzing power.
6181 *3*	(1/2$^+$ to 7/2$^+$)	L nO	XREF: n(6200) E(level): other: 6180 *3* from (α,p). J^{π}: 6180γ to 3/2$^+$ g.s.; primary 1438γ from 7618.7, 5/2$^+$ level in (p,γ). Apart from the γ branch Iγ(6180)=100 observed in (p,γ), the contribution of Iγ to unknown levels adds up to 122% of Iγ(6180).
6224.4 *30*		L n	XREF: n(6200) J^{π}: L(α,d)=6 for a group at 6200 *10*.
6283 *3*	(5/2$^+$)[#]	L R	E(level): weighted average of 6283 *3* from (α,p) and 6284 *4* from (^{3}He,d). J^{π}: other: L=(2) from 0$^+$ in (^{3}He,d).
6329 *4*	(1/2,3/2$^-$)	R	J^{π}: L=(0,1) from 0$^+$ in (^{3}He,d).
6380.2 *9*	(9/2$^-$)[#]	H L R	XREF: R(6377) E(level): weighted average of 6380.8 *8* from (^{16}O,αpγ) and 6379.3 *25* from (α,p). J^{π}: other: L=(2,3) from 0$^+$ in (^{3}He,d) for a group at E=6377 *2* based on very few data points in a very narrow range of angles in the DWBA fit.
6401.2 *22*	(1/2$^-$)[#]	LM	
6428 *2*	(1/2$^+$)[#]	L R	E(level): weighted average of 6429 *3* from (α,p) and 6427 *2* from (^{3}He,d). J^{π}: other: L=(3) from 0$^+$ in (^{3}He,d) is inconsistent.
6469 *3*	(1/2$^-$,3/2$^-$)	L R	E(level): weighted average of 6475 *4* from (α,p) and 6468 *2* from (^{3}He,d). J^{π}: L=(1) from 0$^+$ in (^{3}He,d).
6492 *2*	(3/2$^+$)[#]	L O R	E(level): weighted average of 6492 *3* from (α,p), 6492 *3* from (p,γ), and 6491 *2* from (^{3}He,d). J^{π}: other: L=(2) from 0$^+$ in (^{3}He,d).
6546 *2*	(1/2$^+$)[#]	L R	E(level): weighted average of 6549 *3* from (α,p) and 6545 *2* from (^{3}He,d). J^{π}: other: L=(0,1) from 0$^+$ in (^{3}He,d), indicating (1/2,3/2$^-$).
6643 *2*	(1/2$^-$,3/2$^-$)	D R	E(level): from (^{3}He,d). J^{π}: from L=(1) from 0$^+$ in (^{3}He,d).
6659.1 *30*	(7/2$^+$)[#]	L y	XREF: y(6677)
6660.2 *8*	(11/2$^-$)	F	J^{π}: 2312.4γ to 9/2$^-$, 3497.1γ to 7/2$^-$ in (^{28}Si,αpγ), assuming spin increasing as excitation energy increases.
6675.6 *25*		D L R y	XREF: y(6677) E(level): weighted average of 6679.4 *31* from (α,p) and 6674 *2* from (^{3}He,d).
6761 *2*	3/2$^+$,5/2$^+$	D R WX	XREF: X(6745) E(level): weighted average of 6761 *2* from (^{3}He,d), 6750 *20* from ^{36}Ar(d,^{3}He), and 6745 *12* from ^{36}Ar(pol d,^{3}He). J^{π}: L(d,^{3}He)=2 from 0$^+$.
6779.0 *20*		D L R	E(level): weighted average of 6781.3 *30* from (α,p) and 6778 *2*

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				from (^{3}He,d).
6800 *4*			D L	XREF: D(6802) E(level): from (α,p).
6823 *2*			D R	E(level): from (^{3}He,d).
6842 *2*			D L R	E(level): weighted average of 6842 *4* from (α,p) and 6842 *2* from (^{3}He,d).
6863 *3*	$(9/2^+)$[#]		L	
6866.7 *6*	$(1/2,3/2^-)$		D O R	E(level): other: 6866 *2* from (^{3}He,d). J^{π}: L(^{3}He,d)=(0,1) from 0^+.
6891.6 *30*	$(9/2^+)$[#]		L	
6948.1 *34*	$5/2^+$		L WX	XREF: W(6930) E(level): weighted average of 6948.6 *34* from (α,p) and 6930 *20* from ^{36}Ar(d,^{3}He). Other: 6953 *43* from ^{36}Ar(pol d,^{3}He). J^{π}: L(pol d,^{3}He)=2 and L+1/2 from analyzing power.
6986.3 *34*	$(9/2^+)$[#]		L	
7066.2 *7*	$(5/2^+)$[#]		L O R	E(level): weighted average of 7066.2 *7* from (p,γ), and 7066 *2* from (^{3}He,d).
7103.4 *7*	$3/2^{(-)}$		L O R	XREF: L(7105?) E(level): weighted average of 7103.4 *7* from (p,γ), and 7103 *2* from (^{3}He,d). J^{π}: spin=3/2 from γ(θ) in (p,γ); parity from L(^{3}He,d)=(1,3) from 0^+.
7121.7 *24*			L	
7178.6 *8*	$1/2^{(+)}$		O	T=3/2 J^{π}: spin=1/2 from γ(θ) in (p,γ); parity from IAS in ^{35}S.
7179 *4*	$(7/2)^+$[#]		L N	E(level): weighted average of 7180 *4* from ^{32}S(α,p) and 7170 *10* from ^{33}S(α,d). J^{π}: also L=6 from $3/2^+$ in ^{33}S(α,d), indicating $(7/2:17/2)^+$.
7185	$5/2^+$		O R X	XREF: R(7178)X(7181) J^{π}: L(pol d,^{3}He)=2 and L+1/2 from analyzing power for a group at E=7181 *60*. Other: L(^{3}He,d)=(2) from 0^+ for a group at E=7178 *2* considered the same level.
7194.6 *7*	$1/2^-$	0.027 keV *8*	OP R	J^{π}: from resonance analysis of measured σ(θ) (1971BrXT) in ^{34}S(p,p),(p,p′γ):resonances. Other: 1/2,3/2 from γ(θ) in (p,γ).
7213.7 *25*			L y	XREF: y(7250)
7225.5 *8*	5/2		O y	XREF: y(7250) J^{π}: from γ(θ) in (p,γ).
7228.2 *20*	$(3/2^+)$[#]		L R y	XREF: R(7227)y(7250) E(level): weighted average of 7230.9 *30* from ^{32}S(α,p), 7227 *2* from ^{34}S(^{3}He,d). J^{π}: other: L=(0,1) from 0^+ in (^{3}He,d) is inconsistent.
7234.0 *8*	$5/2^{(+)}$		O	J^{π}: spin=5/2 from γ(θ) in (p,γ); primary 6014.1γ to $1/2^+$.
7269.2 *1*			O	
7272.7 *9*	$1/2^-$	0.014 keV *4*	OP R y	XREF: P(7274)R(7273)y(7250)

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				J^{π}: from resonance analysis of measured $\sigma(\theta)$ (1971BrXT) in ^{34}S(p,p),(p,p′γ):resonances. Other: (1/2,3/2) from $\gamma(\theta)$ in (p,γ).
7348 *3*	(7/2)[#]		L	
7361.9 *8*	$3/2^{(+)}$		L O R	E(level): weighted average of 7362 *3* from (α,p), 7362.0 *8* from (p,γ), and 7361 *2* from (^{3}He,d). J^{π}: spin=3/2 from $\gamma(\theta)$ in (p,γ); 6141.9γ (M1+E2) to $1/2^+$.
7396.4 *10*	$7/2^{(-)}$		O R	E(level): weighted average of 7396.0 *10* from (p,γ), and 7398 *2* from (^{3}He,d). J^{π}: spin=7/2 from $\gamma(\theta)$ in (p,γ); 4233.3γ (M1+E2) to $7/2^-$.
7413.8 *30*			L	
7450.9 *8*	3/2		L O	E(level): weighted average of 7447 *3* from (α,p) and 7451.1 *6* from (p,γ). J^{π}: from $\gamma(\theta)$ in (p,γ).
7496			O	
7501.1 *8*	$(5/2^+,7/2,9/2^+)$		l O	XREF: l(7503) J^{π}: primary 3558.0γ to $9/2^+$, 4498.2γ to $5/2^+$.
7502.9 *7*			l O	XREF: l(7503)
7518.8 *7*	7/2		O	J^{π}: from $\gamma(\theta)$ in (p,γ).
7548.9 *6*	$7/2^-$	<0.69 fs	OP	T=3/2 XREF: P(7550) J^{π}: from resonance analysis of measured $\sigma(\theta)$ in ^{34}S(p,p),(p,p′γ):resonances. Spin=7/2 also from $\gamma(\theta)$ in (p,γ). $T_{1/2}$: lifetime τ<1 fs from (p,γ) (1973Fa07, DSAM). Other: Γ_p=4.6 eV *10* and Γ=5 eV *+4−2* from ^{34}S(p,p) elastic scattering excitation function analysis (1971BrXT). Other Γ_p, Γ_γ, Γ values/ratios are mostly deduced from (p,γ) resonance strengths, which have large discrepancies. See comments in the (p,γ) dataset for details.
7561.3 *6*	1/2,3/2		L O	E(level): other: 7566.5 *35* from (α,p). J^{π}: from $\gamma(\theta)$ in (p,γ).
7565 *20*	$5/2^+$		X	J^{π}: L=2 from 0^+ in (pol d,^{3}He) and L+1/2 transfer from analyzing power.
7590 *4*			L	
7600.8 *8*	$5/2^+$	<13.9 fs	O	J^{π}: spin=5/2 from $\gamma(\theta)$ in (p,γ); 7599.9γ M1+E2 to $3/2^+$. $T_{1/2}$: lifetime τ<20 fs from (p,γ) (1971Wi13, DSAM), which is equivalent to Γ>0.033 eV.
7618.7 *5*	$5/2^+$		O	J^{π}: spin=5/2 from $\gamma(\theta)$ in (p,γ); 7617.8γ M1+E2 to $3/2^+$.
7648.1 *30*			L	
7656.6 *8*	$(1/2,3/2,5/2^+)$		O	J^{π}: primary 3688.8γ to $1/2^+$.
7670 *10*	$(7/2\text{ to }17/2)^+$		N	J^{π}: L(α,d)=6 from $3/2^+$.
7671.9 *8*	$5/2^+$		l Op	XREF: p(7686) J^{π}: spin=5/2 from $\gamma(\theta)$ in (p,γ); 4668.8γ, M1+E2 to $5/2^+$; 5908.2γ, M1+E2 to $5/2^+$; other: 3324.0γ to $9/2^-$ seems to favor $5/2^-$.
7685.5 *8*	$3/2^-$		OP	J^{π}: spin=3/2 from $\gamma(\theta)$ in (p,γ) and $3/2^-$ from R-Matrix analysis in (p,p):resonances. Γ_p=446 eV *15* from (p,p):resonances.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
7693.9 *8*	(1/2$^+$,3/2,5/2)		O	J^{π}: primary 7693.0γ to 3/2$^+$, 3634.6γ to (3/2)$^-$, 4690.9γ to 5/2$^+$.
7706.4 *8*	5/2$^+$		L OP	E(level): other: 7709.9 *25* from (α,p). J^{π}: spin=5/2 from γ(θ) in (p,γ) and 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances. Γ_p=4 eV *1* in (p,p).
7744.8 *9*	7/2$^-$		L O	E(level): other: 7744.1 *30* from (α,p). J^{π}: spin=7/2 from γ(θ) in (p,γ); 4581.6γ M1+E2 to 7/2$^-$.
7748	(7/2 to 17/2)$^+$		NO	XREF: N(7750) E(level): from (p,γ). J^{π}: L(α,d)=6 from 3/2$^+$.
7777.1 *9*	(5/2$^+$)		L O	E(level): from (p,γ). Other: 7770.3 *33* from (α,p). J^{π}: based on primary transitions: 4613.9γ to 7/2$^-$, 5131.0γ to 7/2$^+$, 6557.1γ to 1/2$^+$.
7781.7 *11*	(5/2$^-$)		O	J^{π}: primary 3433.8γ to 9/2$^-$; relatively weak 6561.7γ to 1/2$^+$.
7797.2 *9*	1/2$^-$	0.031 keV *10*	L OP	E(level): weighted average of 7799.8 *30* from (α,p), 7797.1 *9* from (p,γ), and 7796 *3* from ^{34}S(p,p),(p,p′γ): resonances. J^{π}: spin=1/2 from γ(θ) in (p,γ); (1/2)$^-$ from R-Matrix analysis of in (p,p):resonances.
7837.2 *10*	3/2$^-$	3.7 keV *4*	OP	T=3/2 E(level): weighted average of 7837.2 *10* from (p,γ), and 7837 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^{π}: spin=3/2 from γ(θ) in (p,γ) and 3/2$^-$ from R-Matrix analysis in (p,p):resonances. Γ: other: lifetime τ<5 fs from (p,γ) (1971Wi13, DSAM).
7839.0 *9*	(5/2$^+$)		O	J^{π}: primary 6619.0γ to 1/2$^+$, 3895.9γ to 9/2$^+$.
7868.7 *10*	(1/2$^+$,3/2,5/2$^+$)		L O	E(level): other: 7868.5 *35* from (α,p). J^{π}: primary 6648.6γ to 1/2$^+$, 2654.1γ to (5/2$^+$).
7873.03[b] *30*	13/2$^+$		F HI NO	XREF: N(7870) E(level): weighted average of 7873.3 *8* from (^{28}Si,αpγ), 7873.4 *4* from (^{16}O,αpγ), and 7872.78 *30* from (^{14}N,αpnγ). J^{π}: 1946.35γ E1+M2, ΔJ=1 to 5927, (11/2)$^-$; 1786.2γ D(+Q) to 13/2$^-$; band head.
7880.8 *9*	3/2$^+$,5/2$^+$	0.008 keV *4*	OP	J^{π}: from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
7885			O	
7889.1 *15*			O	
7899.2 *7*	(5/2)		l O	XREF: l(7901) J^{π}: primary 2735.8γ to 7/2$^-$, 3018.1γ to

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				$7/2^{(+)}$, 3721.1γ to $3/2^-$, 5204.9γ to $3/2^+$.
7903			l O	XREF: l(7901)
7923.3 *9*	$(3/2^+,5/2^+)$		O	J^{π}: primary 5277.2γ to $7/2^+$, 6703.2γ to $1/2^+$.
7930			O	
7949			O	
7970.3 *10*	$(5/2^-)$		O	J^{π}: primary 3622.3γ to $9/2^-$, weak 6750.2γ to $1/2^+$.
7978.6 *35*			L	
7987.8 *10*	$3/2^+$		O	J^{π}: spin=3/2 from γ(θ) in (p,γ); 7986.8γ M1+E2 to $3/2^+$.
7995.6 *10*	(5/2)		O	J^{π}: primary 2592.2γ to $3/2^-$, 7994.6γ to $3/2^+$, 2832.1γ to $7/2^-$, 5349.5γ to $7/2^+$.
8001.0 *9*	$(7/2^+)$		L O	E(level): other: 8000.7 *41* from (α,p). J^{π}: from primary transitions in (p,γ): 3653.0γ to $9/2^-$, 4057.9 to $9/2^+$, 3827.3γ to $5/2^-$, 6237.4γ to $5/2^+$, 8000.0γ to $3/2^+$.
8005.0 *10*	$5/2^+$	0.011 keV *5*	OP WX	XREF: P(8004)W(8010)X(8007) J^{π}: L(d,^{3}He)=2 from 0^+; spin=5/2 from γ(θ) in (p,γ).
8010 *10*	$(7/2\text{ to }17/2)^+$		N	J^{π}: L(α,d)=6 from $3/2^+$.
8019.2 *35*			L	
8035.7 *10*	$1/2^-,3/2^-$	0.026 keV *7*	OP	E(level): other: 8034 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8038.4 *11*	$1/2^+,3/2^+$	0.30 keV *2*	OP	E(level): weighted average of 8038.5 *11* from (p,γ), and 8038 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^{π}: spin=3/2 from γ(θ) in (p,γ), but $1/2^+$ from R-Matrix analysis in (p,p); 8037.5γ M1+E2 to $3/2^+$.
8075.9 *10*	(5/2)		L O	E(level): weighted average of 8075.1 *46* from (α,p) and 8075.9 *10* from (p,γ). J^{π}: primary 3897.8γ to $3/2^-$, 8074.9γ to $3/2^+$; 2912.4g to $7/2^-$, 3962.0γ to $7/2^+$.
8080			O	
8096.1 *11*	(5/2)		O	J^{π}: primary 2932.6γ to $7/2^-$, 5450.0γ to $7/2^+$; 3918.0γ to $3/2^-$, 4177.4γ to $3/2^+$.
8100 *10*	$(7/2\text{ to }17/2)^+$		N	J^{π}: L(α,d)=6 from $3/2^+$.
8106.4 *11*	$3/2^+$		L O	E(level): other: 8103.8 *42* from (α,p). J^{π}: spin=3/2 from γ(θ) in (p,γ); 8105.4γ M1+E2 to $3/2^+$.
8113.3 *11*	$(1/2^+,3/2,5/2^+)$		L O	XREF: L(8121?) J^{π}: primary 2897.4γ $(5/2^+)$, 4145.4γ to $1/2^+$.
8147.2 *11*	$3/2^-$	0.56 keV *6*	OP	XREF: P(8150) J^{π}: from R-Matrix analysis in (p,p); also proposed by 1976Me12 in (p,γ) based on strengths of primary γ rays. Γ: from ^{34}S(p,p),(p,p′γ):resonances in 1977Ou01.
8148 *3*	$1/2^-$	2.66 keV *27*	P	
8150 *3*	$3/2^-$	0.56 keV *6*	P	
8156.8 *11*	$(5/2^+)$		O	J^{π}: primary 3978.7γ to $3/2^-$, 8155.8γ to $3/2^+$, 4213.6γ to $9/2^+$.
8171.7 *42*			L	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^π[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
8179.4 *12*	$(3/2^-,5/2,7/2^+)$		O	J^π: primary 5016.1γ to $7/2^-$, 5484.9γ to $3/2^+$.
8208.2 *10*	$5/2^+$	0.033 keV *10*	OP WX	T=3/2 XREF: W(8210)X(8204) E(level): other: 8209 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^π: L(pol d,^{3}He)=2 and L+1/2 from analyzing power.
8211 *3*	$1/2^+$	0.094 keV *15*	L OP	XREF: O(8209) E(level): from (p,p):resonances. Other: 8210.2 *46* from (α,p).
8216.3 *10*	$5/2^+$	0.014 keV *3*	l OP wx	XREF: w(8210)x(8204) J^π: from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8242.4 *12*	$3/2^-$	0.140 keV *15*	l OP	XREF: l(8246.0)P(8243) J^π: from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8251 *5*	$(3/2^+,5/2^+)$		l O	XREF: l(8246.0) J^π: primary 4138γ to $7/2^+$, 7031γ to $1/2^+$.
8269.0 *5*	$5/2^+$	0.005 keV *3*	L OP	XREF: L(8270.6) E(level): other: 8271 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^π: $3/2^+,5/2^+$ from R-Matrix analysis in (p,p):resonances; primary 5105.5γ to $7/2^-$ rules out $3/2^+$, since it would require Mult=M2 which is unlikely based on RUL and Γ=0.005 keV.
8277.2 *5*	$(5/2)^+$	0.006 keV *3*	OP	XREF: P(8279) J^π: $3/2^+,5/2^+$ from R-Matrix analysis in (p,p):resonances; primary 3939.2γ to $9/2^-$ disfavors both $3/2^+$ and $5/2^+$, but $3/2^+$ is much less likely.
8282.4 *5*	$(3/2^-,5/2)$		O	J^π: primary 3118.9γ to $7/2^-$, 4104.2γ to $3/2^-$, 8281.4γ to $3/2^+$.
8287.82 *30*	$1/2^-$	0.04 keV *1*	L OP	XREF: L(8292)P(8290) J^π: from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8297.2 *9*	$3/2^-$	0.073 keV *15*	OP	XREF: P(8300)
8298.2 *5*	$3/2^+$		l O	XREF: l(8292) J^π: spin=3/2 from γ(θ) in (p,γ); 8297.1γ M1+E2 to $3/2^+$.
8318.1 *3*	$(5/2^+)$		L O	XREF: L(8321.0) J^π: primary 4350.2γ to $1/2^+$, 5154.8γ to $7/2^-$, 5671.9γ to $7/2^+$.
8319.6[a] *5*	$15/2^-$	12.5 fs *7*	F H	E(level): weighted average of 8319.4 *9* from (^{28}Si,αpγ) and 8319.7 *5* from (^{16}O,αpγ). J^π: 2911.9γ E2, ΔJ=2 to $11/2^-$, 2232.7γ to $13/2^-$; band assignment. $T_{1/2}$: lifetime τ=18 fs *1* from (^{28}Si,αpγ) (2013Bi10, DSAM). $T_{1/2}$: from (HI,xnγ).
8323.1 *13*	$(3/2^+,5/2^+)$		L O	XREF: L(8321.0) J^π: primary 4210.4γ to $7/2^+$, 7103.0γ to $1/2^+$.
8346 *5*			O	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
8381.8 *8*	5/2$^+$	0.023 keV *7*	OP	XREF: P(8383) J^{π}: from $\gamma(\theta)$ in (p,γ) and R-Matrix analysis in (p,p):resonances.
8389.4 *16*			O	
8402.9 *5*	5/2$^-$	0.002 keV *1*	L OP	XREF: L(8398.8)P(8404) E(level): others: 8398.8 *38* from (α,p), and 8404 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^{π}: 5/2$^-$, 7/2$^-$ from R-Matrix analysis in (p,p):resonances; its primary 5708.5γ to 3/2$^+$, 2694 level rules out 7/2$^-$ based on RUL and Γ=0.002 keV.
8405.8 *14*	(5/2$^-$)	0.001 keV *1*	OP	E(level): weighted average of 8405.7 *14* from (p,γ), and 8406 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^{π}: (5/2$^-$,7/2$^-$) from R-Matrix analysis in (p,p):resonances; primary 8404.6γ to 3/2$^+$.
8409.4 *18*	1/2$^-$	0.125 keV *15*	OP	E(level): weighted average of 8409.6 *18* from ^{34}S(p,γ) and 8409 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8416.7 *5*	1/2$^+$	0.026 keV *7*	OP	E(level): weighted average of 8416.7 *5* from (p,γ), and 8418 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^{π}: 1/2$^+$ from R-Matrix analysis in (p,p):resonances; but primary 4473.5γ to 9/2$^+$ and 7196.5γ to 1/2$^+$ in (p,γ) gives (5/2$^+$), which may indicate two different resonances with close energies.
8430.3 *5*			L O	E(level): other: 8429.2 *40* from (α,p).
8434.8 *5*	(3/2)$^+$	0.090 keV *15*	OP	XREF: P(8435)
8464.3 *5*	(1/2$^+$,3/2,5/2$^+$)		O	J^{π}: primary 4496.4γ to 1/2$^+$, 3249.6γ to (5/2$^+$).
8484.4 *5*	(3/2)$^+$	0.012 keV *5*	l OP	E(level): weighted average of 8484.4 *5* from (p,γ), and 8486 *3* from ^{34}S(p,p),(p,p′γ):resonances. J^{π}: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; 3/2$^+$ proposed by 2001Vo24 in (p,γ) based on proposed isobaric analog to 2940, 3/2$^+$ level in ^{35}S.
8486.2 *5*	3/2$^-$	0.150 keV *15*	l OP	E(level): weighted average of 8486.1 *5* from (p,γ), and 8488 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8487.5 *5*	15/2$^-$	<69 fs	F H	E(level): weighted average of 8487.8 *10* from (^{28}Si,αpγ) and 8487.4 *5* from (^{16}O,αpγ). J^{π}: 3080.0γ E2, ΔJ=2 to 11/2$^-$, 2399.8γ D(+Q) to 13/2$^-$. $T_{1/2}$: lifetime τ<0.1 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
8506.5 *5*			L O	E(level): other: 8505.1 *47* from (α,p).
8514.3 *5*	1/2$^-$	0.150 keV *15*	OP	XREF: P(8515)
8533.9 *5*			L O	E(level): weighted average of 8532.4 *43* from (α,p) and 8533.9 *5* from (p,γ).
8558.2? *13*	(13/2$^-$)	<0.14 ps	F	XREF: F(?) J^{π}: 2631γ D(+Q), ΔJ=1 to 11/2$^-$ in

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
				(^{28}Si,αpγ). $T_{1/2}$: lifetime τ<0.2 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
8571.9 *5*	(5/2)$^+$	0.08 keV *1*	L OP X	T=(3/2) XREF: P(8573)X(8580) E(level): weighted average of 8571.9 *5* from (p,γ), and 8573 *3* from ^{34}S(p,p),(p,p$'\gamma$):resonances. Other: 8567.9 *37* from (α,p).
8580.5 *5*	1/2$^+$	0.75 keV *8*	OP	XREF: P(8582)
8585.8 *5*			O	
8590.1 *5*	5/2$^+$	0.003 keV *2*	OP	XREF: P(8592) E(level): weighted average of 8590.0 *5* from (p,γ), and 8592 *3* from ^{34}S(p,p),(p,p$'\gamma$):resonances. J^{π}: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 5426.6γ to 7/2$^-$ ruled out 3/2$^+$ based on RUL and Γ=0.003 keV.
8612.0 *5*	3/2$^+$,5/2$^+$		l O W	XREF: W(8610) J^{π}: L(d,^{3}He)=2 from 0$^+$.
8613.7 *5*	5/2$^+$	0.175 keV *20*	l OP	XREF: P(8616)
8618.2 *5*	3/2$^+$,5/2$^+$	0.002 keV *1*	l OP	XREF: P(8620) E(level): weighted average of 8618.1 *5* from (p,γ), and 8620 *3* from ^{34}S(p,p),(p,p$'\gamma$):resonances.
8630.1 *3*	7/2$^-$	0.001 keV *1*	O	J^{π}: 4282.1γ M1+E2 to 9/2$^-$, 6866.2γ D+Q to 5/2$^+$, spin=3/2,7/2 from $\gamma(\theta)$ in (p,γ).
8633 *3*	(3/2,5/2$^+$)	0.001 keV *1*	P	
8640	(5/2$^+$)		O	J^{π}: spin=(5/2) from $\gamma(\theta)$ in (p,γ); primary 4697γ to 9/2$^+$.
8641.4 *5*	3/2$^+$,5/2$^+$	0.003 keV *2*	OP	XREF: P(8643)
8654 *4*			L	
8686.2 *5*	5/2$^-$	0.001 keV *1*	OP	E(level): weighted average of 8686.2 *5* from (p,γ), and 8687 *3* from ^{34}S(p,p),(p,p$'\gamma$):resonances. J^{π}: 5/2$^-$,7/2$^-$ from R-Matrix analysis in (p,p):resonances; primary 7446.0γ to 1219, 1/2$^+$ disfavors both 5/2$^-$ and 7/2$^-$, but 7/2$^-$ is much less likely.
8688.3 *5*	1/2$^+$	0.20 keV *2*	OP	XREF: P(8689)
8691 *3*	1/2$^-$	6.4 keV *7*	P	
8696.9 *5*	3/2$^-$	0.8 keV *1*	L OP	XREF: L(8698)P(8698)
8700 *10*	(7/2 to 17/2)$^+$		N	J^{π}: L(α,d)=6 from 3/2$^+$.
8706.3 *5*			O	
8717.8 *5*			L O	E(level): weighted average of 8720.9 *37* from (α,p) and 8717.7 *5* from (p,γ).
8750.9 *5*	3/2$^-$	0.30 keV *3*	OP	E(level): weighted average of 8750.9 *5* from (p,γ), and 8750 *3* from ^{34}S(p,p),(p,p$'\gamma$):resonances.
8767.0 *5*			O	
8772.9 *5*	1/2$^-$	0.57 keV *6*	OP	XREF: P(8775)
8779.8 *5*	3/2$^-$	0.214 keV *25*	OP	XREF: P(8781) E(level): weighted average of 8779.8 *5* from (p,γ), and 8781 *3* from ^{34}S(p,p),(p,p$'\gamma$):resonances.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
8787.2 *5*	(5/2$^+$)	0.001 keV *1*	L OP	E(level): others: 8786.3 *38* from (α,p), and 8788 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: (3/2$^+$,5/2$^+$) from R-Matrix analysis in (p,p):resonances; primary 3623.7γ and 5623.8γ to 7/2$^-$.
8788.3[b] *6*	(15/2$^+$)	<0.28 ps	F H	XREF: F(?) J$^{\pi}$: 915.4γ D, ΔJ=1 to 13/2$^+$; band assignment. $T_{1/2}$: lifetime τ<0.4 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
8798.4 *5*	(3/2$^+$,5/2$^+$)	0.001 keV *1*	OP	E(level): weighted average of 8798.4 *5* from (p,γ), and 8799 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8820.9 *5*	(1/2$^+$ to 7/2$^+$)		O	J$^{\pi}$: primary 5817.6γ and 7057.0γ to 5/2$^+$; 6126.3γ and 8819.7γ to 3/2$^+$, which yields (1/2$^+$,3/2±,5/2±,7/2$^+$).
8824.2 *5*	1/2$^+$	1.70 keV *17*	OP	XREF: P(8825) E(level): weighted average of 8824.2 *5* from (p,γ), and 8825 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8829.3 *5*	1/2$^-$	12.2 keV *12*	OP	E(level): weighted average of 8829.3 *5* from (p,γ), and 8829 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: 1/2$^-$ from R-Matrix analysis in (p,p):resonances; but primary 6183.0γ to 7/2$^+$ gives J≥3/2, which may indicate a different level.
8830 *3*	1/2$^+$	0.080 keV *15*	P	
8833.1 *5*	(5/2,7/2)		O	J$^{\pi}$: primary 3618.4γ to (5/2$^+$), 4063.0γ to (5/2$^-$); 4720.4γ to 7/2$^+$, 5669.6γ to 7/2$^-$.
8838.1 *5*	5/2$^-$,7/2$^-$	0.001 keV *1*	l OP	XREF: P(8839)
8844.4[b] *4*	17/2$^+$	5.9 ps *11*	F HI N	XREF: N(8840) E(level): weighted average of 8845.1 *9* from (^{28}Si,αpγ), 8844.6 *5* from (^{16}O,αpγ), and 8844.2 *4* from (^{14}N,αpnγ). J$^{\pi}$: 971.38γ E2, ΔJ=2 to 13/2$^+$, 524.8γ to 15/2$^-$; band assignment. $T_{1/2}$: weighted average of 6.1 ps *11* from (^{16}O,αpγ) (1991Ja11, DSAM) and 5.5 ps *14* from (^{14}N,αpnγ) (1976Wa11, DSAM).
8856.2 *5*	5/2$^+$	0.010 keV *5*	OP	E(level): weighted average of 8856.1 *5* from (p,γ), and 8858 *3* from ^{34}S(p,p),(p,p′γ):resonances. J$^{\pi}$: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 5692.6γ to 7/2$^-$ rules out 3/2$^+$ based on RUL and Γ=0.010 keV.
8868.6 *5*	3/2$^+$,5/2$^+$	0.027 keV *10*	OP	E(level): weighted average of 8868.6 *5* from (p,γ), and 8870 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8884.1 *5*			O	
8886.1 *5*	(5/2)		L O	E(level): other: 8887.5 *41* from (α,p). J$^{\pi}$: primary 3722.6γ to 7/2$^-$, 6239.8γ to 7/2$^+$; 4707.9γ to 3/2$^-$, 4967.2γ to 3/2$^+$.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	Jπ‡	T$_{1/2}$ or Γ&	XREF	Comments
8893.3 *5*	(5/2$^+$,7/2$^+$)		O	Jπ: primary 4950.0γ to 9/2$^+$, 6198.7γ to 3/2$^+$.
8904.8 *5*	(1/2,3/2,5/2$^+$)		O	Jπ: primary 4726.6γ to 3/2$^-$, 6210.2γ to 3/2$^+$, 7684.5γ to 1/2$^+$.
8906.8 *5*	5/2$^+$	0.002 keV *1*	OP x	XREF: P(8909)x(8921) Jπ: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 4963.5γ to 9/2$^+$ rules out 3/2$^+$ based on RUL and Γ=0.002 keV.
8919.8 *5*	(5/2$^-$,7/2,9/2$^+$)		O x	XREF: x(8921) Jπ: primary 4571.7γ to 9/2$^-$, 7155.9γ to 5/2$^+$.
8933.3 *5*			O	
8953.1 *5*	3/2$^+$	0.075 keV *15*	OP	XREF: P(8955) E(level): weighted average of 8953.0 *5* from (p,γ), and 8955 *3* from ^{34}S(p,p),(p,p′γ):resonances. Jπ: spin=3/2 from γ(θ) in (p,γ) and (3/2)$^+$ from R-Matrix analysis in (p,p):resonances.
8957.9 *5*	(1/2$^-$,3/2$^-$)	0.04 keV *1*	L OP	XREF: P(8959) E(level): others: 8957.7 *44* from (α,p) and 8959 *3* from (p,p):resonances.
8981.9 *5*	5/2$^-$,7/2$^-$	0.003 keV *2*	OP	XREF: P(8983) E(level): weighted average of 8981.9 *5* from (p,γ), and 8983 *3* from ^{34}S(p,p),(p,p′γ):resonances.
8984.2 *5*	5/2$^+$	0.025 keV *10*	OP	XREF: P(8985) Jπ: 3/2$^+$,5/2$^+$ from R-Matrix analysis in (p,p):resonances; primary 5820.7γ to 7/2$^-$ rules out 3/2$^+$ based on RUL and Γ=0.025 keV.
8988.4 *5*			O	
8992.4 *5*			O	
8996.7 *5*	5/2$^-$,7/2$^-$	0.002 keV *1*	L OP	XREF: L(8996.9)P(8998)
9001.1 *5*			O	
9019.3 *5*	3/2$^-$	3.27 keV *30*	OP	XREF: P(9021) Γ: weighted average of 3.1 keV *3* from (p,γ) and 3.50 keV *35* from (p,p):resonances.
9024.5 *5*			l O	
9029.9 *5*	(5/2)$^+$	0.04 keV *1*	l OP	XREF: P(9031)
9033.1 *5*			O	
9038.4 *5*	1/2$^-$	0.292 keV *30*	OP	XREF: P(9040) E(level): weighted average of 9038.3 *5* from (p,γ), and 9040 *3* from ^{34}S(p,p),(p,p′γ):resonances.
9048.4 *5*	5/2$^-$,7/2$^-$	0.001 keV *1*	OP	XREF: P(9048)
9050 *3*	(5/2)$^+$	0.095 keV *15*	P	
9081.5 *4*	5/2$^+$	59 eV *10*	L OP	XREF: L(9085.7)P(9084) E(level): weighted average of 9081.4 *4* from (p,γ), and 9084 *3* from ^{34}S(p,p),(p,p′γ):resonances. Jπ: spin=5/2 from γ(θ) in (p,γ) and (5/2)$^+$ from R-Matrix analysis in (p,p):resonances. Γ: weighted average of 65 eV *20* from (p,γ) and 57 eV *10* from (p,p):resonances.
9088.5 *5*			L OP	XREF: L(9085.7)P(9089) E(level): weighted average of 9088.5 *5* from

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	J$^{\pi}$‡	$T_{1/2}$ or Γ&	XREF	Comments
				(p,γ), and 9089 *4* from ^{34}S(p,p),(p,p′γ):resonances.
9099.3 *5*			O	
9100.4 *5*	3/2^{-}	0.20 keV *2*	OP	XREF: P(9101)
9107.4 *5*			L OP	XREF: L(9109.3)P(9108) E(level): weighted average of 9107.4 *5* from (p,γ), and 9108 *4* from ^{34}S(p,p),(p,p′γ):resonances.
9110.1 *5*			L OP	XREF: L(9109.3)P(9114) E(level): weighted average of 9110.0 *5* from (p,γ), and 9114 *4* from ^{34}S(p,p),(p,p′γ):resonances.
9124.0 *5*			OP	XREF: P(9120)
9127 *4*	5/2@		J P	E(level): from (p,p):resonances. Other: 9127 *9* from ^{31}P(α,p):resonances.
9135.1 *5*		2.0 keV *4*	Op	XREF: p(9136) Γ: from (p,γ).
9138.3 *5*			Op	XREF: p(9136)
9147 *4*	(7/2 to 17/2)$^{+}$		N P	XREF: N(9150)P(9147) J$^{\pi}$: L(α,d)=6 from 3/2^{+}.
9155.7 *5*			O	
9157.1 *5*	5/2^{-}		OP	XREF: P(9159) J$^{\pi}$: spin=5/2 from γ(θ) in (p,γ); 5993.7γ M1+E2 to 7/2^{-}.
9161 *4*	1/2@		J l	XREF: l(9160.0)
9163.3 *5*	(3/2^{-})		l OP	XREF: l(9160.0)P(9168) E(level): other: 9168 *4* from (p,p):resonances.
9184.2 *5*			Op	XREF: p(9185)
9188.8 *5*			Op	XREF: p(9185)
9194.1 *5*			L OP	E(level): others: 9194.0 *39* from (α,p), and 9196 *4* from (p,p):resonances.
9207 *4*	9/2^{+}		P	
9222 *4*	5/2^{+}		P X	XREF: X(9220) J$^{\pi}$: L(pol d,^{3}He)=2 and L+1/2 from analyzing power for a group at E=9220 *120*.
9227 *4*	5/2^{-}		P	
9241 *4*	7/2^{+}		P	
9255 *4*	1/2@		J P	E(level): from (p,p):resonances. Other: 9256 *4* from ^{31}P(α,p):resonances.
9264.9 *39*			L	
9271 *4*			P	
9276 *4*			P	
9284 *4*	5/2^{-}		P	
9294 *4*			P	
9299 *4*			P	
9316.2 *39*			L	
9325 *4*	5/2^{+}		P	
9335 *4*	7/2^{+}		L P	E(level): weighted average of 9336 *4* from ^{34}S(p,p),(p,p′γ):resonances and 9334 *5* from ^{32}S(α,p).
9345 *4*			P	
9349 *4*			P	
9355 *4*			P	
9376 *4*			L	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
9400 *9*	1/2[@]		J	
9450 *10*	(7/2 to 17/2)$^+$		N	J^{π}: L(α,d)=6 from 3/2$^+$.
9456 *4*	3/2[@]		J	
9476 *4*			L	
9481.0 *18*	3/2[@]		J	
9508 *4*	(1/2$^-$,3/2,5/2$^+$)		L X	XREF: X(9514) J^{π}: L(pol d,^{3}He)=(1,2) from 0$^+$.
9551 *6*	5/2[@]		J	
9673 *6*	1/2[@]		J	
9712.7 *27*	1/2[@]		J L	E(level): weighted average of 9713.0 *27* from ^{31}P(α,p),(α,n):resonances and 9711.8 *44* from (α,p).
9740 *4*			L	
9751.1 *27*	7/2[@]		J	
9799 *5*			L	
9814.0 *27*	5/2$^{(+)}$		J X	XREF: X(9820) J^{π}: spin=5/2 from angular distributions in ^{31}P(α,p):resonances; L=(1,2) from 0$^+$ in (pol d,^{3}He) gives (1/2$^-$,3/2,5/2$^+$).
9836 *5*			L	
9869.8 *27*	1/2[@]		J L	XREF: L(9870)
9900.8 *27*	3/2[@]		J	
9923 *4*	3/2$^-$[@]		J	
9958 *6*	3/2[@]		J	
9969 *5*	(1/2,3/2)[@]		J L	XREF: J(9968) E(level): from (α,p) and (α,n):resonances.
10031 *5*	(1/2$^+$,3/2$^-$)[@]		J	
10058 *5*	3/2[@]		J	
10075 *5*	(1/2)[@]		J L	E(level): from (α,p) and (α,n):resonances.
10090 *5*	5/2$^+$[@]		J	
10132 *5*	3/2$^-$[@]		J	
10163 *5*	[@]		L	
10172 *5*	3/2$^-$,5/2$^-$[@]		J	
10179 *5*			L	
10181.1[a] *5*	19/2$^-$	42.3 fs *28*	F H	E(level): weighted average of 10181.1 *10* from (^{28}Si,αpγ) and 10181.1 *5* from (^{16}O,αpγ). J^{π}: 1693.5γ E2, ΔJ=2 to 15/2$^-$; 1336.3γ D, ΔJ=1 to 17/2$^+$; band assignment. $T_{1/2}$: lifetime τ=61 fs *4* from (^{28}Si,αpγ) (2013Bi10, DSAM).
10215 *5*	3/2[@]		J L	XREF: L(10218.0)
10221.9 *11*	(17/2$^-$)	<0.28 ps	F H	XREF: F(?) E(level): from (^{16}O,αpγ). J^{π}: 1377.8γ to 17/2$^+$ consistent with ΔJ=0. $T_{1/2}$: lifetime τ<0.4 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
10236 *5*	3/2$^-$[@]		J	
10278 *5*	1/2[@]		J	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
10295 *5*			J	
10322 *5*	$5/2^+$[@]		J	
10340 *5*	(3/2,5/2)[@]		J	
10359 *5*	3/2[@]		J	
10385 *5*			J	
10395 *5*			L	
10399 *5*	$3/2^+,7/2^+$[@]		J	
10431 *5*	$5/2^+$[@]		J	
10465 *5*	$5/2^+$[@]		J L	E(level): weighted average of 10467 *5* from ^{31}P(α,p),(α,n):resonances and 10463.2 *49* from (α,p).
10486 *5*	$5/2^+$[@]		J	
10517 *6*			L	
10534 *5*	$1/2^+,5/2^+$[@]		J	
10548 *6*			L	
10584 *5*	$3/2^+$[@]		J L	E(level): weighted average of 10587 *5* from ^{31}P(α,p),(α,n):resonances and 10579.5 *54* from (α,p).
10642 *5*	$5/2^+$[@]		J L	E(level): weighted average of 10641 *5* from ^{31}P(α,p),(α,n):resonances and 10643.0 *50* from (α,p).
10678 *5*	$3/2^+,5/2^+$[@]		J	
10696 *5*	$7/2^-$[@]		J	
10733 *5*	$3/2^+$[@]		J L	E(level): weighted average of 10734 *5* from ^{31}P(α,p),(α,n):resonances and 10732.1 *50* from (α,p).
10760 *5*	$3/2^+,7/2^-$[@]		J L	E(level): weighted average of 10760 *5* from ^{31}P(α,p),(α,n):resonances and 10759.1 *55* from (α,p).
10800 *5*	$5/2^+$[@]		J	
10816 *5*	$1/2^+,3/2^-$[@]		J	
10844 *5*	$3/2^-$[@]		J	
10858.5[b] *8*	$19/2^+$	<0.28 ps	F H	XREF: F(?) J^{π}: 2069.8γ E2, ΔJ=2 to $15/2^+$, 2014.7γ D, ΔJ=1 to $17/2^+$; band assignment. $T_{1/2}$: lifetime τ<0.4 ps from (^{28}Si,αpγ) (2007Ks01, DSAM).
10870 *5*	(5/2,7/2)[@]		J	
10886 *5*			J	
10909 *5*	$3/2^+,5/2^-$[@]		J	
10928 *5*	7/2[@]		J	
10942 *5*			J	
10964 *5*	$5/2^+$[@]		J	
10973 *5*			J	
10998 *5*	$3/2^+,7/2^-$[@]		J	
11034 *5*	$3/2^+$[@]		J	
11063 *5*	$1/2^+,9/2^+$[@]		J	
11082 *5*	$5/2^+,9/2^+$[@]		J	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)[†]	J^{π}[‡]	$T_{1/2}$ or Γ[&]	XREF	Comments
11105 5			J	
11123 5	3/2[@]		J	
11142 5	$5/2^{+}$[@]		J	
11154 5			J	
11169 5			J	
11179 5	$3/2^{-},5/2^{+}$[@]		J	
11185 5			J	
11198 5	3/2[@]		J	
11230 5			J	
11246 5	3/2[@]		J	
11265 5	$3/2^{-}$[@]		J	
11287 5			J	
11307 5			J	
11314 5	$1/2^{-},5/2^{+}$[@]		J	
11327 5	$1/2^{-},5/2^{+}$[@]		J	
11357 5			J	
11375 5			J	
11390 5			J	
11410 5			J	
11421 5	$3/2^{-}$[@]		J	
11434 5			J	
11459.1[b] 7	$21/2^{+}$	43.0 fs 21	F H J	E(level): weighted average of 11459.2 12 from (^{28}Si,αpγ) and 11459.0 7 from (^{16}O,αpγ). Other: 11460 5 from ^{31}P(α,p),(α,n):resonances. J^{π}: 2614.5γ E2, ΔJ=2 to $17/2^{+}$; band assignment. $T_{1/2}$: lifetime τ=62 fs 3 from (^{28}Si,αpγ) (2013Bi10, DSAM).
11473 5	$1/2^{-},5/2^{-}$[@]		J	
11492 5			J	
11505 5	$3/2^{-},5/2^{-}$[@]		J	
11525 5			J	
11540 5			J	
11551 5	$3/2^{-},5/2^{+}$[@]		J	
11565 5			J	
11589 5			J	
11609 5	$5/2^{+},7/2^{-}$[@]		J	
11627 5	$1/2^{+},5/2^{+}$[@]		J	
11638 5			J	
11651 5			J	
11684 5	$1/2^{+},5/2^{+}$[@]		J	
11773 5	$3/2^{-},7/2^{-}$[@]		J	
11783 5	$3/2^{-}$[@]		J	
1.25×10^{4} 5			J X	T=3/2 XREF: J(12900) E(level): from (pol d,^{3}He).
12572.2[a] 6	$23/2^{-}$	33 fs 4	F H	E(level): weighted average of 12572.2 12 from (^{28}Si,αpγ) and 12572.2 6 from (^{16}O,αpγ). J^{π}: 2390.8γ E2, ΔJ=2 to $19/2^{-}$, 1113.3γ D, ΔJ=1 to $21/2^{+}$; band assignment.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)

E(level)†	$J^{\pi\ddagger}$	$T_{1/2}$ or $\Gamma^{\&}$	XREF	Comments
				$T_{1/2}$: lifetime τ=48 fs *6* from (^{28}Si,αpγ) (2013Bi10, DSAM).
13900			J	T=3/2
16306.4^{a} *16*	(27/2^{-})	<9.0 fs	F	J^{π}: 3734γ to 23/2^{-}; band assignment. $T_{1/2}$: lifetime τ<13 fs from (^{28}Si,αpγ) (2013Bi10, DSAM).

† Most of Adopted Levels are reported in (p,γ) dataset with precise E(level) values, deduced based on measured γ-ray energies which however are not reported in references in (p,γ) dataset, while E(level) values from Eγ values with uncertainties in other datasets are less precise and the numbers of γ rays are much less than that of (p,γ) datasets. Therefore, a least-squares fit to available Eγ values with uncertainties would not produce E(level) values more precise and complete than those already reported in (p,γ) and the evaluators have adopted E(level) values from (p,γ) or averages of values based on Eγ data from all available datasets where applicable, as noted.

‡ For proton-resonance levels populated in (p,p):resonances, J^{π} assignments are from R-Matrix analysis, unless otherwise noted.

From FRESCO-DWBA analysis of measured $\sigma(\theta)$ of (α,p) (2019Se09).

@ From angular distributions in ^{31}P(α,p):resonances.

& Sources of $T_{1/2}$ values are listed under comments; Γ from ^{34}S(p,p),(p,p$'\gamma$):resonances, unless otherwise noted.

[a] Band(A): Band based on $f_{7/2}$ orbital.

[b] Band(B): Band based on 13/2^{+}.

$\gamma(^{35}Cl)$

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	α[a]	Comments
1219.38	1/2+	1219.4 *1*	100	0.0	3/2+	M1+E2	0.093 *10*	3.66×10^{-5} *5*	B(M1)(W.u.)=0.101 +*14*−*11*; B(E2)(W.u.)=2.2 +*6*−*5* E_γ: weighted average of 1219.3 *2* from ^{35}Ar ε decay and 1219.4 *1* from (p,p′γ), and 1219.4 *6* from (α,pγ). Mult.,δ: ΔJ=1 from $\gamma(\theta)$ in (p,p′γ), E1+M2 ruled out by RUL. δ from adopted B(E2)↑=0.00076 *7* measured in Coulomb excitation and adopted lifetime τ=0.172 ps *20*.
1763.15	5/2+	543.77	<0.2	1219.38	1/2+	[E2]		0.000323 *5*	I_γ: from (p,γ). Other: <0.5 from (α,pγ).
		1763.16 *10*	100	0.0	3/2+	M1+E2	−2.85 *10*	0.0002124 *30*	B(M1)(W.u.)=0.00122 +*14*−*12*; B(E2)(W.u.)=12.1 +*11*−*9* E_γ: weighted average of 1763.0 *2* from ^{35}Ar ε decay, 1763.1 *2* from (^{24}Mg,αpγ), 1763.3 *2* from (^{16}O,αpγ), 1763.15 *10* from (^{14}N,αpnγ), and 1763.2 *1* from (p,p′γ). Other: 1763.2 *6* from (α,pγ). Mult.: from $\gamma\gamma$(ADO) in (^{16}O,αpγ) with ΔJ=1 and γ(lin pol) in (^{28}Si,αpγ). δ: weighted average of −3.0 *1* from (α,pγ), −2.66 *12* from (p,p′γ), −2.95 *45* from Coulomb excitation, −2.6 *4* from (^{16}O,αpγ), all from $\gamma(\theta)$ or p$\gamma(\theta)$. Other: −0.42 *17* from $\gamma\gamma(\theta)$ in (p,γ); 1.8 +*10*−*4*; deduced by the evaluators from adopted B(E2)↑=0.0106 *7* and $T_{1/2}$=0.36 ps *3*.
2645.67	7/2+	882.80 *11*	11.4 *10*	1763.15	5/2+	M1+E2	+0.21 *5*	5.29×10^{-5} *9*	B(M1)(W.u.)=0.0196 +*35*−*27*; B(E2)(W.u.)=4.2 +*23*−*18* E_γ: weighted average of 882.9 *1* from (^{24}Mg,αpγ), 882.7 *5* from (^{16}O,αpγ), 882.32 *35* from (^{14}N,αpnγ), 882.4 *10* from (α,pγ), and 882.3 *3* from (p,p′γ). I_γ: unweighted average of 8.7 *6* from (^{28}Si,αpγ), 15.2 *13* from (^{24}Mg,αpγ), 11.08 *28* from (^{16}O,αpγ), 11.1 *22* from (α,pγ), 9.5 *11* from (p,γ), and 13 *5* from (p,p′γ). Mult.: D+Q from p$\gamma(\theta)$ in (α,pγ); E1+M2 ruled out by RUL. δ: weighted average of +0.25 *5* from (α,pγ) and +0.17 *5* from (p,p′γ).
		1426.26	<1.2	1219.38	1/2+	[M3]		7.52×10^{-5} *11*	I_γ: from (α,pγ). Other: <2.2 from (p,γ).
		2645.70 *20*	100.0 *11*	0.0	3/2+	E2		0.000633 *9*	B(E2)(W.u.)=3.6 +*6*−*4* E_γ: weighted average of 2645.5 *4* from (^{24}Mg,αpγ), 2645.8 *5* from (^{16}O,αpγ), 2645.79 *30* from

$^{35}_{17}Cl_{18}$-23

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
									(^{14}N,αpnγ), and 2645.7 *2* from (p,p′γ). Other: 2646.2 *15* from (α,pγ). I_γ: others: 100.0 *31* from (^{24}Mg,αpγ), 100.0 *22* from (^{16}O,αpγ), 100.0 *22* from (α,pγ), and 100.0 *20* from (p,p′γ). Other: 100 *6* from (^{28}Si,αpγ). Mult.: from $\gamma\gamma(\theta)$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ); $\gamma(\theta)$ and γ(lin pol) in (p,p′γ).
2694.00	3/2$^+$	930.9 *2*	16.3 *13*	1763.15	5/2$^+$	M1+E2	+0.09 *3*	4.68×10^{-5} *7*	B(M1)(W.u.)=0.110 *+26−19*; B(E2)(W.u.)=3.9 *+33−22* E_γ: weighted average of 930.7 *2* from ^{35}Ar ε decay, 930.5 *15* from (α,pγ), and 931.3 *3* from (p,p′γ). I_γ: weighted average of 14.3 *18* from ^{35}Ar ε decay, 18.2 *26* from (α,pγ), 16.3 *13* from (p,γ), and 18.6 *25* from (p,p′γ). Mult.,δ: from $\gamma(\theta)$ and γ(lin pol) in (p,p′γ).
		1474.8 *3*	9.5 *10*	1219.38	1/2$^+$	M1+E2	−0.63 *21*	8.4×10^{-5} *4*	B(M1)(W.u.)=0.0116 *+37−29*; B(E2)(W.u.)=8.0 *+45−39* E_γ: weighted average of 1474.8 *3* from ^{35}Ar ε decay and 1475 *2* from (p,p′γ). I_γ: weighted average of 9.3 *10* from ^{35}Ar ε decay, 11.7 *26* from (α,pγ), 9.5 *13* from (p,γ), and 8.3 *25* from (p,p′γ). Mult.,δ: from $\gamma(\theta)$ and γ(lin pol) in (p,p′γ).
		2693.7 *2*	100.0 *17*	0.0	3/2$^+$	M1+E2	+0.26 *2*	0.000553 *8*	B(M1)(W.u.)=0.026 *+6−4*; B(E2)(W.u.)=0.92 *+26−19* E_γ: weighted average of 2693.7 *2* from ^{35}Ar ε decay and 2694.1 *6* from (p,p′γ). Other: 2693.2 *20* from (α,pγ). I_γ: weighted average of 100.0 *17* from ^{35}Ar ε decay, 100 *4* from (α,pγ), 100.0 *25* from (p,γ), and 100.0 *30* from (p,p′γ). Mult.: from $\gamma(\theta)$ and γ(lin pol) in (p,p′γ). δ: weighted average of +0.17 *8* from (α,pγ) and +0.26 *2* from (p,p′γ).
3002.64	5/2$^+$	308.64	<5	2694.00	3/2$^+$	[M1,E2]		0.0014 *9*	I_γ: from (α,pγ).
		356.97	<1	2645.67	7/2$^+$	[M1,E2]		9	I_γ: from (α,pγ).
		1239.47	2.2 *11*	1763.15	5/2$^+$	[M1,E2]		4.4×10^{-5} *6*	I_γ: from (p,p′γ). B(M1)(W.u.)=0.011 *+8−5* if M1, B(E2)(W.u.)=26 *+19−12* if E2.
		1783.21	3.3 *11*	1219.38	1/2$^+$	[E2]		0.0002268 *32*	B(E2)(W.u.)=6.4 *+35−22* I_γ: from (p,p′γ).
		3002.3 *4*	100.0 *10*	0.0	3/2$^+$	M1+E2	+0.08 *2*	0.000671 *9*	B(M1)(W.u.)=0.034 *+14−7*; B(E2)(W.u.)=0.09 *+7−4* E_γ: weighted average of 3002.1 *4* from ^{35}Ar ε decay,

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	α[a]	Comments
									3002.5 *15* from (α,pγ), and 3002.5 *5* from (p,p′γ). I_γ: weighted average of 100.0 *19* from ^{35}Ar ε decay and 100.0 *10* from (p,p′γ). Mult.: from $\gamma(\theta)$ and γ(lin pol) in (p,p′γ). δ: weighted average of +0.09 *3* from (α,pγ) and +0.07 *2* from (p,p′γ). Other: −0.22 *1* from (p,γ).
3162.87	7/2$^-$	160.67 *20*	1.5 *4*	3002.64	5/2$^+$	[E1]		0.00330 *5*	B(E1)(W.u.)=6.6×10^{-5} *17* E_γ: weighted average of 161 *1* from (^{16}O,αpγ) and 160.66 *20* from (^{14}N,αpnγ). I_γ: unweighted average of 1.10 *30* from (^{28}Si,αpγ) and 1.89 *22* from (p,γ).
		517.47 *11*	12.3 *16*	2645.67	7/2$^+$	(E1)		0.0001097 *15*	B(E1)(W.u.)=1.61×10^{-5} *20* E_γ: weighted average of 517.7 *1* from (^{24}Mg,αpγ), 517.3 *5* from (^{16}O,αpγ), 517.26 *10* from (^{14}N,αpnγ), 517.4 *15* from (α,pγ), and 517.0 *8* from (p,p′γ). I_γ: unweighted average of 9.2 *5* from (^{28}Si,αpγ), 14.3 *8* from (^{24}Mg,αpγ), 11.1 *13* from (^{16}O,αpγ), 11.1 *33* from (α,pγ), 8.9 *11* from (p,γ), and 19 *8* from (p,p′γ). Mult.: D from $\gamma\gamma(\theta)$(DCO) in (^{28}Si,αpγ); $\Delta\pi$=yes from level scheme.
		1399.9 *7*	0.34 *5*	1763.15	5/2$^+$	(E1+M2)	+0.44 *12*	0.000181 *12*	B(E1)(W.u.)=1.88×10^{-8} *+31−34*; B(M2)(W.u.)=0.0085 *+43−38* I_γ: weighted average of 0.50 *20* from (^{16}O,αpγ) and 0.33 *5* from (p,γ). Mult.,δ: D+Q from $\gamma(\theta)$ in (p,γ) (1971Pr11); $\Delta\pi$=yes from level scheme.
		1943.43	<0.22	1219.38	1/2$^+$	[E3]		0.0001742 *24*	
		3162.53 *10*	100.0 *11*	0.0	3/2$^+$	M2+E3	+0.17 *2*	0.000523 *7*	B(M2)(W.u.)=0.255 *9*; B(E3)(W.u.)=4.2 *+11−9* E_γ: weighted average of 3162.7 *4* from (^{24}Mg,αpγ), 3162.6 *3* from (^{16}O,αpγ), 3162.43 *10* from (^{14}N,αpnγ), and 3162.6 *1* from (p,p′γ), and 3162.6 *15* from (α,pγ). I_γ: weighted average of 100 *5* from (^{28}Si,αpγ), 100.0 *30* from (^{24}Mg,αpγ), 100.0 *20* from (^{16}O,αpγ), 100.0 *33* from (α,pγ), 100.0 *11* from (p,γ), and 100.0 *20* from (p,p′γ). Mult.: from $\gamma(\theta)$ and γ(lin pol) in (p,γ). δ: weighted average of +0.16 *1* from (^{28}Si,αpγ), +0.18 *7* from (^{16}O,αpγ), and +0.25 *4* from (p,γ). Others: −0.14 *2* from (α,pγ) and −0.22 *5* from (p,p′γ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ [†]	I_γ [‡]	E_f	J_f^π	Mult. [#]	α [a]	Comments
3918.47	3/2+	1272.78	<3.7	2645.67	7/2+	[E2]	5.56×10^{-5} 8	
		2155.1 15	24.2 28	1763.15	5/2+	[M1,E2]	0.00036 4	E_γ: from ^{35}Ar ε decay (1.7755 s). I_γ: weighted average of 29 12 from ^{35}Ar ε decay and 23.9 28 from (p,γ). B(M1)(W.u.)=0.075 +32−18 if M1, B(E2)(W.u.)=61 +26−14 if E2.
		2698.98	19 5	1219.38	1/2+	[M1,E2]	0.00060 6	I_γ: from (α,pγ). B(M1)(W.u.)=0.030 +15−9 if M1, B(E2)(W.u.)=16 +8−5 if E2.
		3918.24	100.0 19	0.0	3/2+	M1+E2	0.00108 7	I_γ: weighted average of 100 6 from ^{35}Ar ε decay, 100 5 from (α,pγ), and 100.0 19 from (p,γ). δ: −0.21 2 or +20 +10−5 from $\gamma(\theta)$ in (p,γ). B(M1)(W.u.)=0.052 +22−11 if M1, B(E2)(W.u.)=13 +6−3 if E2.
3942.9	9/2+	1297.4 15	8.0 21	2645.67	7/2+	[M1,E2]	5.2×10^{-5} 8	E_γ: from (α,pγ). I_γ: weighted average of 6.2 31 from (^{28}Si,αpγ), 8.1 21 from (α,pγ), and 8.7 22 from (p,γ). B(M1)(W.u.)=0.0042 +13−12 if M1, B(E2)(W.u.)=9.3 +29−27 if E2.
		2180.1 10	100.0 21	1763.15	5/2+	E2	0.000414 6	B(E2)(W.u.)=8.7 +17−13 E_γ: weighted average of 2180 1 from (^{16}O,αpγ) and 2180.2 15 from (α,pγ). I_γ: weighted average of 100.0 21 from (α,pγ) and 100.0 22 from (p,γ). Other: 100 11 from (^{28}Si,αpγ).
3967.6	1/2+	1273.6	2.7 6	2694.00	3/2+	[M1,E2]	4.9×10^{-5} 7	B(M1)(W.u.)=0.019 +8−6 if M1, B(E2)(W.u.)=45 +18−13 if E2.
		2204.4	≤1.3	1763.15	5/2+	[E2]	0.000426 6	
		2748.5 6	100 6	1219.38	1/2+	[M1]	0.000569 8	B(M1)(W.u.)=0.071 +23−14 E_γ: from ^{35}Ar ε decay (1.7755 s). I_γ: other: 100 7 from ^{35}Ar ε decay.
		3967.4	23 4	0.0	3/2+	[M1,E2]	0.00109 7	I_γ: weighted average of 22 5 from ^{35}Ar ε decay and 24 4 from (p,γ). B(M1)(W.u.)=0.0054 +19−13 if M1, B(E2)(W.u.)=1.31 +46−30 if E2.
4059.1	(3/2)−	1365.1	1.3 4	2694.00	3/2+	[E1]	0.0001824 26	B(E1)(W.u.)=2.1×10^{-4} +8−7
		2295.9	5.5 16	1763.15	5/2+	[E1]	0.000848 12	B(E1)(W.u.)=1.9×10^{-4} +7−6 I_γ: weighted average of 8 4 from (α,pγ) and 5.1 16 from (p,γ).
		2839.6	100.0 21	1219.38	1/2+	[E1]	1.17×10^{-3} 2	B(E1)(W.u.)=0.00179 +34−22 I_γ: weighted average of 100 4 from (α,pγ) and 100.0 21 from (p,γ).
		4058.9	<2.1	0.0	3/2+	[E1]	1.70×10^{-3} 2	

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
4112.4	7/2+	2349.2	89 *6*	1763.15	5/2+	M1+E2	−0.16 *2*	0.000405 *6*	B(M1)(W.u.)=0.0159 +*48*−*31*; B(E2)(W.u.)=0.28 +*12*−*8* I_γ: other: 16 *5* from (α,pγ).
		4112.1	100 *6*	0.0	3/2+	E2(+M3)	0.00 *10*	1.21×10^{-3} *2*	B(E2)(W.u.)=0.76 +*22*−*15* I_γ: other: 100 *5* from (α,pγ).
4173.43	(5/2+)	1479.40	45 *14*	2694.00	3/2+	[M1,E2]		9.1×10^{-5} *13*	B(M1)(W.u.)=0.048 +*20*−*12* if M1, B(E2)(W.u.)=82 +*34*−*21* if E2. B(E2)(W.u.)=82 +*34*−*21* upper bound exceeds RUL=100 if E2.
		1527.72	<17	2645.67	7/2+	[M1,E2]		0.000106 *15*	
		2410.19	28 *9*	1763.15	5/2+	[M1,E2]		0.00048 *5*	B(M1)(W.u.)=0.0068 +*33*−*19* if M1, B(E2)(W.u.)=4.5 +*21*−*13* if E2.
		2953.92	≤5.2	1219.38	1/2+	[E2]		0.000771 *11*	
		4173.16	100 *17*	0.0	3/2+	[M1,E2]		0.00116 *7*	B(M1)(W.u.)=0.0047 +*15*−*7* if M1, B(E2)(W.u.)=1.02 +*33*−*14* if E2.
4177.9	3/2−	1175.2	<0.82	3002.64	5/2+	[E1]		6.06×10^{-5} *8*	
		1483.9	13.1 *33*	2694.00	3/2+	[E1]		0.000265 *4*	B(E1)(W.u.)=5.7×10^{-4} +*16*−*14*
		1532.2	<0.82	2645.67	7/2+	[M2]		6.18×10^{-5} *9*	
		2414.7	<1.6	1763.15	5/2+	[E1]		0.000921 *13*	
		2958.4	51 *8*	1219.38	1/2+	(E1+M2)	+0.11 *4*	1.22×10^{-3} *2*	B(E1)(W.u.)=2.8×10^{-4} +*5*−*4*; B(M2)(W.u.)=1.7 +*16*−*11* Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level scheme. B(M2)(W.u.)=1.7 +*16*−*11* upper bound exceeds RUL=3.
		4177.6	100 *8*	0.0	3/2+	(E1+M2)	+0.06 *3*	1.74×10^{-3} *3*	B(E1)(W.u.)=1.93×10^{-4} +*28*−*20*; B(M2)(W.u.)=0.18 +*24*−*14* Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level scheme.
4347.76	9/2−	1184.90 *20*	100.0 *29*	3162.87	7/2−	M1+E2	−0.36 *3*	3.57×10^{-5} *5*	B(M1)(W.u.)=0.007 +*7*−*3*; B(E2)(W.u.)=2.5 +*27*−*10* E_γ: weighted average of 1185.0 *2* from (^{24}Mg,αpγ), 1184.9 *3* from (^{16}O,αpγ), and 1184.80 *20* from (^{14}N,αpnγ). Other: 1184.6 *10* from (α,pγ). I_γ: weighted average of 100 *4* from (^{28}Si,αpγ), 100.0 *31* from (^{24}Mg,αpγ), 100 *4* from (^{16}O,αpγ), 100.0 *31* from (α,pγ), and 100.0 *29* from (p,γ). Mult.: from γγ(DCO) and γγ(lin pol) in (^{28}Si,αpγ), ΔJ=1. δ: weighted average of −0.36 *3* from (α,pγ), −0.3 *1* from (^{28}Si,αpγ), and −0.65 *30* from (^{16}O,αpγ).
		1701.88 *25*	46.8 *24*	2645.67	7/2+	E1+M2	−0.018 *12*	0.000434 *6*	B(E1)(W.u.)=3.7×10^{-5} +*39*−*14*; B(M2)(W.u.)=0.019

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	α[a]	Comments
								+47−17 E_γ: weighted average of 1702.0 *3* from (^{24}Mg,αpγ), 1701.7 *5* from (^{16}O,αpγ), and 1701.85 *25* from (^{14}N,αpnγ). Other: 1702.1 *15* from (α,pγ). I_γ: unweighted average of 45.6 *24* from (^{28}Si,αpγ), 39.5 *24* from (^{24}Mg,αpγ), 52.0 *14* from (^{16}O,αpγ), 52 *7* from (α,pγ), and 44.9 *29* from (p,γ). Mult.: from $\gamma\gamma$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ) with ΔJ=1 and $\gamma(\theta)$ and γ(lin pol) in (α,pγ). δ: from (p,γ). Others: +0.5 *+5−3* from (^{28}Si,αpγ), +0.5 *4* from (^{16}O,αpγ).
4347.76	9/2$^-$	2584.51	≤4.4	1763.15	5/2$^+$	[M2]	0.000329 *5*	
		4347.8 *8*	12.4 *11*	0.0	3/2$^+$	[E3]	0.001000 *14*	B(E3)(W.u.)=44 *+46−17* E_γ,I_γ: from (^{16}O,αpγ).
4624.3	3/2,5/2	4624.0	100	0.0	3/2$^+$	D+Q		E_γ: other: 4618 *2* from (α,pγ). Mult.,δ: from p$\gamma(\theta)$ with −0.08 *14* for J=3/2; +0.36 *15* for J=5/2 in (α,pγ). If J(4624=5/2), Mult would be M1+E2 based on RUL.
4769.86	(5/2$^-$)	1606.95	100 *16*	3162.87	7/2$^-$	[M1,E2]	0.000133 *19*	B(M1)(W.u.)=0.040 *+23−5* if M1, B(E2)(W.u.)=58 *+34−7* if E2.
		1767.17	54 *16*	3002.64	5/2$^+$	[E1]	0.000483 *7*	B(E1)(W.u.)=4.6×10^{-4} *+30−9*
		2075.79	<15	2694.00	3/2$^+$	[E1]	0.000702 *10*	
		3550.29	<15	1219.38	1/2$^+$	[M2]	0.000645 *9*	
		4769.51	<15	0.0	3/2$^+$	[E1]	1.95×10^{-3} *3*	
4839.10	(1/2$^+$,3/2)	1836.41	<12	3002.64	5/2$^+$			
		2145.03	<6.8	2694.00	3/2$^+$			
		2193.36[b]	<5.1	2645.67	7/2$^+$			
		3075.81	100 *9*	1763.15	5/2$^+$			
		3619.52	<12	1219.38	1/2$^+$			
		4838.74	70 *9*	0.0	3/2$^+$			
4854.4	(1/2,3/2)	3634.8	100 *7*	1219.38	1/2$^+$			
		4854.0	33 *7*	0.0	3/2$^+$			
4880.96	7/2$^{(+)}$	1878.27	15 *5*	3002.64	5/2$^+$	[M1,E2]	0.000240 *30*	B(M1)(W.u.)=0.051 *+30−16* if M1, B(E2)(W.u.)=55 *+32−17* if E2.
		2186.89	<16	2694.00	3/2$^+$	[E2]	0.000417 *6*	
		2235.21	47 *8*	2645.67	7/2$^+$	[M1,E2]	0.00040 *4*	I_γ: other: 52 *9* from (α,pγ) for a doublet. B(M1)(W.u.)=0.095 *+43−18* if M1, B(E2)(W.u.)=72 *+33−14* if E2. B(E2)(W.u.)=72 *+33−14* upper bound exceeds RUL=100 if E2.
		3117.4 *20*	100 *8*	1763.15	5/2$^+$	[M1,E2]	0.00078 *6*	E_γ: from (α,pγ). I_γ: weighted average of 100 *9* from (α,pγ) and 100 *8* from (p,γ). B(M1)(W.u.)=0.075 *+33−12* if M1, B(E2)(W.u.)=29 *+13−5* if E2.
		4880.60	<8.1	0.0	3/2$^+$	[E2]	1.46×10^{-3} *2*	

$^{35}_{17}Cl_{18}$-28

Adopted Levels, Gammas (continued)

$\gamma(^{35}\mathrm{Cl})$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
5009.1	(1/2,3/2)	835.7	<25	4173.43	$(5/2^+)$				
		950.0	<10	4059.1	$(3/2)^-$				
		1846.2[b]	<50	3162.87	$7/2^-$				
		2006.4	<20	3002.64	$5/2^+$				
		2315.0	<25	2694.00	$3/2^+$				
		2363.3[b]	<5	2645.67	$7/2^+$				
		3245.8	<10	1763.15	$5/2^+$				
		3789.5	<7	1219.38	$1/2^+$				
		5008.7	100	0.0	$3/2^+$				
5163.34	$7/2^-$	2000.41	93 *12*	3162.87	$7/2^-$	D+Q	+0.44 *20*		
		2160.63	23 *9*	3002.64	$5/2^+$				
		3400.01	100 *12*	1763.15	$5/2^+$	D(+Q)	0.00 *3*		
		3943.72[b]	<23	1219.38	$1/2^+$				
		5162.93	<23	0.0	$3/2^+$				
5214.5	$(5/2^+)$	205.4	<23	5009.1	(1/2,3/2)				
		1041.1	<60	4173.43	$(5/2^+)$				
		1155.4	<10	4059.1	$(3/2)^-$				
		2051.6	<15	3162.87	$7/2^-$				
		2211.8	<25	3002.64	$5/2^+$				
		2520.4	<15	2694.00	$3/2^+$				
		2568.7	<15	2645.67	$7/2^+$				
		3451.2	<6	1763.15	$5/2^+$				
		3994.9	<4	1219.38	$1/2^+$				
		5214.1	100	0.0	$3/2^+$	[M1,E2]		0.00147 *8*	
5403.3	$3/2^-$	2400.6	46 *9*	3002.64	$5/2^+$	[E1]		0.000913 *13*	B(E1)(W.u.)=0.0011 *+6−2*
		3640.0	<18	1763.15	$5/2^+$	[E1]		1.54×10^{-3} *2*	
		4183.7	100 *18*	1219.38	$1/2^+$	[E1]		1.75×10^{-3} *2*	B(E1)(W.u.)=4.5×10^{-4} *+20−6*
		5402.9	<13	0.0	$3/2^+$	[E1]		2.14×10^{-3} *3*	
5407.07	$11/2^-$	1059.32 *20*	14.5 *22*	4347.76	$9/2^-$	M1+E2	+0.14 *3*	3.67×10^{-5} *5*	B(M1)(W.u.)=0.0082 *+30−20*; B(E2)(W.u.)=0.54 *+35−23* E_γ: weighted average of 1059.3 *2* from (^{24}Mg,αpγ), 1059.5 *2* from (^{16}O,αpγ), and 1059.17 *20* from (^{14}N,αpnγ). Other: 1059.3 *10* from (α,pγ). I_γ: unweighted average of 12.0 *12* from (^{28}Si,αpγ), 16.9 *17* from (^{24}Mg,αpγ), 19.2 *5* from (^{16}O,αpγ), and 9.9 *22* from (α,pγ). Mult.: from γγ(DCO) and γγ(lin pol) in (^{28}Si,αpγ), ΔJ=1. δ: weighted average of +0.12 *3* from (^{16}O,αpγ),

$\gamma(^{35}$Cl) (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	α[a]	Comments
									+0.2 *1* from (^{28}Si,αpγ), and +0.25 *8* from (α,pγ).
5407.07	11/2$^-$	2243.71 *23*	100.0 *22*	3162.87	7/2$^-$	E2		0.000444 *6*	B(E2)(W.u.)=4.6 *+15−9* E_γ: weighted average of 2244.2 *3* from (^{24}Mg,αpγ), 2243.3 *2* from (^{16}O,αpγ), 2244.00 *25* from (^{14}N,αpnγ), and 2244 *3* from (α,pγ). I_γ: weighted average of 100.0 *33* from (^{28}Si,αpγ), 100.0 *32* from (^{24}Mg,αpγ), 100.0 *23* from (^{16}O,αpγ), and 100.0 *22* from (α,pγ). Mult.: from $\gamma\gamma$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ), ΔJ=2.
5586.0	5/2$^+$	2940.2	67 *17*	2645.67	7/2$^+$				
		3822.6	100 *17*	1763.15	5/2$^+$				
		4366.3	<17	1219.38	1/2$^+$				
		5585.5	<17	0.0	3/2$^+$				
5598.4	3/2$^+$,5/2$^+$	3835.0	37 *14*	1763.15	5/2$^+$	[M1,E2]		0.00105 *7*	B(M1)(W.u.)=0.050 *+29−19* if M1, B(E2)(W.u.)=13 *+8−5* if E2.
		5597.9	100 *14*	0.0	3/2$^+$	[M1,E2]		0.00157 *8*	B(M1)(W.u.)=0.044 *+23−12* if M1, B(E2)(W.u.)=5.3 *+28−14* if E2.
5646	(5/2,7/2,9/2$^+$)	2483	100	3162.87	7/2$^-$				
		2643		3002.64	5/2$^+$				
		3883		1763.15	5/2$^+$				
		5646	<8	0.0	3/2$^+$				
5655	(3/2)$^+$	2652	100 *6*	3002.64	5/2$^+$	[M1,E2]		0.00058 *5*	B(M1)(W.u.)=0.076 *+30−14* if M1, B(E2)(W.u.)=41 *+16−7* if E2.
		3892	7.5 *25*	1763.15	5/2$^+$	[M1,E2]		0.00107 *7*	B(M1)(W.u.)=0.0018 *+10−6* if M1, B(E2)(W.u.)=0.45 *+25−15* if E2.
		4435	<7.5	1219.38	1/2$^+$	[M1,E2]		0.00125 *8*	
5682.2	1/2$^-$,3/2$^-$	1504.3	100	4177.9	3/2$^-$				
5723.52	5/2$^+$	3029.38	63 *8*	2694.00	3/2$^+$				
		3960.13	100 *11*	1763.15	5/2$^+$				
5758	(1/2,3/2$^-$)	1580	67 *22*	4177.9	3/2$^-$				
		4538	<13	1219.38	1/2$^+$				
		5758	100 *22*	0.0	3/2$^+$				
5806.6	(1/2,3/2,5/2)	5806.1	100	0.0	3/2$^+$				
5830	(5/2$^-$)	2667	100	3162.87	7/2$^-$	D+Q			I_γ: from (α,pγ). δ: −2.2 *5* or −0.6 *1* for J=5/2 from $\gamma(\theta)$ in (α,pγ).
5926.65	(11/2)$^-$	1579.09 *30*	100 *11*	4347.76	9/2$^-$	M1+E2	−0.2 *1*	0.0001074 *22*	E_γ: weighted average of 1578.9 *5* from (^{16}O,αpγ), 1579.15 *30* from (^{14}N,αpnγ), and 1579.3 *15* from (α,pγ).

$^{35}_{17}$Cl$_{18}$-30

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	α[a]	Comments
									Mult.,δ: D+Q and δ from $\gamma\gamma(\theta)$(DCO) in (^{28}Si,αpγ) with ΔJ=1; E1+M2 ruled out by RUL. Other: −0.8 *4* from $\gamma(\theta)$ in (α,pγ).
5926.65	(11/2)$^-$	1983.7	11.1 *22*	3942.9	9/2$^+$	[E1]		0.000639 *9*	I_γ: from (^{28}Si,αpγ).
		2763.66		3162.87	7/2$^-$				
6087.31	13/2$^-$	680.35 *10*	100.0 *21*	5407.07	11/2$^-$	M1+E2	+0.020 *15*	8.70×10^{-5} *12*	B(M1)(W.u.)=0.0107 *10* E_γ: weighted average of 680.4 *1* from (^{24}Mg,αpγ), 680.5 *5* from (^{16}O,αpγ), and 680.22 *15* from (^{14}N,αpnγ). Other: 679.9 *10* from (α,pγ). I_γ: weighted average of 100.0 *31* from (^{28}Si,αpγ), 100.0 *21* from (^{16}O,αpγ), and 100 *6* from (α,pγ). Mult.,δ: from $\gamma(\theta)$ and γ(lin pol) in (α,pγ). Other: δ=+0.1 *1* from $\gamma\gamma$(DCO) in (^{28}Si,αpγ), ΔJ=1.
		1739.4 *4*	6.9 *9*	4347.76	9/2$^-$	E2		0.0002076 *29*	B(E2)(W.u.)=0.054 *+9−8* E_γ: weighted average of 1739.6 *5* from (^{16}O,αpγ) and 1739.3 *4* from (^{14}N,αpnγ). I_γ: weighted average of 6.3 *6* from (^{28}Si,αpγ), 7.9 *9* from (^{16}O,αpγ), and 18 *6* from (α,pγ). Mult.: from $\gamma\gamma(\theta)$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,αpγ), ΔJ=2.
6106.2	(3/2,5/2$^+$)	4342.8	10.3 *35*	1763.15	5/2$^+$				
		4886.5	62 *17*	1219.38	1/2$^+$				
		6105.6	100 *17*	0.0	3/2$^+$				
6181	(1/2$^+$ to 7/2$^+$)	6180	100	0.0	3/2$^+$				
6380.2	(9/2$^-$)	296 *1*		6087.31	13/2$^-$				I_γ: from (^{16}O,αpγ).
		971 *1*		5407.07	11/2$^-$				I_γ: from (^{16}O,αpγ).
6492	(3/2$^+$)	6491	100	0.0	3/2$^+$				
6660.2	(11/2$^-$)	2312.4		4347.76	9/2$^-$				
		3497.1		3162.87	7/2$^-$				
7066.2	(5/2$^+$)	1902.8	1.8	5163.34	7/2$^-$				
		2892.6	1.0	4173.43	(5/2$^+$)				
		2953.7	1.2	4112.4	7/2$^+$				
		3007.0	2.9	4059.1	(3/2)$^-$				
		3903.1	12	3162.87	7/2$^-$				
		4063.3	6.1	3002.64	5/2$^+$				

$^{35}_{17}Cl_{18}$-31

Adopted Levels, Gammas (continued)

γ(^{35}Cl) (continued)

E_i(level)	J_i^π	E_γ†	I_γ‡	E_f	J_f^π	Mult.#	δ#
7066.2	(5/2$^+$)	4371.9	10	2694.00	3/2$^+$		
		4420.2	3.3	2645.67	7/2$^+$		
		5302.6	29	1763.15	5/2$^+$		
		5846.3	37	1219.38	1/2$^+$		
		7065.4	100	0.0	3/2$^+$		
7103.4	3/2$^{(-)}$	2925.4	2.3	4177.9	3/2$^-$		
		3184.8	2.3	3918.47	3/2$^+$		
		4100.5	4.7	3002.64	5/2$^+$		
		4409.1	17	2694.00	3/2$^+$	D(+Q)	0.00 *3*
		5339.8	6.3	1763.15	5/2$^+$		
		5883.5	100	1219.38	1/2$^+$	D(+Q)	0.00 *3*
		7102.6	23	0.0	3/2$^+$		
7178.6	1/2$^{(+)}$	1072.4	7.9	6106.2	(3/2,5/2$^+$)		
		1775.3	4.5	5403.3	3/2$^-$		
		2339.4	4	4839.10	(1/2$^+$,3/2)		
		3005.0	2.1	4173.43	(5/2$^+$)		
		3119.4	58	4059.1	(3/2)$^-$		
		3210.8	24	3967.6	1/2$^+$		
		3260.0	7.9	3918.47	3/2$^+$		
		4484.3	11	2694.00	3/2$^+$		
		5958.7	45	1219.38	1/2$^+$		
		7177.8	100	0.0	3/2$^+$		
7185	5/2$^+$	1970	<1.1	5214.5	(5/2$^+$)		
		2176	<1.1	5009.1	(1/2,3/2)		
		3011	<11	4173.43	(5/2$^+$)		
		3126	57 *13*	4059.1	(3/2)$^-$		
		4022	<6.5	3162.87	7/2$^-$		
		4182	<6.5	3002.64	5/2$^+$		
		4491	4.4 *22*	2694.00	3/2$^+$		
		4539	13 *5*	2645.67	7/2$^+$		
		5421	9 *7*	1763.15	5/2$^+$		
		5965	35 *15*	1219.38	1/2$^+$		
		7184	100 *22*	0.0	3/2$^+$		
7194.6	1/2$^-$	2185.4	1.2	5009.1	(1/2,3/2)		
		2340.1	3.0	4854.4	(1/2,3/2)		
		2355.4	2.7	4839.10	(1/2$^+$,3/2)		
		3016.6	24	4177.9	3/2$^-$		
		3135.4	4.5	4059.1	(3/2)$^-$		
		3226.8	2.7	3967.6	1/2$^+$		
		3276.0	6	3918.47	3/2$^+$		
		4500.3	1.6	2694.00	3/2$^+$		
		5431.0	0.75	1763.15	5/2$^+$		

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ‡	E_f	J^π_f	Mult.#	δ#	Comments
7194.6	$1/2^-$	5974.7	100	1219.38	$1/2^+$			
		7193.8	3.0	0.0	$3/2^+$			
7225.5	5/2	1119.3	3.1	6106.2	$(3/2,5/2^+)$			
		1418.9		5806.6	(1/2,3/2,5/2)			
		1543.3	3.4	5682.2	$1/2^-,3/2^-$			
		1571	1.4	5655	$(3/2)^+$			
		1822.2	1.5	5403.3	$3/2^-$			
		2010.9	12	5214.5	$(5/2^+)$			
		3047.5	14	4177.9	$3/2^-$			
		3166.3	3.4	4059.1	$(3/2)^-$			
		3306.9	2.5	3918.47	$3/2^+$			
		4222.6		3002.64	$5/2^+$			
		4531.2	17	2694.00	$3/2^+$	D+Q	+0.36 *4*	
		5461.9	12	1763.15	$5/2^+$	D+Q	−0.36 *7*	
		7224.7	100	0.0	$3/2^+$	D+Q	+0.36 *4*	
7234.0	$5/2^{(+)}$	1579	0.22	5655	$(3/2)^+$			
		3056.0		4177.9	$3/2^-$			
		4539.7	1.9	2694.00	$3/2^+$			
		4588.0	3.2	2645.67	$7/2^+$			
		5470.4	0.22	1763.15	$5/2^+$			
		6014.1	1.9	1219.38	$1/2^+$			
		7233.2	100	0.0	$3/2^+$	D+Q	−0.10	
7272.7	$1/2^-$	1515	1.7	5758	$(1/2,3/2^-)$			
		1869.4	0.29	5403.3	$3/2^-$			
		2263.5	2.0	5009.1	(1/2,3/2)			
		2433.5	1.5	4839.10	$(1/2^+,3/2)$			
		3099.1	0.58	4173.43	$(5/2^+)$			
		3213.4	1.3	4059.1	$(3/2)^-$			
		3304.9	1.2	3967.6	$1/2^+$			
		4578.4	1.6	2694.00	$3/2^+$			
		6052.8	35	1219.38	$1/2^+$			
		7271.9	100	0.0	$3/2^+$			
7361.9	$3/2^{(+)}$	1707	0.41	5655	$(3/2)^+$			
		2352.7	0.82	5009.1	(1/2,3/2)			
		2522.7	1.4	4839.10	$(1/2^+,3/2)$			
		3183.8	0.69	4177.9	$3/2^-$			
		3302.6	2.7	4059.1	$(3/2)^-$			
		3394.1	4.1	3967.6	$1/2^+$			
		3443.3	1.1	3918.47	$3/2^+$			
		4359.0	0.41	3002.64	$5/2^+$			
		4667.6	0.69	2694.00	$3/2^+$			
		5598.3	11	1763.15	$5/2^+$	D+Q		δ: >+2.5 or −0.01 *10* from $\gamma(\theta)$ in (p,γ).

$^{35}_{17}Cl_{18}$-33

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ [†]	I_γ [‡]	E_f	J^π_f	Mult. [#]	δ [#]	α [a]	Comments
7361.9	$3/2^{(+)}$	6141.9	100	1219.38	$1/2^+$	(M1+E2)			Mult.: D+Q from $\gamma(\theta)$ in (p,γ); E1+M2 is less likely for this strong primary transition. δ: −0.05 *3* or +2.5 *3* in (p,γ).
		7361.1	14	0.0	$3/2^+$	D+Q			δ: $-0.16<\delta(Q/D)<-0.12$ or $+7.5<\delta(Q/D)<+10$ in (p,γ).
7396.4	$7/2^{(-)}$	3048.5	23	4347.76	$9/2^-$				
		3283.8	17	4112.4	$7/2^+$				
		3453.3	11	3942.9	$9/2^+$				
		4233.3	100	3162.87	$7/2^-$	(M1+E2)	+0.28 *25*	1.12×10^{-3} *3*	Mult.: D+Q from $\gamma(\theta)$ in (p,γ); E1+M2 is less likely for this strong primary transition.
		4393.5	30	3002.64	$5/2^+$	D(+Q)	+0.011 *18*		
		4750.4	28	2645.67	$7/2^+$				
		5632.8	4.3	1763.15	$5/2^+$				
7450.9	3/2	3532.2	6.9 *34*	3918.47	$3/2^+$				
		4448.0	13.7 *14*	3002.64	$5/2^+$				
		4756.6	11.0 *11*	2694.00	$3/2^+$				
		5687.3	5.5 *28*	1763.15	$5/2^+$				
		6230.9	100 *10*	1219.38	$1/2^+$				
7501.1	$(5/2^+,7/2,9/2^+)$	3558.0	46 *6*	3942.9	$9/2^+$				
		4498.2	23.6 *24*	3002.64	$5/2^+$				
		4855.1	100 *11*	2645.67	$7/2^+$				
		5737.5	12.7 *13*	1763.15	$5/2^+$				
7502.9		1396.7 [@]	14 [@]	6106.2	$(3/2,5/2^+)$				
		1904.4 [@]	4 [@]	5598.4	$3/2^+,5/2^+$				
		2493.7 [@]	4 [@]	5009.1	(1/2,3/2)				
		3324.8 [@]	6 [@]	4177.9	$3/2^-$				
		3443.6 [@]	100 [@]	4059.1	$(3/2)^-$				
		3559.8 [@]	4 [@]	3942.9	$9/2^+$				
		4339.7 [@]	8 [@]	3162.87	$7/2^-$				
		4500.0 [@]	8 [@]	3002.64	$5/2^+$				
		4856.9 [@]	14 [@]	2645.67	$7/2^+$				
		5739.2 [@]	1.4 [@]	1763.15	$5/2^+$				
		6282.9 [@]	36 [@]	1219.38	$1/2^+$				
		7502.0 [@]	0.6 [@]	0.0	$3/2^+$				
7518.8	7/2	1920.3	2.9 *15*	5598.4	$3/2^+,5/2^+$				
		1932.7	4.4 *22*	5586.0	$5/2^+$				
		3170.9	2.9 *15*	4347.76	$9/2^-$				
		3575.7	2.9 *15*	3942.9	$9/2^+$				

$^{35}_{17}Cl_{18}$-34

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
7518.8	7/2	4355.6	100 *10*	3162.87	$7/2^-$	D(+Q)	+0.1 *2*		
		4872.8	4.4 *22*	2645.67	$7/2^+$				
		5755.1	29.4 *30*	1763.15	$5/2^+$	D+Q	+0.098 *22*		
7548.9	$7/2^-$	1903	0.59 *7*	5646	$(5/2,7/2,9/2^+)$				
		1962.8	0.11	5586.0	$5/2^+$	[E1]		0.000624 *9*	
		2778.9	1.49 *11*	4769.86	$(5/2^-)$	[M1,E2]		0.00064 *6*	
		4385.7	100.0 *11*	3162.87	$7/2^-$	M1+E2	−0.07 *2*	1.16×10^{-3} *2*	
		4545.9	2.13 *11*	3002.64	$5/2^+$	(E1+M2)	+0.6 *4*	0.00163 *22*	Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme.
		4902.9	0.72 *8*	2645.67	$7/2^+$	[E1]		1.99×10^{-3} *3*	
		5785.2	0.35 *4*	1763.15	$5/2^+$	[E1]		2.24×10^{-3} *3*	
		7548.0	0.287 *32*	0.0	$3/2^+$	(M2)			Mult.: Q(+O) with δ(O/Q)=0.1 *2* from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme; E3 component is less likely based on RUL.
7561.3	1/2,3/2	3383.2	2.6	4177.9	$3/2^-$				
		3502.0	17	4059.1	$(3/2)^-$				
		3593.5	1.4	3967.6	$1/2^+$				
		3642.6	4.6	3918.47	$3/2^+$				
		4866.9	63	2694.00	$3/2^+$				
		6341.3	100	1219.38	$1/2^+$				
		7560.4	97	0.0	$3/2^+$				
7600.8	$5/2^+$	2437.4	5.8	5163.34	$7/2^-$	[E1]		0.000934 *13*	
		2719.7	6.1	4880.96	$7/2^{(+)}$				
		2761.6	6.1	4839.10	$(1/2^+,3/2)$				
		2830.8	3.9	4769.86	$(5/2^-)$				
		2976.4	5.5	4624.3	3/2,5/2				
		3422.7	9.1	4177.9	$3/2^-$	[E1]		1.44×10^{-3} *2*	
		3427.2		4173.43	$(5/2^+)$				
		3541.5	6.1	4059.1	$(3/2)^-$	[E1]		1.50×10^{-3} *2*	
		4437.6	15	3162.87	$7/2^-$	[E1]		1.85×10^{-3} *3*	
		4597.8	24	3002.64	$5/2^+$	M1+E2	+0.50 *22*	1.25×10^{-3} *3*	Mult.: D+Q from $\gamma(\theta)$ in (p,γ); E1+M2 is less likely for this strong primary transition.
		4906.4	55	2694.00	$3/2^+$	(M1+E2)		0.00139 *8*	Mult.: D+Q from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=no from level scheme.
		4954.8	5.8	2645.67	$7/2^+$	[M1,E2]		0.00140 *8*	
		5837.1	58	1763.15	$5/2^+$	M1+E2	+0.31 *7*	1.57×10^{-3} *2*	
		6380.8	3.3	1219.38	$1/2^+$	[E2]			
		7599.9	100	0.0	$3/2^+$	M1+E2	−0.19 *4*		
7618.7	$5/2^+$	1438	0.77	6181	$(1/2^+$ to $7/2^+)$				
		2404.1	2.6	5214.5	$(5/2^+)$				
		2455.3	0.9	5163.34	$7/2^-$				

$^{35}_{17}Cl_{18}$-35

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	α[a]	Comments
7618.7	5/2$^+$	2779.5	0.9	4839.10	(1/2$^+$,3/2)				
		3440.6	3.9	4177.9	3/2$^-$				
		3559.4	2.6	4059.1	(3/2)$^-$				
		4455.5	3.9	3162.87	7/2$^-$				
		5855.0	13	1763.15	5/2$^+$	M1+E2	+0.48 *11*	1.59×10^{-3} *3*	
		7617.8	100	0.0	3/2$^+$	M1+E2	+2.2 *3*		
7656.6	(1/2,3/2,5/2$^+$)	1550.4	5.3 *26*	6106.2	(3/2,5/2$^+$)				
		1974.3	1.8 *9*	5682.2	1/2$^-$,3/2$^-$				
		2002	1.8 *9*	5655	(3/2)$^+$				
		2817.4	5.3 *26*	4839.10	(1/2$^+$,3/2)				
		3478.5	1.8 *9*	4177.9	3/2$^-$				
		3597.3	56 *6*	4059.1	(3/2)$^-$				
		3688.8	3.5 *18*	3967.6	1/2$^+$				
		7655.7	100 *11*	0.0	3/2$^+$				
7671.9	5/2$^+$	2073.4	2.1	5598.4	3/2$^+$,5/2$^+$				
		2508.5	7	5163.34	7/2$^-$				
		2790.8	2.1	4880.96	7/2$^{(+)}$				
		2901.9	1.8	4769.86	(5/2$^-$)				
		3324.0	11	4347.76	9/2$^-$				
		3498.3	8.8	4173.43	(5/2$^+$)				
		4508.7	3.2	3162.87	7/2$^-$				
		4668.9	16	3002.64	5/2$^+$	M1+E2	−1.4 *5*	0.00134 *4*	
		5025.8	23	2645.67	7/2$^+$	D(+Q)	+0.3 *22*		
		5908.2	100	1763.15	5/2$^+$	M1+E2	+0.86 *14*	1.64×10^{-3} *3*	
		7671.0	1.4	0.0	3/2$^+$				
7685.5	3/2$^-$	1927		5758	(1/2,3/2$^-$)				
		2087.0	2.9	5598.4	3/2$^+$,5/2$^+$				
		2282.1	2.9	5403.3	3/2$^-$				
		2470.9	1	5214.5	(5/2$^+$)				
		2676.3	0.86	5009.1	(1/2,3/2)				
		3507.4	4.3	4177.9	3/2$^-$				
		3511.9	5.7	4173.43	(5/2$^+$)				
		3626.2	8.6	4059.1	(3/2)$^-$				
		4682.5	14	3002.64	5/2$^+$	D(+Q)	−0.01 *5*		
		5921.8	1	1763.15	5/2$^+$				
		6465.5	1.4	1219.38	1/2$^+$				
		7684.6	100	0.0	3/2$^+$	D+Q	+6.0 *7*		Mult.: D+Q with δ=+6.0 *7* from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme; but E1+M2 with such a large δ(M2/E1) is unlikely for this strong primary transition based on RUL, which might suggest a different level in (p,γ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	δ[#]	α[a]	Comments
7693.9	(1/2$^+$,3/2,5/2)	2095.4	1.0 *5*	5598.4	3/2$^+$,5/2$^+$				
		3634.6	1.0 *5*	4059.1	(3/2)$^-$				
		4690.9	2.1 *11*	3002.64	5/2$^+$				
		7693.0	100 *11*	0.0	3/2$^+$				
7706.4	5/2$^+$	2051	4.8	5655	(3/2)$^+$	M1+E2		0.00032 *4*	δ: <−2 or ≥13 from $\gamma(\theta)$ in (p,γ).
		2825.3	0.48	4880.96	7/2$^{(+)}$				
		2867.2	1.6	4839.10	(1/2$^+$,3/2)				
		3082.0	0.96	4624.3	3/2,5/2				
		3528.3	2.4	4177.9	3/2$^-$				
		3593.8	2.4	4112.4	7/2$^+$				
		3787.7	0.72	3918.47	3/2$^+$				
		4703.4	1.2	3002.64	5/2$^+$				
		5060.3	4.8	2645.67	7/2$^+$	M1+E2		0.00143 *8*	δ: −1.01 *10* or ≥13 from (p,γ).
		5942.7	0.48	1763.15	5/2$^+$				
		6486.4	0.6	1219.38	1/2$^+$				
		7705.5	100	0.0	3/2$^+$	D(+Q)	+0.1 *1*		
7744.8	7/2$^-$	2863.7	47	4880.96	7/2$^{(+)}$				
		2974.8	4.7	4769.86	(5/2$^-$)				
		3632.2	14	4112.4	7/2$^+$				
		3801.7	23	3942.9	9/2$^+$	(E1)		1.60×10^{-3} *2*	Mult.: D(+Q) with δ=−0.16 *17* from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme; M2 component is less likely for the primary transition.
		4581.6	37	3162.87	7/2$^-$	M1+E2	+0.18 *8*	1.23×10^{-3} *2*	
		5098.7	100	2645.67	7/2$^+$	(E1)		2.05×10^{-3} *3*	Mult.: D(+Q) with δ=+0.1 *30* from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme; M2 component is less likely for the primary transition.
		5981.1	5.6	1763.15	5/2$^+$				
		6524.8[b]	1.4	1219.38	1/2$^+$	[E3]			This transition is considered questionable by the evaluators since it would require Mult=E3 which is unlikely for this primary transition based on RUL.
7777.1	(5/2$^+$)	2122	1.3	5655	(3/2)$^+$				
		2613.7	2.8	5163.34	7/2$^-$				
		2937.9	5	4839.10	(1/2$^+$,3/2)				
		3007.1	7.5	4769.86	(5/2$^-$)				
		3599.0	18	4177.9	3/2$^-$				
		3717.8	2.8	4059.1	(3/2)$^-$				
		4613.9	5	3162.87	7/2$^-$				
		4774.1	10	3002.64	5/2$^+$				
		5082.7	23	2694.00	3/2$^+$				
		5131.0	3.3	2645.67	7/2$^+$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
7777.1	(5/2$^+$)	6013.4	100	1763.15	5/2$^+$				
		6557.1	28	1219.38	1/2$^+$				
		7776.2	45	0.0	3/2$^+$				
7781.7	(5/2$^-$)	1675.5	3.9	6106.2	(3/2,5/2$^+$)				
		2183.2		5598.4	3/2$^+$,5/2$^+$				
		2378.3		5403.3	3/2$^-$				
		2567.1	4.2	5214.5	(5/2$^+$)				
		3011.7	19	4769.86	(5/2$^-$)				
		3433.8	5.6	4347.76	9/2$^-$				
		3608.1	100	4173.43	(5/2$^+$)				
		3669.1	17	4112.4	7/2$^+$				
		3722.4	3.3	4059.1	(3/2)$^-$				
		3863.0	31	3918.47	3/2$^+$				
		4618.5	14	3162.87	7/2$^-$				
		5135.6	22	2645.67	7/2$^+$				
		6018.0	50	1763.15	5/2$^+$				
		6561.7	3.3	1219.38	1/2$^+$				
		7780.8	4.7	0.0	3/2$^+$				
7797.2	1/2$^-$	2039		5758	(1/2,3/2$^-$)				
		2142	1.6	5655	(3/2)$^+$				
		2393.8	2.4	5403.3	3/2$^-$				
		2788.0	2.4	5009.1	(1/2,3/2)				
		2942.7	1.6	4854.4	(1/2,3/2)				
		2958.0	1.3	4839.10	(1/2$^+$,3/2)				
		3737.9	0.36	4059.1	(3/2)$^-$				
		6577.2	11	1219.38	1/2$^+$				
		7796.3	100	0.0	3/2$^+$				
7837.2	3/2$^-$	1656[&]	1.4[&]	6181	(1/2$^+$ to 7/2$^+$)				
		2238.7[&]	1.4[&]	5598.4	3/2$^+$,5/2$^+$				
		2622.6[&]	3.8[&]	5214.5	(5/2$^+$)				
		3659.1[&]	76[&]	4177.9	3/2$^-$	(M1+E2)	−0.05 *3*	0.000917 *13*	Mult.: D+Q from (p,γ); $\Delta\pi$=no from level scheme. B(M1)(W.u.)=1.03×10^3 *19* exceeds RUL=10. B(E2)(W.u.)=7×10^2 *+11−6* exceeds RUL=100.
		3663.6[&]	8.1[&]	4173.43	(5/2$^+$)				
		3777.9[&]	5.4[&]	4059.1	(3/2)$^-$				
		3894.1[&b]	1.4[&]	3942.9	9/2$^+$				This γ is a primary γ from the unresolved 7837+7839 doublet goes to 9/2$^+$, which is unlikely from a 3/2$^-$, 7837 level. It is likely from the (5/2$^+$), 7839.0 level.
		3918.5[&]	0.54[&]	3918.47	3/2$^+$				

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
7837.2	3/2$^-$	4834.2[&]	11[&]	3002.64	5/2$^+$				
		6073.5[&]	5.1[&]	1763.15	5/2$^+$				
		6617.2[&]	100[&]	1219.38	1/2$^+$	(E1)			Mult.: D(+Q) with δ=+0.06 *3* from (p,γ); $\Delta\pi$=yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=6.5 *+10−8* exceeds RUL=0.1.
		7836.3[&]	57[&]	0.0	3/2$^+$	(E1)			Mult.: D(+Q) with δ=+0.02 *3* from (p,γ); $\Delta\pi$=yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=2.24 *+47−44* exceeds RUL=0.1.
7839.0	(5/2$^+$)	1658[&]	1.4[&]	6181	(1/2$^+$ to 7/2$^+$)				
		2240.5[&]	1.4[&]	5598.4	3/2$^+$,5/2$^+$				
		2624.4[&]	3.8[&]	5214.5	(5/2$^+$)				
		3660.9[&]	76[&]	4177.9	3/2$^-$				
		3665.4[&]	8.1[&]	4173.43	(5/2$^+$)				
		3779.7[&]	5.4[&]	4059.1	(3/2)$^-$				
		3895.9[&]	1.4[&]	3942.9	9/2$^+$				
		3920.3[&]	0.54[&]	3918.47	3/2$^+$				
		4836.0[&]	11[&]	3002.64	5/2$^+$				
		6075.3[&]	5.1[&]	1763.15	5/2$^+$				
		6619.0[&]	100[&]	1219.38	1/2$^+$				
		7838.1[&]	57[&]	0.0	3/2$^+$				
7868.7	(1/2$^+$,3/2,5/2$^+$)	2654.1	1.6	5214.5	(5/2$^+$)				
		2859.5	4.9	5009.1	(1/2,3/2)				
		3695.1	8.2	4173.43	(5/2$^+$)				
		6105.0	28	1763.15	5/2$^+$				
		6648.6	21	1219.38	1/2$^+$				
		7867.8	100	0.0	3/2$^+$				
7873.03	13/2$^+$	1212.8		6660.2	(11/2$^-$)				E_γ: from (^{28}Si,αpγ) only.
		1786.2 *5*	9.6 *4*	6087.31	13/2$^-$	D(+Q)	−0.6 *6*		I_γ: weighted average of 8.2 *18* from (^{28}Si,αpγ) and 9.7 *4* from (^{16}O,αpγ). Mult.,δ: from $\gamma\gamma$(DCO) in (^{28}Si,αpγ).
		1946.35 *30*	48.0 *26*	5926.65	(11/2)$^-$	E1+M2	+0.2 *1*	0.000594 *22*	E_γ: weighted average of 1946.2 *5* from (^{16}O,αpγ) and 1946.40 *30* from (^{14}N,αpnγ). I_γ: weighted average of 46 *5* from (^{28}Si,αpγ)

$^{35}_{17}Cl_{18}$-39

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
									and 48.5 *26* from (^{16}O,$\alpha p\gamma$). Mult.,δ: from $\gamma\gamma$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,$\alpha p\gamma$), ΔJ=1.
7873.03	$13/2^+$	2465.94 *30*	100.0 *20*	5407.07	$11/2^-$	E1+M2	−0.25 *10*	0.00091 *4*	E_γ: weighted average of 2466.2 *5* from (^{16}O,$\alpha p\gamma$) and 2465.85 *30* from (^{14}N,$\alpha pn\gamma$). I_γ: weighted average of 100 *9* from (^{28}Si,$\alpha p\gamma$) and 100.0 *20* from (^{16}O,$\alpha p\gamma$). Mult.,δ: from $\gamma\gamma$(DCO) and $\gamma\gamma$(lin pol) in (^{28}Si,$\alpha p\gamma$), ΔJ=1.
7880.8	$3/2^+,5/2^+$	2226	40	5655	$(3/2)^+$				
		3041.6	5.7	4839.10	$(1/2^+,3/2)$				
		3256.3	6.7	4624.3	3/2,5/2				
		3702.7	10	4177.9	$3/2^-$				
		3768.2	4.3	4112.4	$7/2^+$				
		3821.5	6.7	4059.1	$(3/2)^-$				
		3962.1	6.7	3918.47	$3/2^+$				
		4877.8	37	3002.64	$5/2^+$				
		5186.4	83	2694.00	$3/2^+$				
		6117.1	100	1763.15	$5/2^+$				
		7879.9	33	0.0	$3/2^+$				
7899.2	(5/2)	2253	2.2	5646	$(5/2,7/2,9/2^+)$				
		2735.8	1.3	5163.34	$7/2^-$				
		3018.1	1.2	4880.96	$7/2^{(+)}$				
		3274.7	0.74	4624.3	3/2,5/2				
		3721.1	12	4177.9	$3/2^-$				
		3725.6	12	4173.43	$(5/2^+)$				
		3839.9	4.4	4059.1	$(3/2)^-$				
		4736.0	5.9	3162.87	$7/2^-$				
		5204.8	1.9	2694.00	$3/2^+$				
		6135.5	100	1763.15	$5/2^+$				
		7898.2	5.9	0.0	$3/2^+$				
7923.3	$(3/2^+,5/2^+)$	3084.1	3.4	4839.10	$(1/2^+,3/2)$				
		3153.3	4.9	4769.86	$(5/2^-)$				
		3298.8	4.6	4624.3	3/2,5/2				
		3745.2	7.3	4177.9	$3/2^-$				
		3749.7	12	4173.43	$(5/2^+)$				
		3864.0	4.2	4059.1	$(3/2)^-$				
		3955.5	3.7	3967.6	$1/2^+$				
		4004.6	2.9	3918.47	$3/2^+$				
		4920.3	2.2	3002.64	$5/2^+$				
		5228.9	27	2694.00	$3/2^+$				

$^{35}_{17}Cl_{18}$-40

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ†	I_γ‡	E_f	J_f^π	Mult.#	Comments
7923.3	$(3/2^+,5/2^+)$	5277.2	2.2	2645.67	$7/2^+$		
		6159.6	3.7	1763.15	$5/2^+$		
		6703.2	66	1219.38	$1/2^+$		
		7922.3	100	0.0	$3/2^+$		
7970.3	$(5/2^-)$	2324	15	5646	$(5/2,7/2,9/2^+)$		
		2384.2	15	5586.0	$5/2^+$		
		2755.7	15	5214.5	$(5/2^+)$		
		3089.2	50	4880.96	$7/2^{(+)}$		
		3200.3	25	4769.86	$(5/2^-)$		
		3622.3	40	4347.76	$9/2^-$		
		3792.2	30	4177.9	$3/2^-$		
		3857.7	30	4112.4	$7/2^+$		
		4807.1	50	3162.87	$7/2^-$		
		4967.3	25	3002.64	$5/2^+$		
		5324.2	100	2645.67	$7/2^+$		
		6206.6	100	1763.15	$5/2^+$		
		6750.2	5	1219.38	$1/2^+$		
7987.8	$3/2^+$	1881.6	0.83	6106.2	$(3/2,5/2^+)$		
		2333	2.8	5655	$(3/2)^+$		
		3809.7	<1.3	4177.9	$3/2^-$		I_γ: total $I\gamma$=1.3 for γ decays to 4173 and 4178 doublet final levels.
		3814.2	<1.3	4173.43	$(5/2^+)$		I_γ: total $I\gamma$=1.3 for γ decays to 4173 and 4178 doublet final levels.
		3928.5	1.7	4059.1	$(3/2)^-$		
		6767.7	100	1219.38	$1/2^+$	M1+E2	δ: +0.21 *4* or −3.1 *4* from (p,γ).
		7986.8	60	0.0	$3/2^+$	M1+E2	δ: −0.36 *2* or −11 *3* from (p,γ).
7995.6	(5/2)	2188.9	0.77	5806.6	(1/2,3/2,5/2)		
		2397.1	1.4	5598.4	$3/2^+,5/2^+$		
		2592.2	3.9	5403.3	$3/2^-$		
		2832.1	1.9	5163.34	$7/2^-$		
		3817.5	5.1	4177.9	$3/2^-$		
		3936.3	1.4	4059.1	$(3/2)^-$		
		4832.4	9	3162.87	$7/2^-$		
		4992.6	2.6	3002.64	$5/2^+$		
		5301.2	0.26	2694.00	$3/2^+$		
		5349.5	0.77	2645.67	$7/2^+$		
		6231.9	1.2	1763.15	$5/2^+$		
		7994.6	100	0.0	$3/2^+$		
8001.0	$(7/2^+)$	2355	2.6	5646	$(5/2,7/2,9/2^+)$		
		2786.4	1.1	5214.5	$(5/2^+)$		
		3119.9	1	4880.96	$7/2^{(+)}$		
		3231.0	11	4769.86	$(5/2^-)$		
		3653.0	2.7	4347.76	$9/2^-$		
		3827.4	14	4173.43	$(5/2^+)$		

γ(^{35}Cl) (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
8001.0	(7/2$^+$)	4057.9	4.3	3942.9	9/2$^+$				
		4837.8	1.1	3162.87	7/2$^-$				
		5354.9	2.3	2645.67	7/2$^+$				
		6237.3	100	1763.15	5/2$^+$				
		8000.0	2	0.0	3/2$^+$				
8005.0	5/2$^+$	2350	3.3	5655	(3/2)$^+$				
		3123.9	1.1	4880.96	7/2$^{(+)}$				
		3165.8	5.5	4839.10	(1/2$^+$,3/2)				
		3826.9	<2.4	4177.9	3/2$^-$				I_γ: total Iγ=2.4 for γ decays to 4173 and 4178 doublet final levels.
		3831.3	<2.4	4173.43	(5/2$^+$)				I_γ: total Iγ=2.4 for γ decays to 4173 and 4178 doublet final levels.
		4841.8	27	3162.87	7/2$^-$				
		5002.0	3.6	3002.64	5/2$^+$				
		5310.6	3.1	2694.00	3/2$^+$				
		5358.9	7.3	2645.67	7/2$^+$				
		6241.3	27	1763.15	5/2$^+$				
		6784.9	1.1	1219.38	1/2$^+$				
		8004.0	100	0.0	3/2$^+$				
8035.7	1/2$^-$,3/2$^-$	2229.0	10.7 *11*	5806.6	(1/2,3/2,5/2)				
		3857.6	26.8 *27*	4177.9	3/2$^-$				
		3976.4	19.6 *20*	4059.1	(3/2)$^-$				
		5341.3	9 *5*	2694.00	3/2$^+$				
		6272.0	9 *5*	1763.15	5/2$^+$				
		6815.6	100 *11*	1219.38	1/2$^+$				
		8034.7	3.6 *18*	0.0	3/2$^+$				
8038.4	1/2$^+$,3/2$^+$	3183.8	1.1	4854.4	(1/2,3/2)				
		3199.1	2.8	4839.10	(1/2$^+$,3/2)				
		3413.9	5.3	4624.3	3/2,5/2				
		3860.3	2.8	4177.9	3/2$^-$				
		3979.1	5.3	4059.1	(3/2)$^-$				
		4070.6	1.4	3967.6	1/2$^+$				
		4119.7	2.5	3918.47	3/2$^+$				
		5035.4	7	3002.64	5/2$^+$				
		5344.0	100	2694.00	3/2$^+$	M1+E2	−0.20 *2*	1.44×10^{-3} *2*	Mult.,δ: D+Q with δ=−0.20 *2* or +19 *8* from (p,γ); E1+M2 and the larger δ(E2/M1) is ruled out by RUL. B(M1)(W.u.)=52.1 *+43−37* exceeds RUL=10. B(E2)(W.u.)=2.8×10^2 *+6−5* exceeds RUL=100.
		6274.7	26	1763.15	5/2$^+$				
		6818.3	7	1219.38	1/2$^+$				
		8037.4	14	0.0	3/2$^+$	M1+E2	−0.13 *4*		B(M1)(W.u.)=2.20 *44*; B(E2)(W.u.)=2.2 *+16−12*

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	α[a]	Comments
								Mult.,δ: D+Q with δ=−0.13 *4* or +8.1 *26* from (p,γ); E1+M2 and the larger δ(E2/M1) is ruled out by RUL.
8075.9	(5/2)	2430	20	5646	(5/2,7/2,9/2$^+$)			
		2489.8	20	5586.0	5/2$^+$			
		2861.3	14	5214.5	(5/2$^+$)			
		2912.4		5163.34	7/2$^-$			
		3194.8		4880.96	7/2$^{(+)}$			
		3305.9	7.1	4769.86	(5/2$^-$)			
		3897.8	14	4177.9	3/2$^-$			
		3963.3	3.6	4112.4	7/2$^+$			
		5072.9	3.6	3002.64	5/2$^+$			
		6312.1	100	1763.15	5/2$^+$			
		8074.9		0.0	3/2$^+$			
8096.1	(5/2)	2289.4	5.1	5806.6	(1/2,3/2,5/2)			
		2450	13	5646	(5/2,7/2,9/2$^+$)			
		2497.6	13	5598.4	3/2$^+$,5/2$^+$			
		2881.5	3.3	5214.5	(5/2$^+$)			
		2932.6	5.1	5163.34	7/2$^-$			
		3256.8	7.7	4839.10	(1/2$^+$,3/2)			
		3326.1	5.1	4769.86	(5/2$^-$)			
		3918.0	4.9	4177.9	3/2$^-$			
		4036.8	5.1	4059.1	(3/2)$^-$			
		4177.4	4.6	3918.47	3/2$^+$			
		4932.9	13	3162.87	7/2$^-$			
		5401.7	13	2694.00	3/2$^+$			
		5450.0	44	2645.67	7/2$^+$			
		6332.3	21	1763.15	5/2$^+$			
		8095.1	100	0.0	3/2$^+$			
8106.4	3/2$^+$	2507.9	1.4	5598.4	3/2$^+$,5/2$^+$			
		3251.8	1.6	4854.4	(1/2,3/2)			
		3267.1	1.4	4839.10	(1/2$^+$,3/2)			
		3928.3	4.3	4177.9	3/2$^-$			
		4047.1	3.5	4059.1	(3/2)$^-$			
		4138.5	8.1	3967.6	1/2$^+$			
		5103.4	24	3002.64	5/2$^+$			
		5412.0	24	2694.00	3/2$^+$	M1+E2	0.00153 *8*	δ: +4.9 *10* or −0.05 *4*.
		6342.6	8.1	1763.15	5/2$^+$			
		6886.3	95	1219.38	1/2$^+$			
		8105.4	100	0.0	3/2$^+$	M1+E2		δ: +1.2 *1* or −0.46 *4*.
8113.3	(1/2$^+$,3/2,5/2$^+$)	1621	19	6492	(3/2$^+$)			
		2007.0	19	6106.2	(3/2,5/2$^+$)			
		2306.6	3.4	5806.6	(1/2,3/2,5/2)			

$^{35}_{17}Cl_{18}$-43

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ†	I_γ‡	E_f	J_f^π
8113.3	$(1/2^+,3/2,5/2^+)$	2458	2.8	5655	$(3/2)^+$
		2514.8		5598.4	$3/2^+,5/2^+$
		2898.7	5.6	5214.5	$(5/2^+)$
		3939.6	22	4173.43	$(5/2^+)$
		4054.0	4.1	4059.1	$(3/2)^-$
		4145.4	16	3967.6	$1/2^+$
		5110.3	5.9	3002.64	$5/2^+$
		5418.9	41	2694.00	$3/2^+$
		6349.5	16	1763.15	$5/2^+$
		6893.2	100	1219.38	$1/2^+$
		8112.3	78	0.0	$3/2^+$
8147.2	$3/2^-$	2464.9	1.6	5682.2	$1/2^-,3/2^-$
		3138.0	4.8	5009.1	(1/2,3/2)
		3969.1	13	4177.9	$3/2^-$
		4087.8	4.8	4059.1	$(3/2)^-$
		4179.3	3.2	3967.6	$1/2^+$
		4228.5	4.8	3918.47	$3/2^+$
		5452.7	4.8	2694.00	$3/2^+$
		6927.1	24	1219.38	$1/2^+$
		8146.2	100	0.0	$3/2^+$
8156.8	$(5/2^+)$	2511	7.8	5646	$(5/2,7/2,9/2^+)$
		2570.7	16	5586.0	$5/2^+$
		2942.2	12	5214.5	$(5/2^+)$
		3386.8	2.0	4769.86	$(5/2^-)$
		3978.7	9.8	4177.9	$3/2^-$
		4044.2	14	4112.4	$7/2^+$
		4097.4	2.0	4059.1	$(3/2)^-$
		4213.6	3.9	3942.9	$9/2^+$
		4993.6	9.8	3162.87	$7/2^-$
		5153.8	12	3002.64	$5/2^+$
		5510.7	100	2645.67	$7/2^+$
		6393.0	2.0	1763.15	$5/2^+$
		8155.8	5.9	0.0	$3/2^+$
8179.4	$(3/2^-,5/2,7/2^+)$	3554.9	13.2 *13*	4624.3	3/2,5/2
		5016.1	11.8 *12*	3162.87	$7/2^-$
		5176.4	5.9 *30*	3002.64	$5/2^+$
		5484.9	4.4 *22*	2694.00	$3/2^+$
		6415.6	100 *10*	1763.15	$5/2^+$
		8178.4	11.8 *12*	0.0	$3/2^+$
8208.2	$5/2^+$	2553	0.9	5655	$(3/2)^+$
		3327.1	1.0	4880.96	$7/2^{(+)}$
		3368.9	0.51	4839.10	$(1/2^+,3/2)$

$^{35}_{17}Cl_{18}$-44

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^{π}_i	E_{γ}[†]	I_{γ}[‡]	E_f	J^{π}_f	Mult.[#]	δ[#]	α[a]	Comments
8208.2	5/2$^+$	4148.8	1.9	4059.1	(3/2)$^-$				
		4289.5		3918.47	3/2$^+$				
		5205.1	1.4	3002.64	5/2$^+$				
		5513.7	0.64	2694.00	3/2$^+$				
		6444.4	18	1763.15	5/2$^+$	M1+E2	+0.33 *4*		B(M1)(W.u.)=0.75 *+35−22*; B(E2)(W.u.)=7.5 *+41−25*
		6988.1	3.9	1219.38	1/2$^+$				
		8207.2	100	0.0	3/2$^+$	M1+E2	−0.36 *4*		B(M1)(W.u.)=2.0 *+9−5*; B(E2)(W.u.)=15 *+7−4*
8216.3	5/2$^+$	2561	1.8	5655	(3/2)$^+$				
		3446.3	1.8	4769.86	(5/2$^-$)				
		4038.2	<6.7	4177.9	3/2$^-$				
		4042.6	<6.7	4173.43	(5/2$^+$)				
		5053.0	91	3162.87	7/2$^-$	(E1)		2.04×10^{-3} *3*	B(E1)(W.u.)=0.062 *+21−12* Mult.: D+Q with δ=+0.11 *2* from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme; but M2 component is unlikely for this strong primary transition.
		5213.2	11	3002.64	5/2$^+$				
		5521.8	6.7	2694.00	3/2$^+$				
		6452.5	3.1	1763.15	5/2$^+$				
		8215.3	100	0.0	3/2$^+$	M1+E2	+0.06 *3*		B(M1)(W.u.)=0.55 *+18−9*; B(E2)(W.u.)=0.11 *+16−8*
8242.4	3/2$^-$	4068.7	7.7	4173.43	(5/2$^+$)				
		5239.3	2.2	3002.64	5/2$^+$				
		7022.3	100	1219.38	1/2$^+$				
8251	(3/2$^+$,5/2$^+$)	4138	4.9	4112.4	7/2$^+$				
		5248	7.3	3002.64	5/2$^+$				
		7031	100	1219.38	1/2$^+$				
		8250	9.8	0.0	3/2$^+$				
8269.0	5/2$^+$	3499.0	4.4	4769.86	(5/2$^-$)				
		4090.8	27	4177.9	3/2$^-$				
		4350.2	6.7	3918.47	3/2$^+$				
		5105.7	16	3162.87	7/2$^-$				
		5265.9	29	3002.64	5/2$^+$				
		5574.5	13	2694.00	3/2$^+$				
		5622.9	8.9	2645.67	7/2$^+$				
		6505.2	18	1763.15	5/2$^+$				
		8268.0	100	0.0	3/2$^+$				
8277.2	(5/2)$^+$	2622	5.2	5655	(3/2)$^+$				
		3396.1	17	4880.96	7/2$^{(+)}$				
		3437.9	38	4839.10	(1/2$^+$,3/2)				
		3507.2	4.1	4769.86	(5/2$^-$)				
		3652.7	10	4624.3	3/2,5/2				
		3929.2	6.2	4347.76	9/2$^-$				
		4103.5	6.9	4173.43	(5/2$^+$)				

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ‡	E_f	J^π_f	Mult.#	Comments
8277.2	$(5/2)^+$	4217.8	5.2	4059.1	$(3/2)^-$		
		4358.4	17	3918.47	$3/2^+$		
		5274.1	35	3002.64	$5/2^+$		
		5582.7	10	2694.00	$3/2^+$		
		5631.0	10	2645.67	$7/2^+$		
		6513.4	79	1763.15	$5/2^+$		
		8276.2	100	0.0	$3/2^+$		
8282.4	$(3/2^-,5/2)$	2696.3	29	5586.0	$5/2^+$		
		3067.8	29	5214.5	$(5/2^+)$		
		3118.9	11	5163.34	$7/2^-$		
		4104.2	50	4177.9	$3/2^-$		
		4223.0	21	4059.1	$(3/2)^-$		
		5119.1	93	3162.87	$7/2^-$		
		6518.6	25	1763.15	$5/2^+$		
		8281.4	100	0.0	$3/2^+$		
8287.82	$1/2^-$	2633	4.6	5655	$(3/2)^+$		
		3278.6	6.8	5009.1	(1/2,3/2)		
		4109.7	21	4177.9	$3/2^-$		
		4228.5	4.6	4059.1	$(3/2)^-$		
		4319.9	6.8	3967.6	$1/2^+$		
		4369.06	2.3	3918.47	$3/2^+$		
		5593.3	52	2694.00	$3/2^+$		
		7067.67	100	1219.38	$1/2^+$	(E1)	B(E1)(W.u.)=0.069 *+24−14* Mult.: D+Q withδ(Q/D)=+0.21 *2* or −3.2 *2* in (p,γ); but M2 component is unlikely for this primary transition based on resonance Γ=0.04 keV.
		8286.77	30	0.0	$3/2^+$		
8298.2	$3/2^+$	3288.9	13	5009.1	(1/2,3/2)		
		3417.1	17	4880.96	$7/2^{(+)}$		
		3443.6	21	4854.4	(1/2,3/2)		
		4124.5	25	4173.43	$(5/2^+)$		
		4238.8	42	4059.1	$(3/2)^-$		
		5134.9	4.2	3162.87	$7/2^-$		This γ is questionable if J^π(8298)=$3/2^+$, since it would require Mult=M2 which is probably unlikely; however J^π(8298)=$3/2^-$ would make the strong 7078.0γ and 8297.1γ less likely.
		5295.1	4.2	3002.64	$5/2^+$		
		5603.7	42	2694.00	$3/2^+$		
		5652.0	25	2645.67	$7/2^+$		
		6534.4	33	1763.15	$5/2^+$		
		7078.1	100	1219.38	$1/2^+$	M1+E2	δ: −1.7 *1* or −0.02 *2* from $\gamma(\theta)$ in (p,γ).
		8297.1	54	0.0	$3/2^+$	M1+E2	δ: −0.15 *4* or +9.5 *36* from $\gamma(\theta)$ in (p,γ).
8318.1	$(5/2^+)$	2511.4	0.9	5806.6	(1/2,3/2,5/2)		

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ‡	E_f	J^π_f	Mult.#	δ#	αa	Comments
8318.1	(5/2$^+$)	2672	0.9	5646	(5/2,7/2,9/2$^+$)				
		2719.6	1.9	5598.4	3/2$^+$,5/2$^+$				
		3548.1	0.39	4769.86	(5/2$^-$)				
		3693.6	2.3	4624.3	3/2,5/2				
		4139.9	<3.9	4177.9	3/2$^-$				
		4144.4	<3.9	4173.43	(5/2$^+$)				
		4258.7	1.5	4059.1	(3/2)$^-$				
		4350.2	3.9	3967.6	1/2$^+$				
		5154.8	0.64	3162.87	7/2$^-$				
		5315.0	3.9	3002.64	5/2$^+$				
		5623.6	2.1	2694.00	3/2$^+$				
		5671.9	1.2	2645.67	7/2$^+$				
		6554.3	2.3	1763.15	5/2$^+$				
		7098.0	2.6	1219.38	1/2$^+$				
		8317.0	100	0.0	3/2$^+$				
8319.6	15/2$^-$	2232.7 *6*	57.2 *28*	6087.31	13/2$^-$	[M1,E2]		0.00040 *4*	E_γ,I_γ: from (^{16}O,αpγ). B(M1)(W.u.)=0.0577 *41* if M1, B(E2)(W.u.)=43.8 *31* if E2.
		2911.9 *8*	100 *5*	5407.07	11/2$^-$	E2		0.000753 *11*	B(E2)(W.u.)=20.3 *12* E_γ,I_γ,Mult.: from (^{16}O,αpγ). Mult=Q, ΔJ=2 from $\gamma\gamma$(DCO); M2 ruled out by RUL.
8323.1	(3/2$^+$,5/2$^+$)	4210.4	1.2	4112.4	7/2$^+$				
		5320.0	3.7	3002.64	5/2$^+$				
		5676.9	4.9	2645.67	7/2$^+$				
		6559.3	<2.4	1763.15	5/2$^+$				
		7103.0	9.8	1219.38	1/2$^+$				
		8322.0	100	0.0	3/2$^+$				
8381.8	5/2$^+$	3611.7		4769.86	(5/2$^-$)				
		3757.3		4624.3	3/2,5/2				
		4269.1	23.5 *24*	4112.4	7/2$^+$				
		4463.0	71 *7*	3918.47	3/2$^+$	M1+E2	+0.15 *3*	1.19×10^{-3} *2*	B(M1)(W.u.)=2.9 *+13−7*; B(E2)(W.u.)=13 *+8−5*
		5378.7	74 *8*	3002.64	5/2$^+$	M1+E2	+0.02 *3*	1.44×10^{-3} *2*	B(M1)(W.u.)=1.8 *+14−7*
		5687.3	2.9 *15*	2694.00	3/2$^+$	M1+E2	−0.72 *31*	0.00157 *4*	B(M1)(W.u.)=0.039 *+32−21*; B(E2)(W.u.)=2.4 *+23−18*
		5735.6	15 *8*	2645.67	7/2$^+$				
		6618.0	100 *10*	1763.15	5/2$^+$	M1+E2	+0.06 *3*		B(M1)(W.u.)=1.3 *+6−3*; B(E2)(W.u.)=0.4 *+6−3*
		8380.7	9 *5*	0.0	3/2$^+$	D+Q	1.65		
8402.9	5/2$^-$	3632.8	16 *8*	4769.86	(5/2$^-$)				
		4224.7	29 *16*	4177.9	3/2$^-$				

$^{35}_{17}Cl_{18}$-47

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ†	I_γ‡	E_f	J^π_f
8402.9	$5/2^-$	4343.5	74 *20*	4059.1	$(3/2)^-$
		5708.4	32 *16*	2694.00	$3/2^+$
		5756.7	23 *13*	2645.67	$7/2^+$
		6639.1	48 *13*	1763.15	$5/2^+$
		8401.8	100 *26*	0.0	$3/2^+$
8405.8	$(5/2^-)$	3524.7	35 *9*	4880.96	$7/2^{(+)}$
		4232.1	3.7 *19*	4173.43	$(5/2^+)$
		4293.1	3.3 *16*	4112.4	$7/2^+$
		5402.7	35 *9*	3002.64	$5/2^+$
		6642.0	100 *26*	1763.15	$5/2^+$
		8404.7	56 *14*	0.0	$3/2^+$
8416.7	$1/2^+$	3407.4	14	5009.1	(1/2,3/2)
		4243.0	31	4173.43	$(5/2^+)$
		4473.5	8.6	3942.9	$9/2^+$
		5413.6	20	3002.64	$5/2^+$
		5770.5	23	2645.67	$7/2^+$
		6652.9	63	1763.15	$5/2^+$
		7196.5	26	1219.38	$1/2^+$
		8415.6	100	0.0	$3/2^+$
8464.3	$(1/2^+,3/2,5/2^+)$	2809	47 *24*	5655	$(3/2)^+$
		3249.6	34 *17*	5214.5	$(5/2^+)$
		3455.0	47 *24*	5009.1	(1/2,3/2)
		3609.7	19 *10*	4854.4	(1/2,3/2)
		3839.8	25 *13*	4624.3	3/2,5/2
		4286.1	31 *15*	4177.9	$3/2^-$
		4290.6	32 *17*	4173.43	$(5/2^+)$
		4496.4	31 *15*	3967.6	$1/2^+$
		4545.5	15 *8*	3918.47	$3/2^+$
		5461.2	100 *24*	3002.64	$5/2^+$
		5769.8	94 *24*	2694.00	$3/2^+$
		7244.1	53 *24*	1219.38	$1/2^+$
		8463.2	59 *18*	0.0	$3/2^+$
8484.4	$(3/2)^+$	2829	12 *6*	5655	$(3/2)^+$
		3603.2	15 *8*	4880.96	$7/2^{(+)}$
		3859.9	2.2 *11*	4624.3	3/2,5/2
		4516.5	2.6 *13*	3967.6	$1/2^+$
		4565.6	11 *6*	3918.47	$3/2^+$
		5481.3	15 *9*	3002.64	$5/2^+$
		5789.9	44 *11*	2694.00	$3/2^+$
		5838.2	6.1 *31*	2645.67	$7/2^+$
		6720.6	100 *26*	1763.15	$5/2^+$
		8483.3	9 *5*	0.0	$3/2^+$

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ [†]	I_γ [‡]	E_f	J^π_f	Mult. [#]	δ [#]	α [a]	Comments
8486.2	3/2⁻	2728	2.4 *13*	5758	(1/2,3/2⁻)				
		3861.7	3.9 *20*	4624.3	3/2,5/2				
		4308.0	<15	4177.9	3/2⁻				I_γ: total $I\gamma$=10 *5* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4312.5	<15	4173.43	(5/2⁺)				I_γ: total $I\gamma$=10 *5* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4426.8	4.1 *22*	4059.1	(3/2)⁻				
		4518.3	15 *9*	3967.6	1/2⁺				
		5483.1	17 *9*	3002.64	5/2⁺				
		5791.7	41 *11*	2694.00	3/2⁺				
		6722.4	8 *4*	1763.15	5/2⁺				
		7266.0	14 *7*	1219.38	1/2⁺				
		8485.1	100 *26*	0.0	3/2⁺				
8487.5	15/2⁻	2399.8 *8*	100.0 *34*	6087.31	13/2⁻	(M1(+E2))	+0.2 +*3*−*2*	0.000427 *17*	Mult.,δ: D+Q and δ from $\gamma\gamma$(DCO) in (^{28}Si,$\alpha p\gamma$); $\Delta\pi$=no from level scheme.
		3080.0 *7*	31.8 *20*	5407.07	11/2⁻	E2		0.000825 *12*	E_γ,I_γ,Mult.: from (^{16}O,$\alpha p\gamma$), with Mult=Q, ΔJ=2 from $\gamma\gamma$(ADO) and M2 ruled out by RUL.
8514.3	1/2⁻	3659.7	16 *8*	4854.4	(1/2,3/2)				
		4336.1	7 *4*	4177.9	3/2⁻				
		4454.9	100 *26*	4059.1	(3/2)⁻				
		4546.4	85 *22*	3967.6	1/2⁺				
		4595.5	24 *12*	3918.47	3/2⁺				
		7294.1	37 *11*	1219.38	1/2⁺				
		8513.2	100 *26*	0.0	3/2⁺				
8558.2?	(13/2⁻)	2631.4 [b]	100	5926.65	(11/2)⁻	D(+Q)	−0.2 +*2*−*3*		E_γ,Mult.,δ: from (^{28}Si,$\alpha p\gamma$); $\Delta\pi$=no from level scheme; Mult=Q, ΔJ=1.
8571.9	(5/2)⁺	2848.3	3.1 *16*	5723.52	5/2⁺				
		2917	1.9 *10*	5655	(3/2)⁺				
		2973.4	1.0 *6*	5598.4	3/2⁺,5/2⁺				
		3408.4	2.6 *13*	5163.34	7/2⁻				
		3690.7	1.3 *7*	4880.96	7/2⁽⁺⁾				
		4398.2	1.0 *6*	4173.43	(5/2⁺)				
		4459.2	1.6 *9*	4112.4	7/2⁺				
		4653.1	1.6 *9*	3918.47	3/2⁺				
		5408.6	2.3 *12*	3162.87	7/2⁻				
		5568.8	6.4 *33*	3002.64	5/2⁺				
		5877.4	0.9 *4*	2694.00	3/2⁺				
		5925.7	11 *6*	2645.67	7/2⁺				
		7351.7	8 *4*	1219.38	1/2⁺				
		8570.8	100 *10*	0.0	3/2⁺				
8580.5	1/2⁺	2088	8 *4*	6492	(3/2⁺)				

$^{35}_{17}Cl_{18}$-49

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f
8580.5	$1/2^+$	3741.2	2.4 *13*	4839.10	$(1/2^+,3/2)$
		4402.3	8 *4*	4177.9	$3/2^-$
		4521.1	3.7 *19*	4059.1	$(3/2)^-$
		4661.7	4.3 *22*	3918.47	$3/2^+$
		5886.0	10 *5*	2694.00	$3/2^+$
		7360.3	100 *11*	1219.38	$1/2^+$
		8579.4	48 *13*	0.0	$3/2^+$
8590.1	$5/2^+$	2832	22 *11*	5758	$(1/2,3/2^-)$
		2935	17 *9*	5655	$(3/2)^+$
		3820.0	73 *18*	4769.86	$(5/2^-)$
		4416.4	59 *14*	4173.43	$(5/2^+)$
		4477.4	16 *8*	4112.4	$7/2^+$
		5426.8	36 *18*	3162.87	$7/2^-$
		5587.0	100 *27*	3002.64	$5/2^+$
		5943.9	25 *13*	2645.67	$7/2^+$
		6826.2	64 *18*	1763.15	$5/2^+$
		8589.0	46 *14*	0.0	$3/2^+$
8612.0	$3/2^+,5/2^+$	6848.1	100 *10*	1763.15	$5/2^+$
		7391.8	34 *8*	1219.38	$1/2^+$
		8610.9	66 *16*	0.0	$3/2^+$
8613.7	$5/2^+$	3732.5	2.9 *14*	4880.96	$7/2^{(+)}$
		4554.3	10 *5*	4059.1	$(3/2)^-$
		4670.5	7.1	3942.9	$9/2^+$
		4694.9	13 *7*	3918.47	$3/2^+$
		5610.6	4.5 *23*	3002.64	$5/2^+$
		5919.2	8 *4*	2694.00	$3/2^+$
		5967.5	41 *11*	2645.67	$7/2^+$
		6849.8	16	1763.15	$5/2^+$
		7393.5	3.6	1219.38	$1/2^+$
		8612.6	100 *11*	0.0	$3/2^+$
8618.2	$3/2^+,5/2^+$	2894.6	6.4 *33*	5723.52	$5/2^+$
		2963	5.2 *27*	5655	$(3/2)^+$
		3763.6	4.2 *21*	4854.4	$(1/2,3/2)$
		3993.7	4.6 *24*	4624.3	$3/2,5/2$
		4650.3	9 *5*	3967.6	$1/2^+$
		4699.4	7 *4*	3918.47	$3/2^+$
		5615.1	49 *12*	3002.64	$5/2^+$
		5923.7	42 *12*	2694.00	$3/2^+$
		5972.0	5.2 *27*	2645.67	$7/2^+$
		6854.3	70 *18*	1763.15	$5/2^+$
		8617.1	100 *24*	0.0	$3/2^+$
8630.1	$7/2^-$	2906.5	0.6 *4*	5723.52	$5/2^+$

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^{π}_i	E_γ[†]	I_γ[‡]	E_f	J^{π}_f	Mult.[#]	δ[#]	α[a]	Comments
8630.1	7/2⁻	3466.6	17 *9*	5163.34	7/2⁻	M1+E2	+0.60 *10*	0.000884 *15*	Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); E1+M2 ruled out by RUL.
		3860.0	3.0 *15*	4769.86	(5/2⁻)				
		4282.1	26 *7*	4347.76	9/2⁻	M1+E2	−0.184 *14*	1.13×10^{-3} *2*	Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); E1+M2 ruled out by RUL.
		4456.4	8 *4*	4173.43	(5/2⁺)				
		4517.4	21 *11*	4112.4	7/2⁺	(E1+M2)	−0.06 *2*	1.86×10^{-3} *3*	Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme.
		4686.9	21 *11*	3942.9	9/2⁺				
		5466.8	7 *4*	3162.87	7/2⁻				
		5627.0	3.0 *15*	3002.64	5/2⁺				
		5983.9	5.5 *28*	2645.67	7/2⁺				
		6866.2	100 *26*	1763.15	5/2⁺	(E1+M2)	−0.011 *3*		Mult.,δ: D+Q and δ from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme.
		8629.0	1.1 *7*	0.0	3/2⁺				
8640	(5/2⁺)	4697	61	3942.9	9/2⁺				
		5477	53	3162.87	7/2⁻				
		5637	16	3002.64	5/2⁺				
		5946	34	2694.00	3/2⁺				
		6876	100	1763.15	5/2⁺				
8641.4	3/2⁺,5/2⁺	2986	10 *6*	5655	(3/2)⁺				
		3055.3	9 *5*	5586.0	5/2⁺				
		3237.9	17 *9*	5403.3	3/2⁻				
		3802.1	9 *5*	4839.10	(1/2⁺,3/2)				
		3871.3	86 *22*	4769.86	(5/2⁻)				
		4016.9	7 *4*	4624.3	3/2,5/2				
		4463.2	<64	4177.9	3/2⁻				I_γ: total Iγ=50 *14* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4467.7	<64	4173.43	(5/2⁺)				I_γ: total Iγ=50 *14* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4582.0	32 *18*	4059.1	(3/2)⁻				
		4722.6	21 *11*	3918.47	3/2⁺				
		5946.9	19 *9*	2694.00	3/2⁺				
		8640.3	100 *25*	0.0	3/2⁺				
8686.2	5/2⁻	3100.1	2.8 *15*	5586.0	5/2⁺				
		3282.7	4.1 *21*	5403.3	3/2⁻				
		4508.0	2.6 *13*	4177.9	3/2⁻				
		4626.8	4.6 *24*	4059.1	(3/2)⁻				
		5522.9	4.9 *25*	3162.87	7/2⁻				
		5991.7	3.7 *19*	2694.00	3/2⁺				
		6922.3	6.3 *32*	1763.15	5/2⁺				
		7466.0	2.5 *13*	1219.38	1/2⁺				

$^{35}_{17}Cl_{18}$-51

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	α[a]	Comments
8686.2	5/2−	8685.0	100 *11*	0.0	3/2+			
8696.9	3/2−	2890.2	3.6 *19*	5806.6	(1/2,3/2,5/2)			
		3098.4	6.0 *30*	5598.4	3/2+,5/2+			
		7476.7	15 *4*	1219.38	1/2+			
		8695.7	100 *10*	0.0	3/2+			
8750.9	3/2−	2944.2	4.2 *21*	5806.6	(1/2,3/2,5/2)			
		3347.4	9 *5*	5403.3	3/2−			
		3536.2	4.5 *23*	5214.5	(5/2+)			
		4572.7	4.0 *21*	4177.9	3/2−			
		4577.2	4.3 *23*	4173.43	(5/2+)			
		6056.3	5.7 *28*	2694.00	3/2+			
		6987.0	57 *15*	1763.15	5/2+			
		7530.7	3.8	1219.38	1/2+			
		8749.7	100 *10*	0.0	3/2+			
8779.8	3/2−	3181.2	16 *8*	5598.4	3/2+,5/2+			
		3565.1	24 *12*	5214.5	(5/2+)			
		4601.6	<71	4177.9	3/2−			I_γ: total Iγ=56 *15* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4606.0	<71	4173.43	(5/2+)			I_γ: total Iγ=56 *15* for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4861.0	16 *8*	3918.47	3/2+			
		5776.7	100 *26*	3002.64	5/2+			
		6085.2	44 *11*	2694.00	3/2+			
		6133.6	48 *11*	2645.67	7/2+			
		7015.9	25 *13*	1763.15	5/2+			
		8778.6	41 *11*	0.0	3/2+			
8787.2	(5/2+)	3063.5	27 *14*	5723.52	5/2+			
		3623.7	76 *20*	5163.34	7/2−			
		4017.1	27 *14*	4769.86	(5/2−)			
		4609.0	36 *20*	4177.9	3/2−			
		5623.8	23 *12*	3162.87	7/2−			
		5784.1	80 *20*	3002.64	5/2+			
		7023.3	100 *24*	1763.15	5/2+			
		8786.0	32 *16*	0.0	3/2+			
8788.3	(15/2+)	915.4 *4*	100	7873.03	13/2+	(M1)	4.82×10^{-5} *7*	E_γ,Mult.: from (^{16}O,αpγ), with Mult=D, ΔJ=1 from $\gamma\gamma$(ADO); $\Delta\pi$=(no) from level scheme.
8798.4	(3/2+,5/2+)	4739.0	13 *7*	4059.1	(3/2)−			
		5795.2	26 *7*	3002.64	5/2+			
		7034.5	11 *6*	1763.15	5/2+			
		7578.1	11 *6*	1219.38	1/2+			
		8797.2	100 *10*	0.0	3/2+			
8820.9	(1/2+ to 7/2+)	5817.7	17 *10*	3002.64	5/2+			

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	α[a]	Comments
8820.9	($1/2^+$ to $7/2^+$)	6126.3	35 *10*	2694.00	$3/2^+$			
		7057.0	40 *10*	1763.15	$5/2^+$			
		8819.7	100 *10*	0.0	$3/2^+$			
8824.2	$1/2^+$	5821.0	100 *24*	3002.64	$5/2^+$			
		6129.6	16.2 *8*	2694.00	$3/2^+$			
		7060.3	43 *11*	1763.15	$5/2^+$			
		7603.9	60 *16*	1219.38	$1/2^+$			
		8823.0	51 *14*	0.0	$3/2^+$			
8829.3	$1/2^-$	3174	30 *15*	5655	$(3/2)^+$			
		3990.0	3.9 *21*	4839.10	($1/2^+$,3/2)			
		4059.2	4.6 *24*	4769.86	($5/2^-$)			
		4910.5	14 *7*	3918.47	$3/2^+$			
		5826.1	67 *18*	3002.64	$5/2^+$			
		6134.7	39 *9*	2694.00	$3/2^+$			
		6183.0	100 *24*	2645.67	$7/2^+$			
		7065.4	27 *15*	1763.15	$5/2^+$			
		7609.0	8 *4*	1219.38	$1/2^+$			
		8828.1	12 *6*	0.0	$3/2^+$			
8833.1	(5/2,7/2)	3618.4	30 *15*	5214.5	($5/2^+$)			
		4063.0	11 *6*	4769.86	($5/2^-$)			
		4659.3	61 *18*	4173.43	($5/2^+$)			
		4720.4	20 *11*	4112.4	$7/2^+$			
		4773.7	48 *13*	4059.1	$(3/2)^-$			
		4914.3	48 *13*	3918.47	$3/2^+$			
		5669.7	16 *8*	3162.87	$7/2^-$			
		5829.9	29 *15*	3002.64	$5/2^+$			
		6138.5	20 *10*	2694.00	$3/2^+$			
		6186.8	100 *26*	2645.67	$7/2^+$			
		7069.2	39 *22*	1763.15	$5/2^+$			
		8831.9	13 *7*	0.0	$3/2^+$			
8838.1	$5/2^-$,$7/2^-$	3674.6		5163.34	$7/2^-$			
		3956.9		4880.96	$7/2^{(+)}$			
		4068.0	13 *7*	4769.86	($5/2^-$)			
		4490.0		4347.76	$9/2^-$			
		4778.7		4059.1	$(3/2)^-$			
		5674.7	46 *11*	3162.87	$7/2^-$			
		7074.2	100 *10*	1763.15	$5/2^+$			
8844.4	$17/2^+$	524.8		8319.6	$15/2^-$			
		971.38 *20*	100.0 *30*	7873.03	$13/2^+$	E2	6.16×10^{-5} *9*	B(E2)(W.u.)=16.3 *+37−26* E_γ: weighted average of 971.4 *5* from (^{16}O,$\alpha p\gamma$) and 971.38 *20* from (^{14}N,$\alpha pn\gamma$).

$^{35}_{17}Cl_{18}$-53

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^{π}_i	E_{γ}†	I_{γ}‡	E_f	J^{π}_f	Comments
8844.4	17/2$^+$	2757.0		6087.31	13/2$^-$	
8856.2	5/2$^+$	3975.0	6.7 *33*	4880.96	7/2$^{(+)}$	
		4231.6	10 *5*	4624.3	3/2,5/2	
		4678.0	7 *4*	4177.9	3/2$^-$	
		4937.4	22 *6*	3918.47	3/2$^+$	
		5692.8	100 *10*	3162.87	7/2$^-$	
		6161.6	5.5 *28*	2694.00	3/2$^+$	
		6209.9		2645.67	7/2$^+$	
		7092.3	14 *8*	1763.15	5/2$^+$	
		8855.0	31 *8*	0.0	3/2$^+$	
8868.6	3/2$^+$,5/2$^+$	4690.4	5.3 *27*	4177.9	3/2$^-$	
		5865.4	33 *8*	3002.64	5/2$^+$	
		6174.0	25 *6*	2694.00	3/2$^+$	
		7104.7	9 *5*	1763.15	5/2$^+$	
		7648.3	33 *8*	1219.38	1/2$^+$	
		8867.4	100 *25*	0.0	3/2$^+$	
8886.1	(5/2)	3231	23 *12*	5655	(3/2)$^+$	
		3287.5	25 *13*	5598.4	3/2$^+$,5/2$^+$	
		3671.4	9 *4*	5214.5	(5/2$^+$)	
		3722.6	35 *16*	5163.34	7/2$^-$	
		4116.0	17 *9*	4769.86	(5/2$^-$)	
		4707.9	31 *16*	4177.9	3/2$^-$	
		4967.3	16 *8*	3918.47	3/2$^+$	
		5882.9	31 *16*	3002.64	5/2$^+$	
		6239.8	100 *27*	2645.67	7/2$^+$	
		7122.2	46 *12*	1763.15	5/2$^+$	
		8884.9	54 *16*	0.0	3/2$^+$	
8893.3	(5/2$^+$,7/2$^+$)	3307.1	4.3 *22*	5586.0	5/2$^+$	
		4780.6	24 *11*	4112.4	7/2$^+$	
		4950.0	11 *6*	3942.9	9/2$^+$	
		5729.9	78 *19*	3162.87	7/2$^-$	
		6198.7	100 *24*	2694.00	3/2$^+$	
		7129.4	51 *14*	1763.15	5/2$^+$	
		8892.1	2.7 *14*	0.0	3/2$^+$	
8904.8	(1/2,3/2,5/2$^+$)	2724	15 *8*	6181	(1/2$^+$ to 7/2$^+$)	
		3250	12 *6*	5655	(3/2)$^+$	
		3895.5	13 *7*	5009.1	(1/2,3/2)	
		4050.2	21 *11*	4854.4	(1/2,3/2)	
		4065.5	16 *8*	4839.10	(1/2$^+$,3/2)	
		4726.6	62 *17*	4177.9	3/2$^-$	
		4845.3	20 *10*	4059.1	(3/2)$^-$	
		6210.2	52 *14*	2694.00	3/2$^+$	

$^{35}_{17}Cl_{18}$-54

Adopted Levels, Gammas (continued)

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ †	I_γ ‡	E_f	J^π_f	Mult.#	δ#	Comments
8904.8	(1/2,3/2,5/2$^+$)	7684.5	35 *10*	1219.38	1/2$^+$			
		8903.6	100 *24*	0.0	3/2$^+$			
8906.8	5/2$^+$	3252		5655	(3/2)$^+$			
		3261	5.4 *28*	5646	(5/2,7/2,9/2$^+$)			
		3308.2	3.2 *16*	5598.4	3/2$^+$,5/2$^+$			
		3320.6	5.8	5586.0	5/2$^+$			
		3376	3.6 *19*	5531				
		3692.1	6.4 *32*	5214.5	(5/2$^+$)			
		4136.7	8 *4*	4769.86	(5/2$^-$)			
		4963.5	19 *5*	3942.9	9/2$^+$			
		7142.9	100 *10*	1763.15	5/2$^+$			
8919.8	(5/2$^-$,7/2,9/2$^+$)	3274	10 *5*	5646	(5/2,7/2,9/2$^+$)			
		4571.7	14 *7*	4347.76	9/2$^-$			
		5756.4	100 *11*	3162.87	7/2$^-$			
		6273.5	46 *12*	2645.67	7/2$^+$			
		7155.9	7 *4*	1763.15	5/2$^+$			
8953.1	3/2$^+$	7732.8		1219.38	1/2$^+$	M1+E2	−0.34 *2*	Mult.,δ: or δ=−5.0 *4* from $\gamma(\theta)$ in (p,γ). E1+M2 ruled out by RUL.
		8951.9		0.0	3/2$^+$	M1+E2	−0.39 *5*	Mult.,δ: or δ=−8.2 *7* from $\gamma(\theta)$ in (p,γ). E1+M2 ruled out by RUL.
8984.2	5/2$^+$	3260.5	1.3 *7*	5723.52	5/2$^+$			
		4144.8	2.0 *10*	4839.10	(1/2$^+$,3/2)			
		4214.1	0.8 *5*	4769.86	(5/2$^-$)			
		4359.6	0.8 *5*	4624.3	3/2,5/2			
		4806.0	2.5 *13*	4177.9	3/2$^-$			
		4810.4	2.5 *13*	4173.43	(5/2$^+$)			
		5065.3	15 *8*	3918.47	3/2$^+$			
		5820.8	10 *5*	3162.87	7/2$^-$			
		6289.6	3.0 *15*	2694.00	3/2$^+$			
		6337.9	2.0 *10*	2645.67	7/2$^+$			
		7220.3	27 *7*	1763.15	5/2$^+$			
		8983.0	100 *10*	0.0	3/2$^+$			
9081.5	5/2$^+$	3357.8	1.7	5723.52	5/2$^+$			
		4200.3	1.7	4880.96	7/2$^{(+)}$			
		4242.1		4839.10	(1/2$^+$,3/2)			
		4311.4		4769.86	(5/2$^-$)			
		4456.9		4624.3	3/2,5/2			
		4903.2	3.3	4177.9	3/2$^-$			
		4907.7	3.3	4173.43	(5/2$^+$)			
		5162.6	15	3918.47	3/2$^+$	D(+Q)	−0.002 *9*	
		5918.1	10	3162.87	7/2$^-$			
		6386.9	3.3	2694.00	3/2$^+$			
		6435.2	1.7	2645.67	7/2$^+$			

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J^π_i	E_γ[†]	I_γ[‡]	E_f	J^π_f	Mult.[#]	δ[#]	α[a]	Comments
9081.5	5/2⁺	7317.5	27	1763.15	5/2⁺	M1+E2	+0.11 *2*		B(M1)(W.u.)=1.16 *+31−25*; B(E2)(W.u.)=0.99 *+50−38*
		9080.2	100	0.0	3/2⁺	M1+E2	+0.089 *9*		B(M1)(W.u.)=2.26 *+49−33*; B(E2)(W.u.)=0.82 *+27−19*
9157.1	5/2⁻	3942.4	17 *5*	5214.5	(5/2⁺)				
		4386.9	16 *4*	4769.86	(5/2⁻)				
		4978.8	<37	4177.9	3/2⁻				I_γ: total $I\gamma$=30 *7* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4983.3	<37	4173.43	(5/2⁺)				I_γ: total $I\gamma$=30 *7* for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		5993.7	22 *6*	3162.87	7/2⁻	M1+E2	+0.10 *2*	1.59×10^{-3} *2*	
		6153.9	20 *5*	3002.64	5/2⁺				
		6462.5	3 *2*	2694.00	3/2⁺				
		6510.8	7 *4*	2645.67	7/2⁺				
		7393.1	17 *5*	1763.15	5/2⁺				
		9155.8	100 *9*	0.0	3/2⁺	(E1)			Mult.: D+Q with δ=+0.09 *1* from $\gamma(\theta)$ in (p,γ); $\Delta\pi$=yes from level scheme; M2 component is less likely.
10181.1	19/2⁻	1336.3 *5*	100.0 *34*	8844.4	17/2⁺	(E1)		0.0001637 *23*	B(E1)(W.u.)=0.00235 *+21−18* I_γ: weighted average of 100 *17* from (^{28}Si,αpγ) and 100.0 *34* from (^{16}O,αpγ). E_γ,Mult.: from (^{16}O,αpγ), with Mult=D from $\gamma\gamma$(ADO) with ΔJ=1; $\Delta\pi$=(yes) from level scheme.
		1693.5 *5*	84 *9*	8487.5	15/2⁻	E2		0.0001880 *26*	B(E2)(W.u.)=44.4 *46* E_γ,I_γ,Mult.: from (^{16}O,αpγ), with Mult=Q, ΔJ=2 from $\gamma\gamma$(ADO) and M2 ruled out by RUL.
		1861.3 *5*	83 *6*	8319.6	15/2⁻	E2		0.000262 *4*	B(E2)(W.u.)=27.4 *+26−24* E_γ,I_γ,Mult.: from (^{16}O,αpγ), with Mult=Q, ΔJ=2 from $\gamma\gamma$(ADO) and M2 ruled out by RUL.
10221.9	(17/2⁻)	1377.8 *10*	100	8844.4	17/2⁺				
10858.5	19/2⁺	2014.7 *9*	100 *5*	8844.4	17/2⁺	(M1)		0.000265 *4*	E_γ,I_γ,Mult.: from (^{16}O,αpγ), with Mult=D from $\gamma\gamma$(ADO) with ΔJ=1; $\Delta\pi$=no from level scheme.
		2069.8 *10*	32.5 *22*	8788.3	(15/2⁺)	E2		0.000361 *5*	E_γ,I_γ,Mult.: from (^{16}O,αpγ) with Mult=Q, ΔJ=2 from $\gamma\gamma$(ADO); M2 ruled out by RUL.
11459.1	21/2⁺	2614.5 *5*	100	8844.4	17/2⁺	E2		0.000619 *9*	B(E2)(W.u.)=15.9 *8* E_γ,Mult.: from (^{16}O,αpγ), with Mult=Q, ΔJ=2 from $\gamma\gamma$(ADO); M2 ruled out by RUL.
12572.2	23/2⁻	1113.3 *5*	24.4 *18*	11459.1	21/2⁺	(E1)		3.60×10^{-5} *5*	B(E1)(W.u.)=0.00272 *+41−35* E_γ,I_γ,Mult.: from (^{16}O,αpγ), with Mult=D from $\gamma\gamma$(ADO) with ΔJ=1; $\Delta\pi$=yes from level scheme.

$\gamma(^{35}Cl)$ (continued)

E_i(level)	J_i^π	E_γ[†]	I_γ[‡]	E_f	J_f^π	Mult.[#]	α[a]	Comments
12572.2	$23/2^-$	2390.8 *5*	100 *4*	10181.1	$19/2^-$	E2	0.000515 *7*	B(E2)(W.u.)=26.0 *+36−29* E_γ,I_γ,Mult.: from (^{16}O,αpγ), with Mult=Q, ΔJ=2 from $\gamma\gamma$(ADO); M2 ruled out by RUL.
16306.4	$(27/2^-)$	3734.0		12572.2	$23/2^-$			

[†] Sources of values with uncertainties are listed under comments and those without uncertainties are deduced from level-energy differences, unless otherwise noted.

[‡] From (p,γ), unless otherwise noted.

[#] From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ in (p,γ) for γ rays originating from low-spin levels (J<9/2) and from $\gamma\gamma(\theta)$(DCO) and γ(lin pol) in (^{28}Si,αpγ) for γ rays originating from high-spin (J≥9/2) levels, unless otherwise noted. Magnetic/electric characters are determined based on γ(lin pol) or RUL with measured $T_{1/2}$, unless otherwise noted. For primary γ transitions from proton resonances where widths or resonance strengths are provided in (p,γ), a D+Q with definite δ(Q/D)>0 from $\gamma(\theta)$ is firmly assigned as M1+E2. This assignment is due to the fact that for these resonances, the proton width Γ_p is dominant over Γ_γ and thus the total width Γ is much greater than Γ_γ. The total width Γ typically falls within the same scale as the resonance strength, usually a few eV in the ^{34}S(p,γ) dataset.

[@] Primary transitions probably from the unresolved 7501+7503 doublet initial levels in 1976Me12 in (p,γ).

[&] Primary transitions probably from the unresolved 7837+7839 doublet initial levels in 1976Me12 in (p,γ).

[a] Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

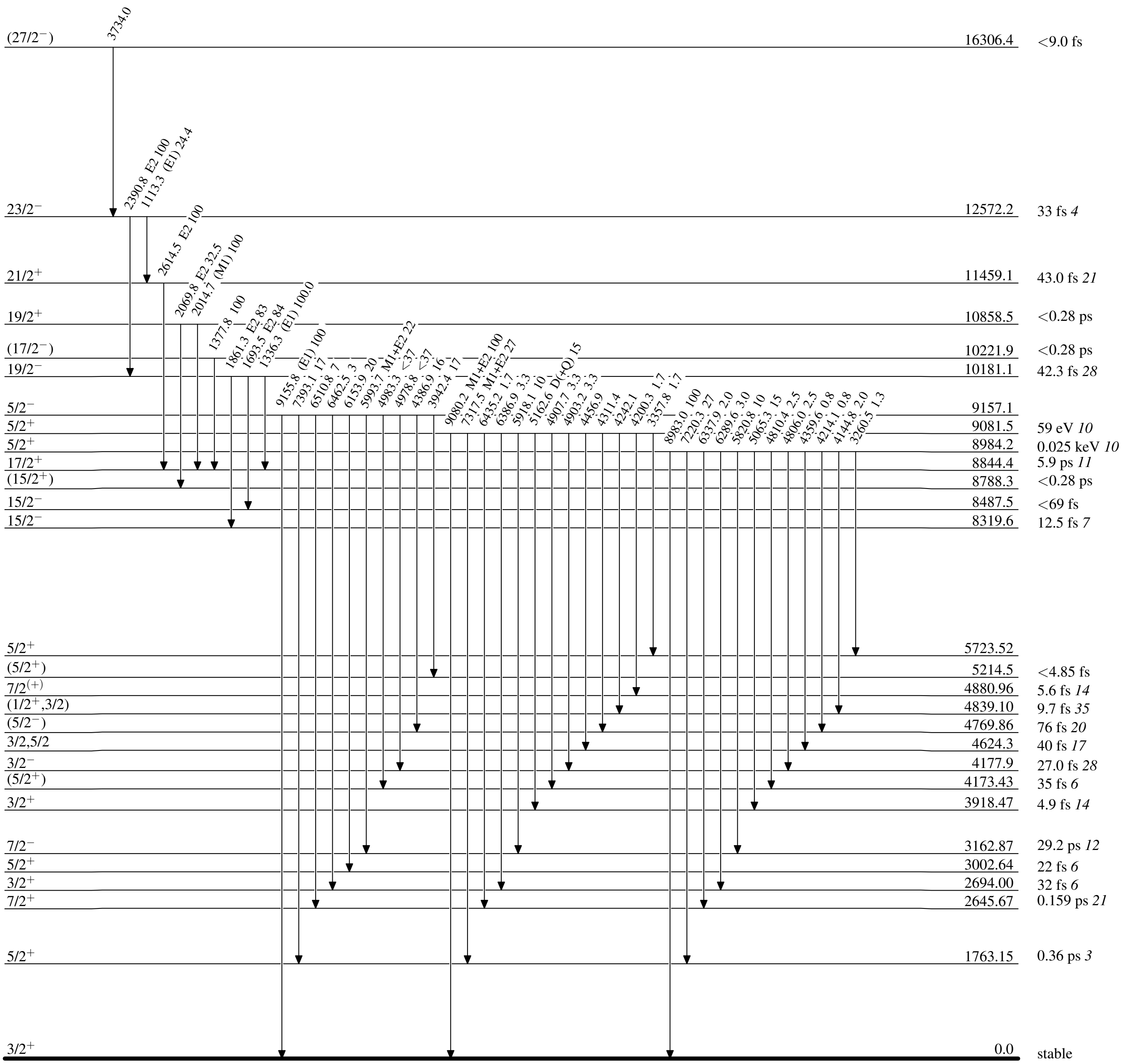
[b] Placement of transition in the level scheme is uncertain.

$^{35}_{17}Cl_{18}$-57

Adopted Levels, Gammas

Level Scheme

Intensities: Relative photon branching from each level

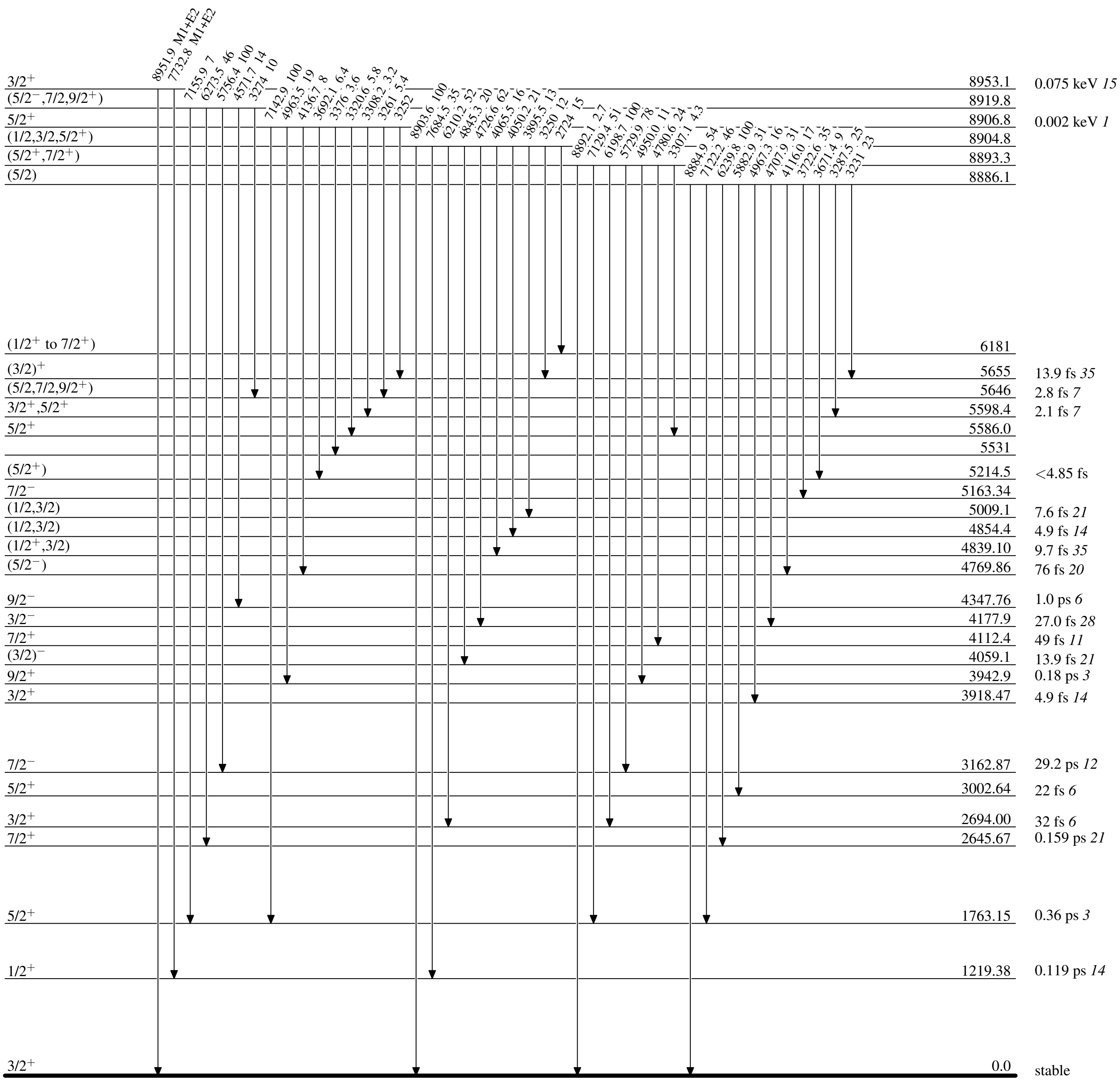


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

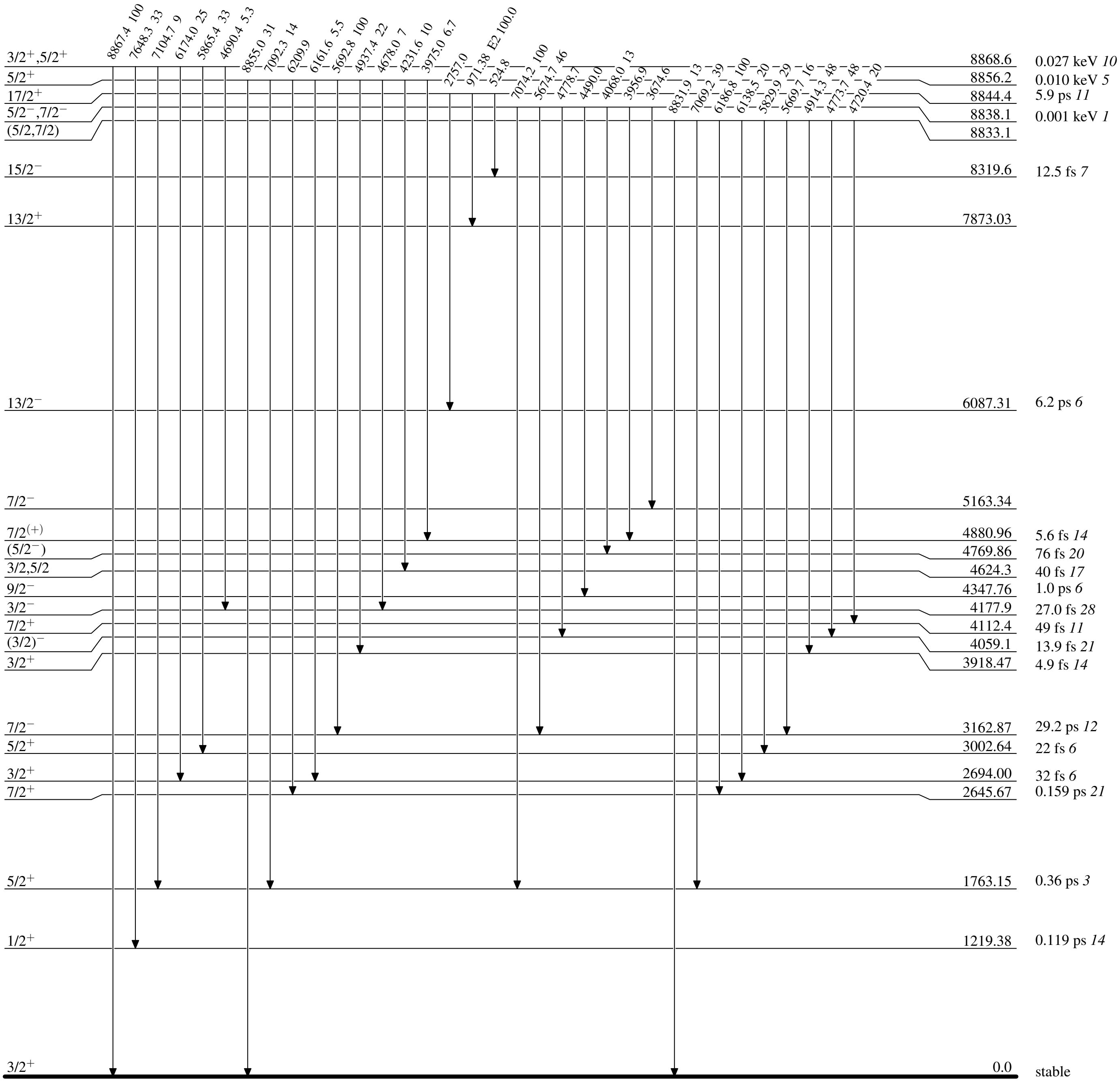


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

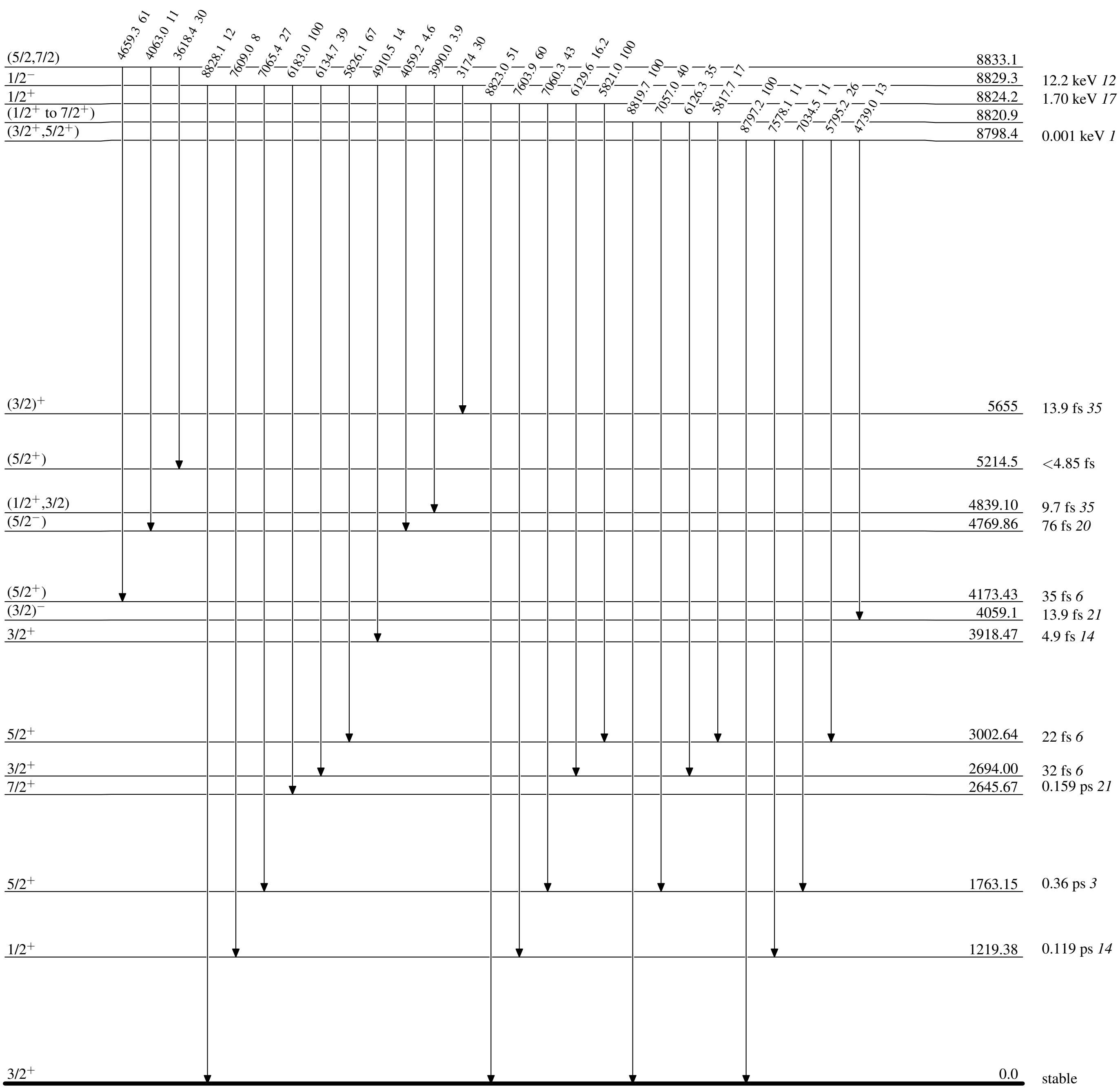


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

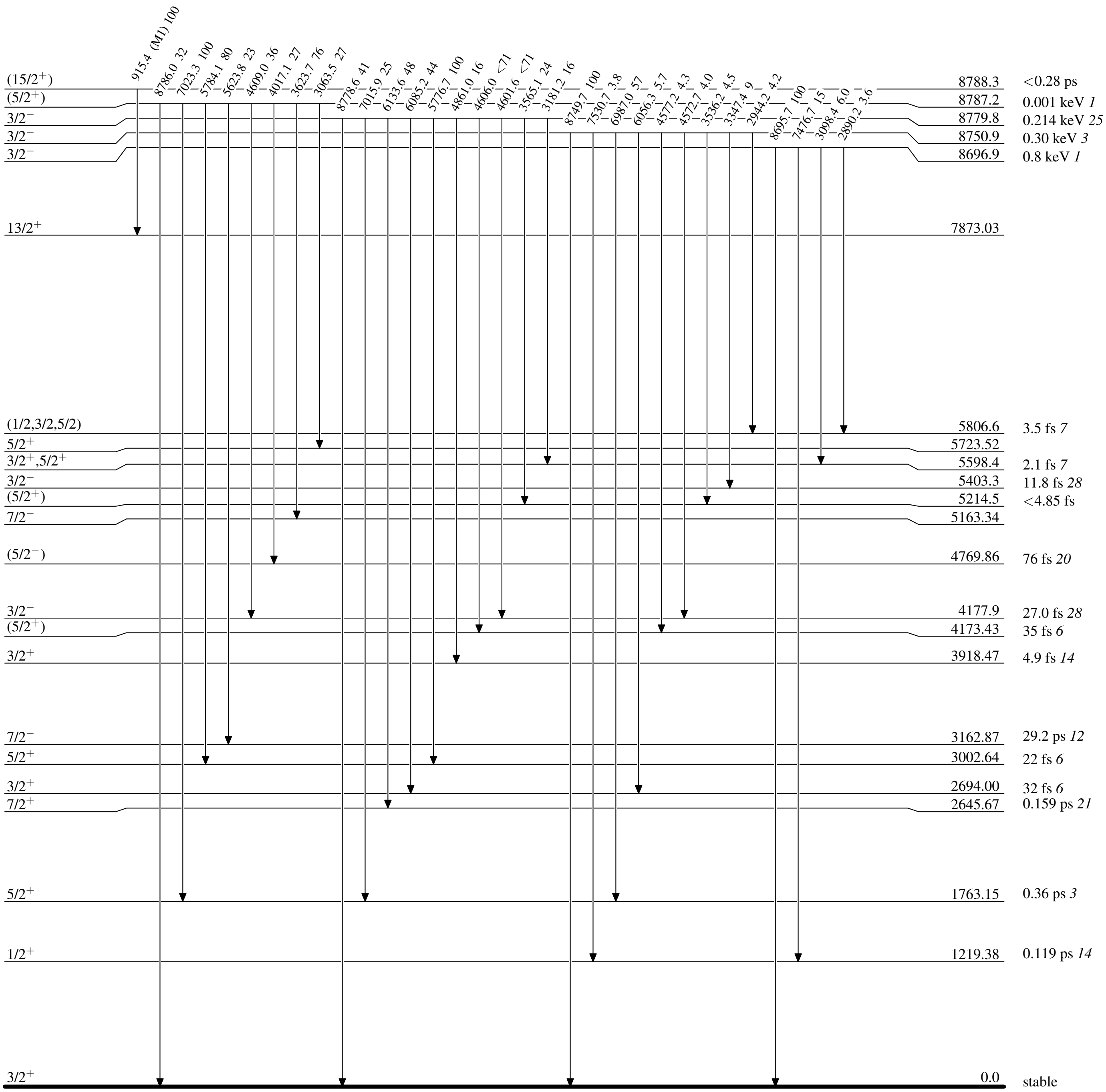


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

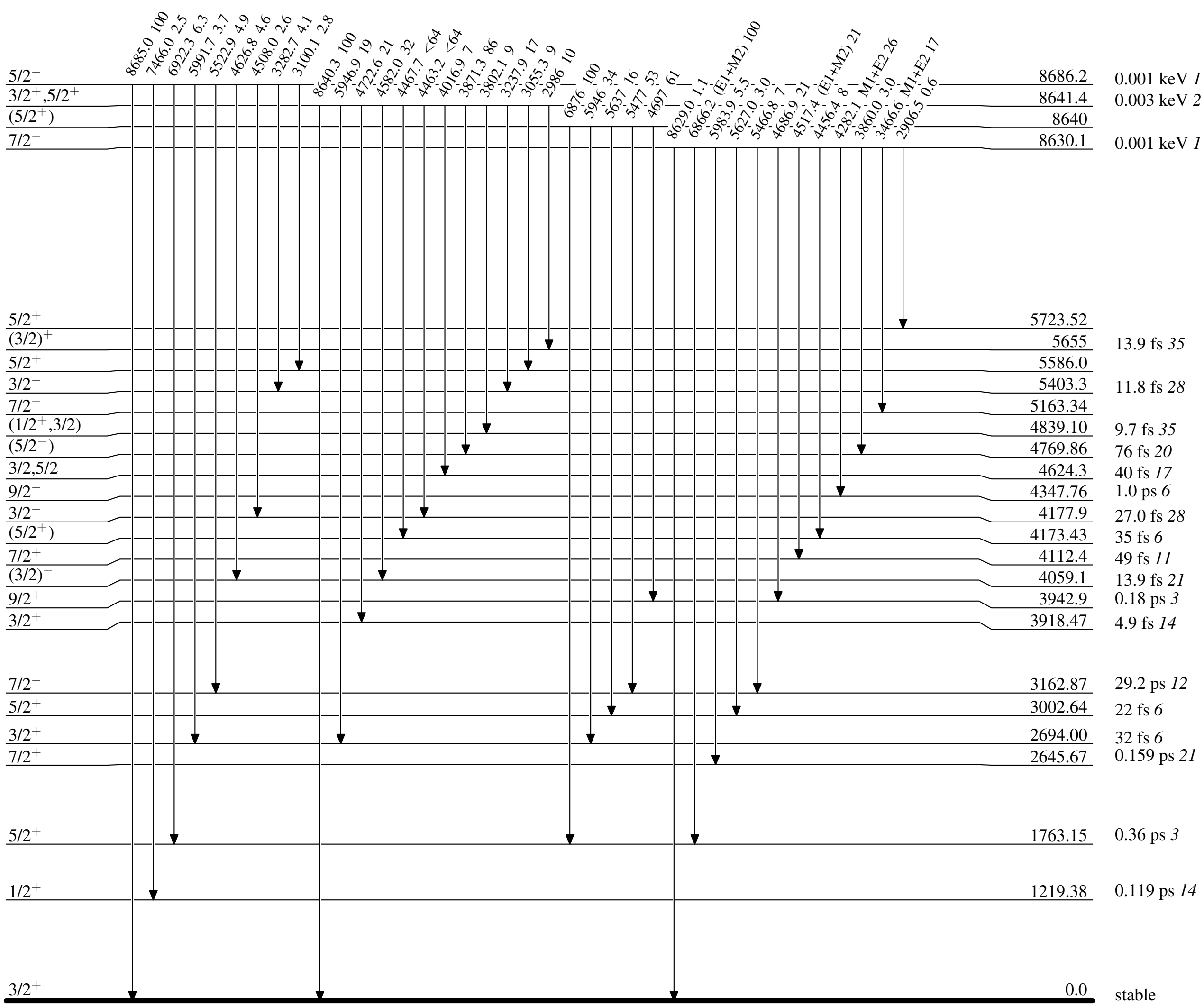


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

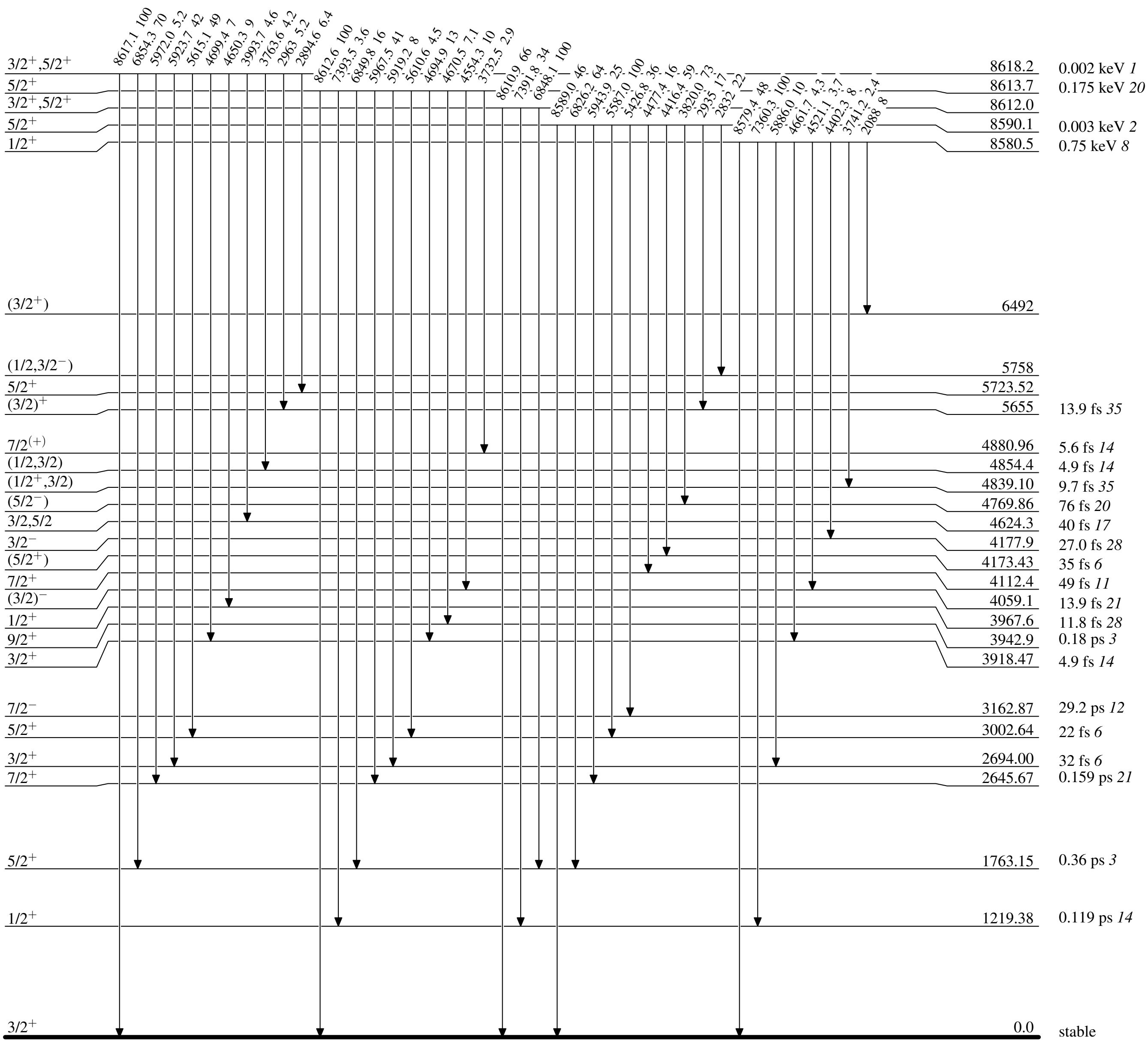


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



$^{35}_{17}Cl_{18}$

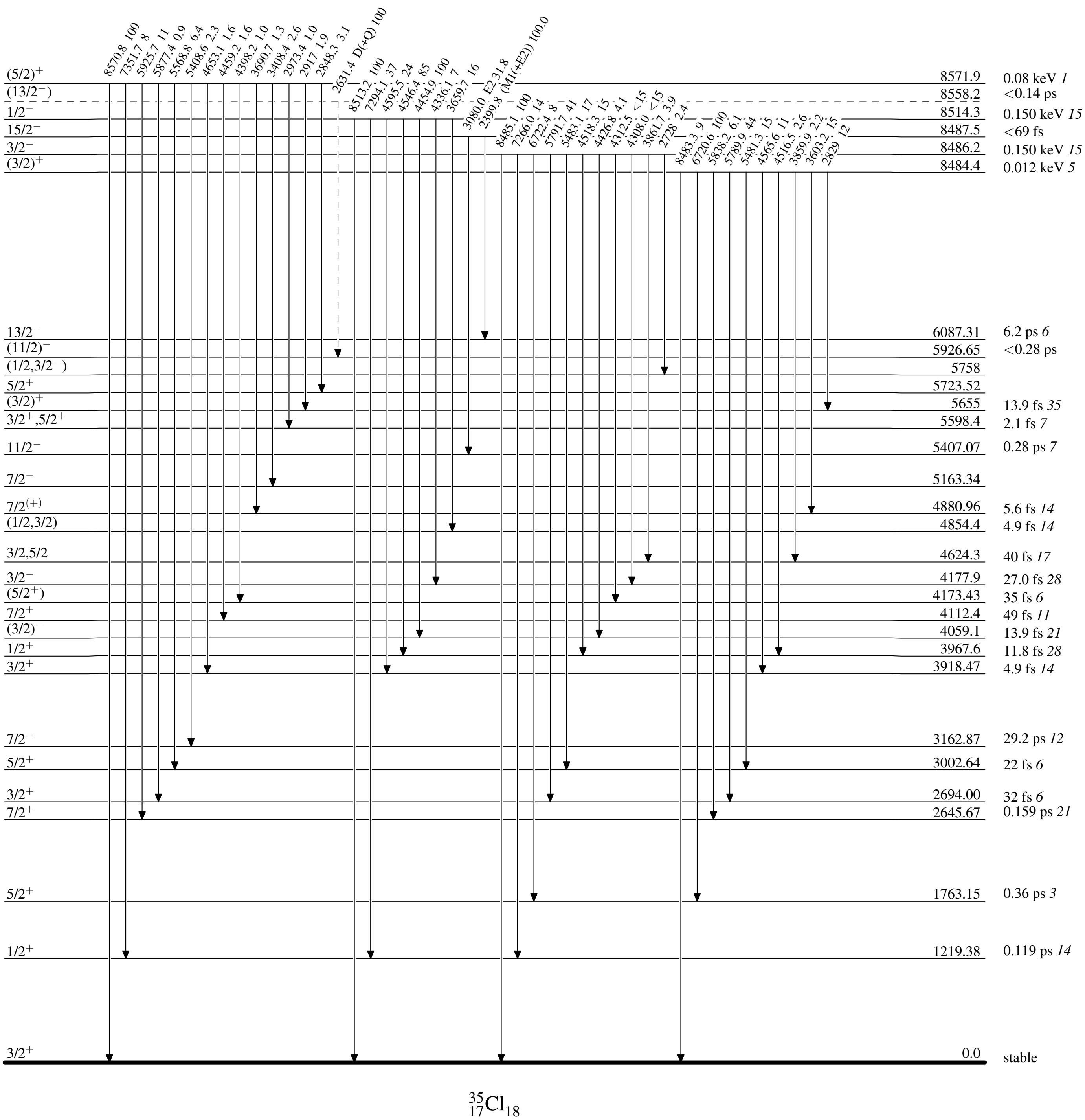
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

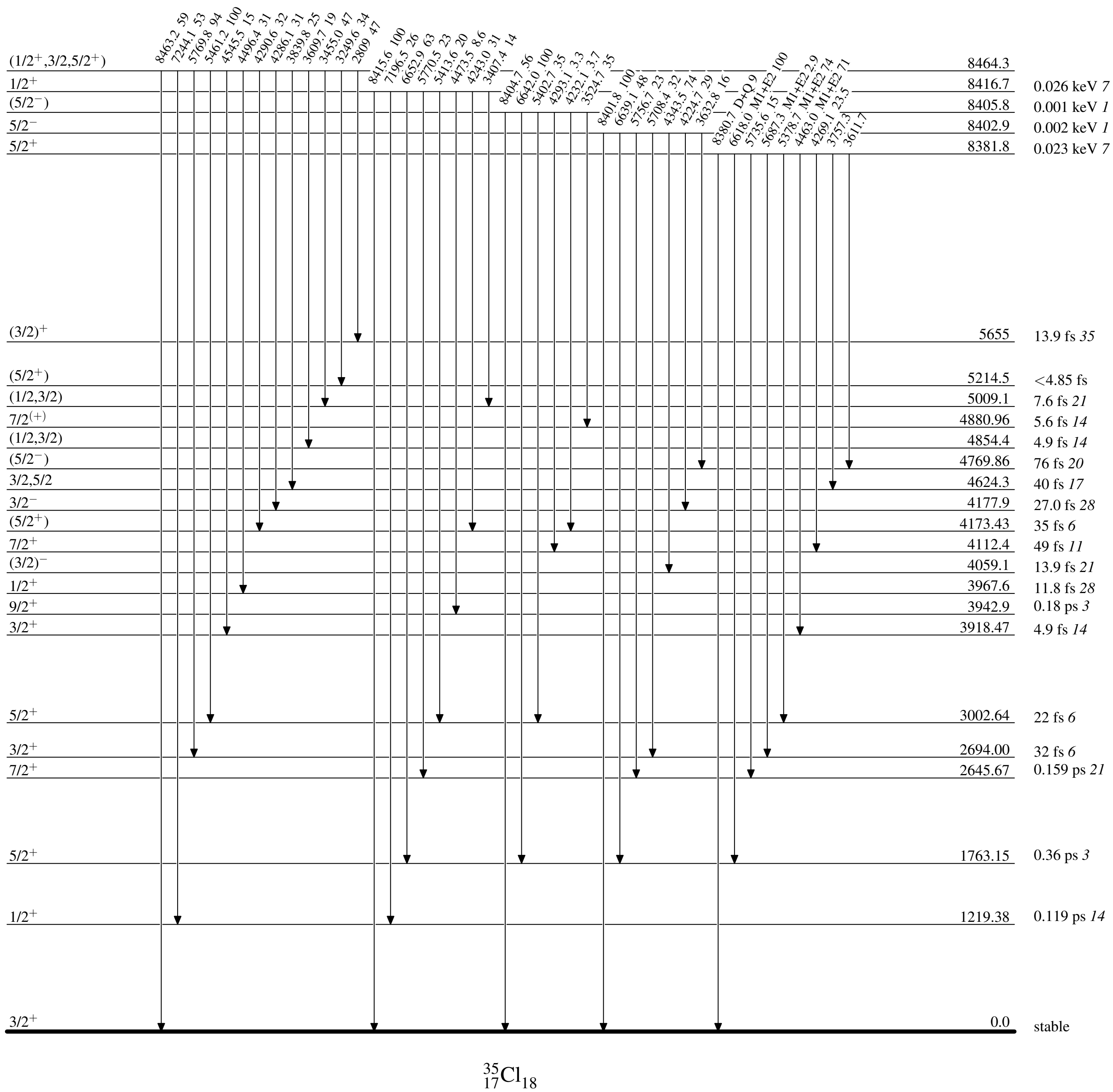
γ Decay (Uncertain)



Adopted Levels, Gammas

Level Scheme (continued)

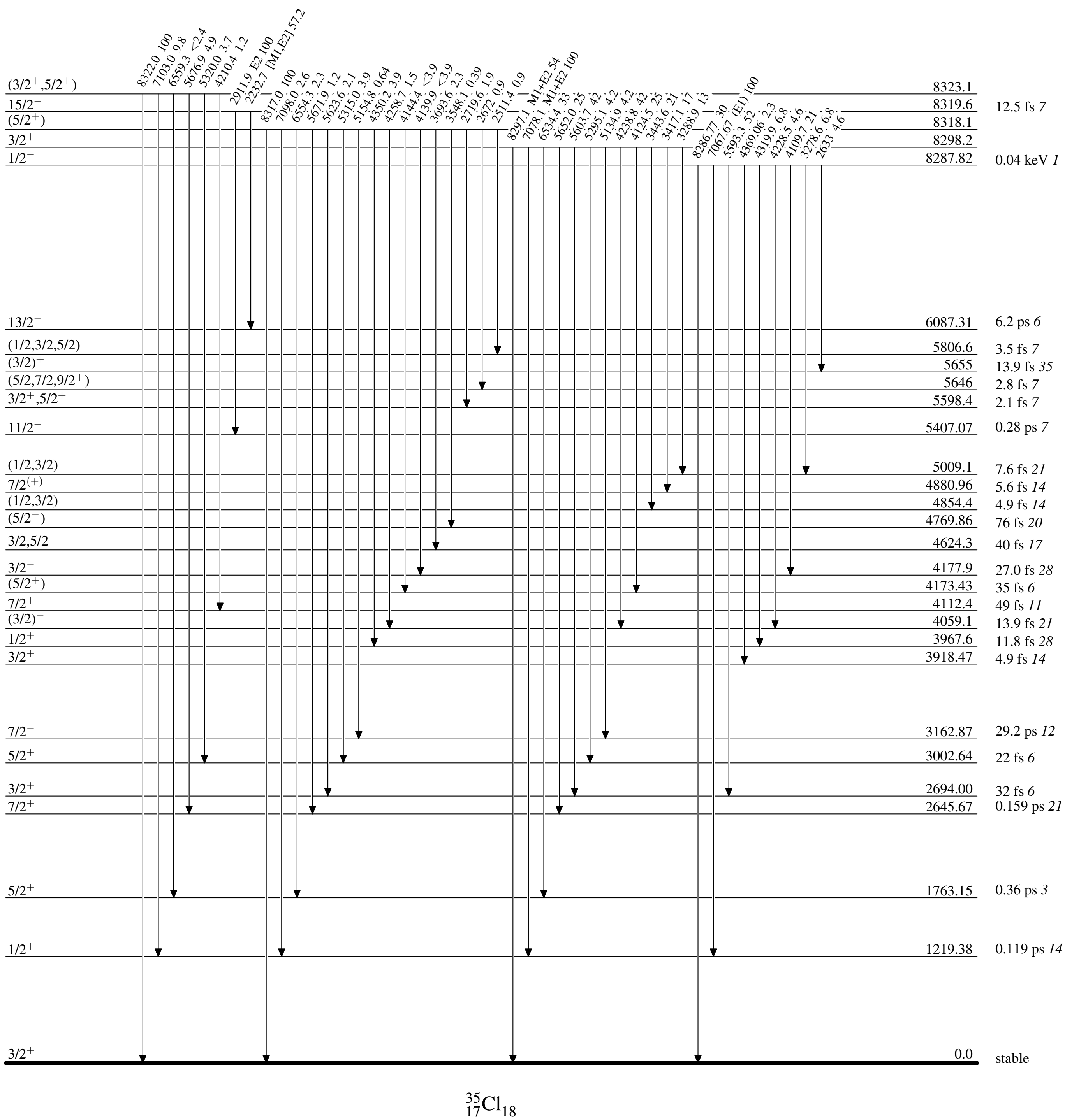
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

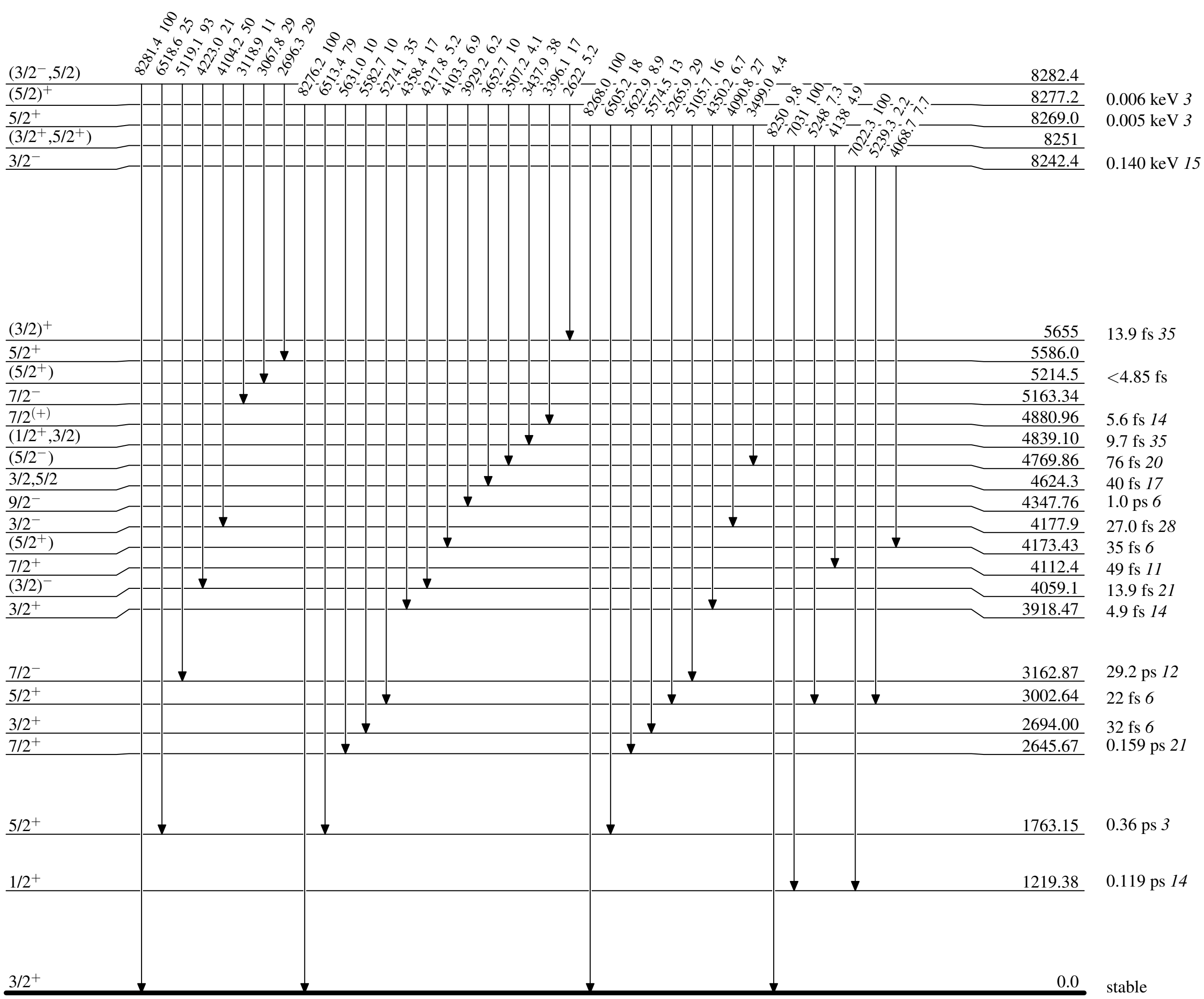
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

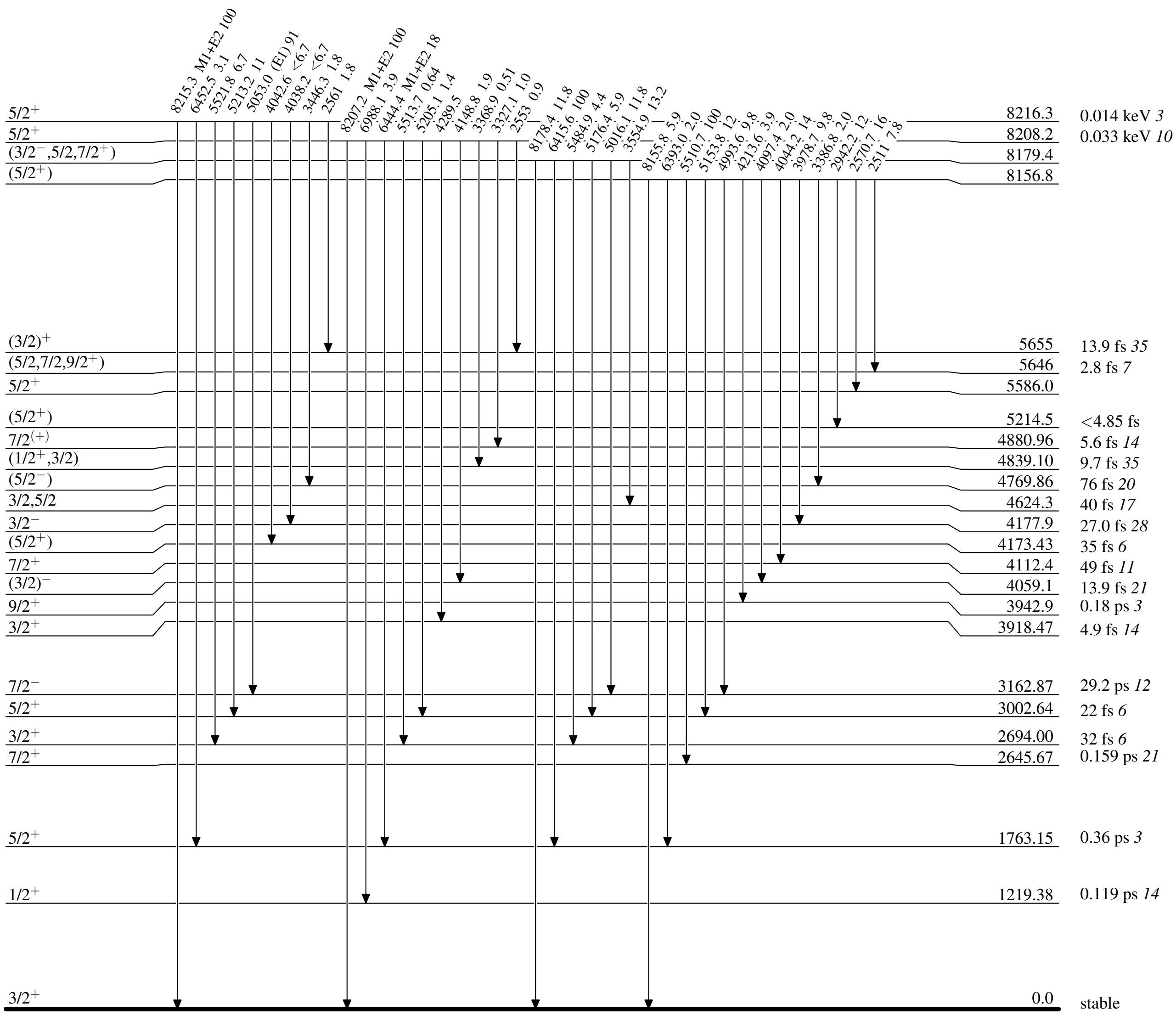
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

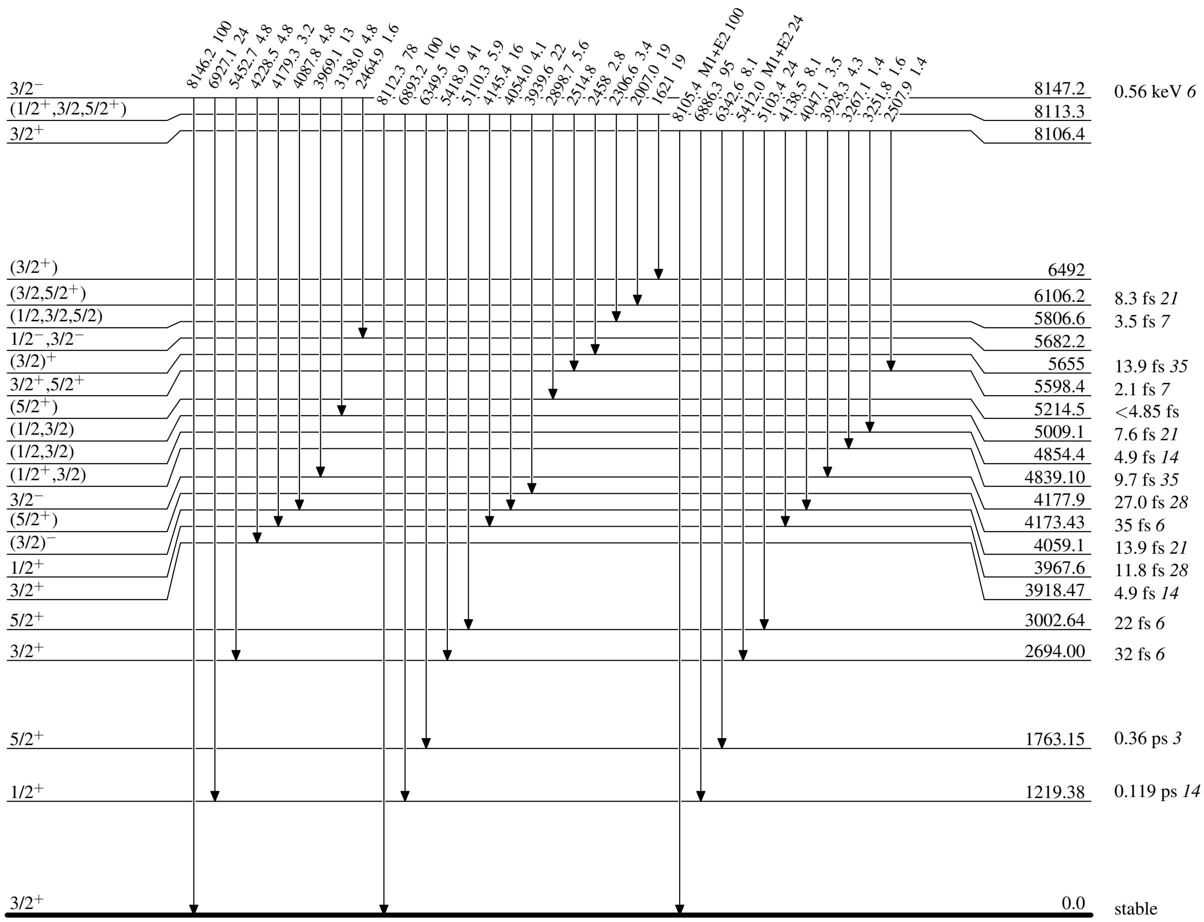


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

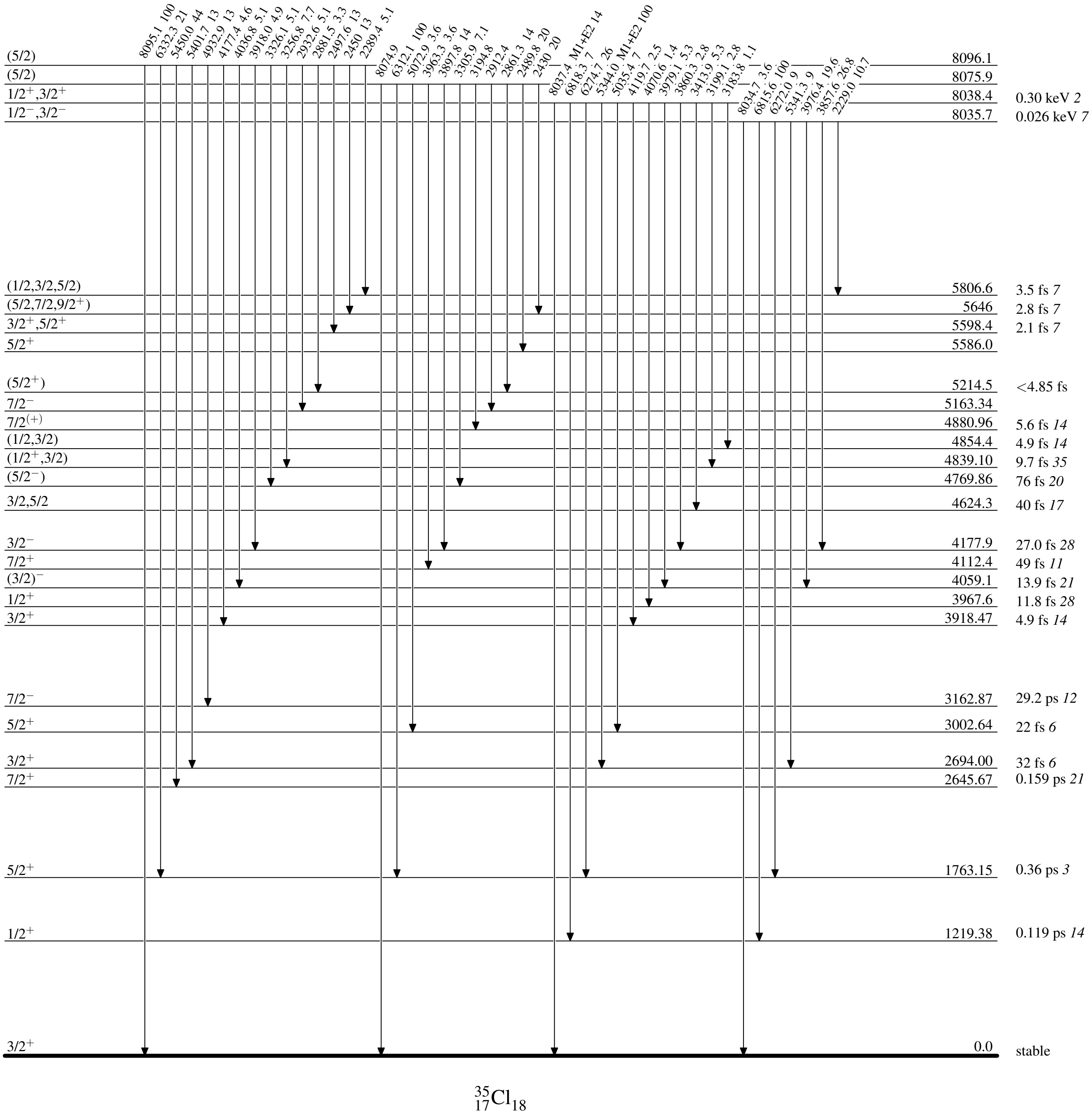


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

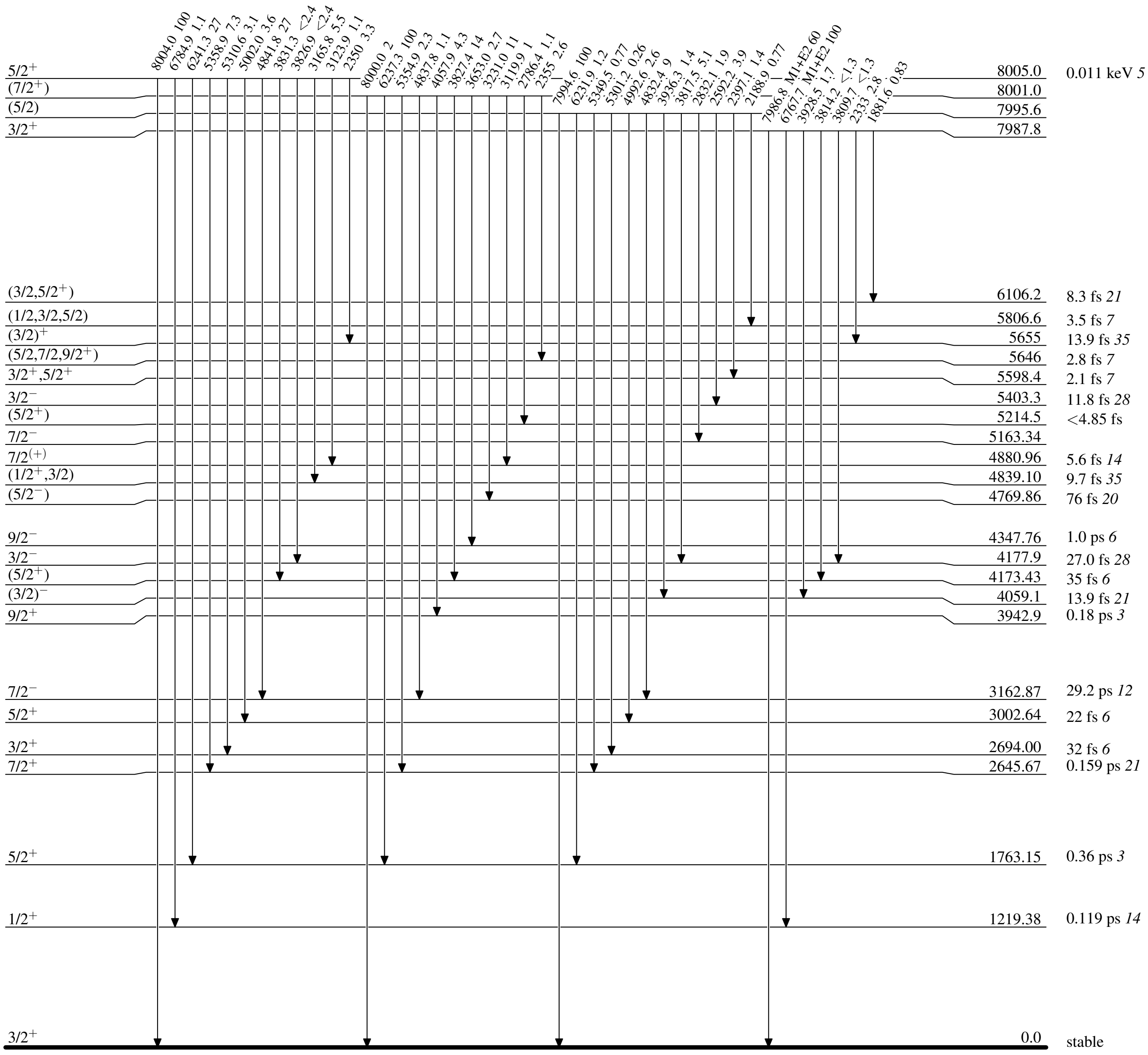
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

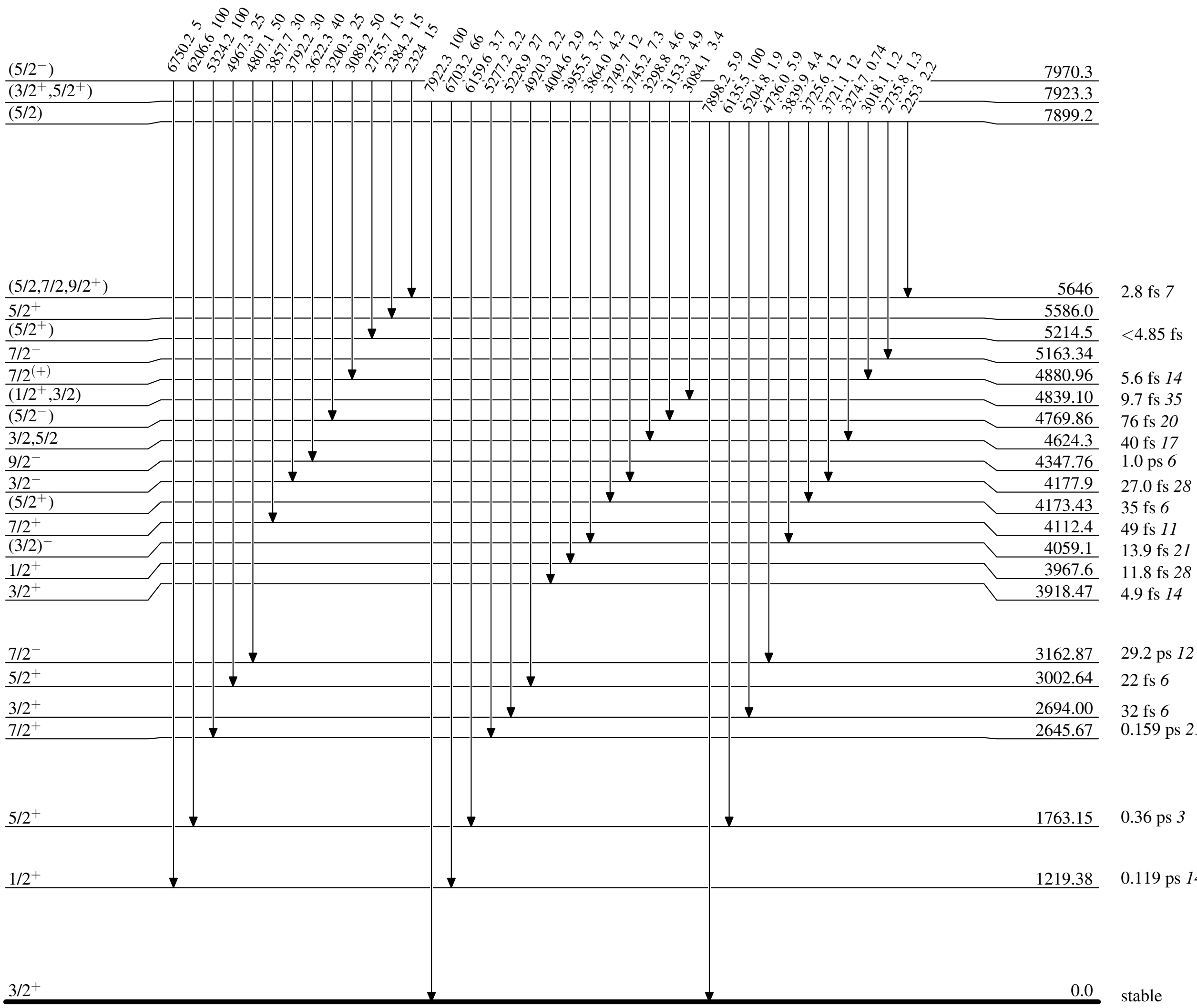


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

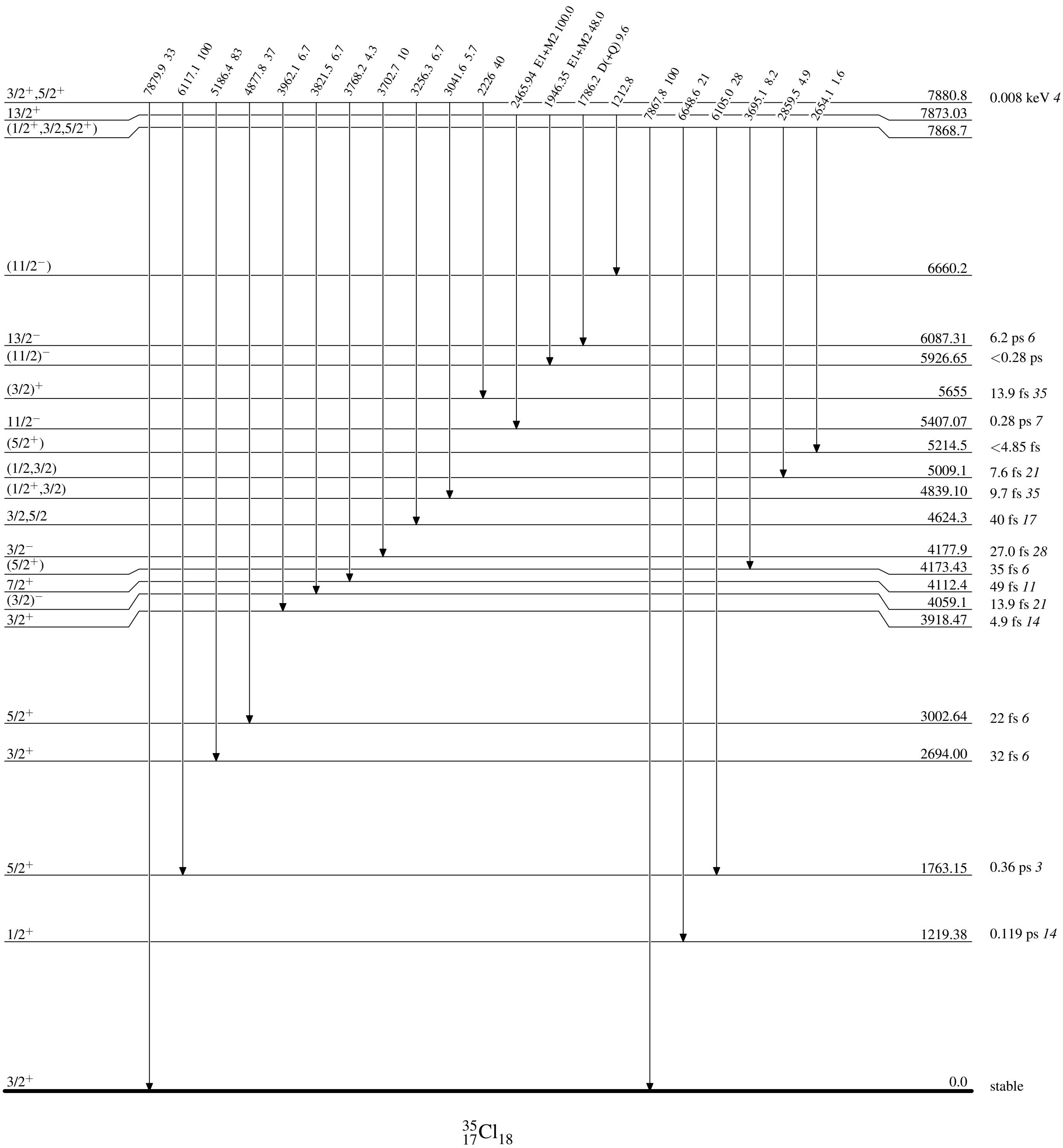


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

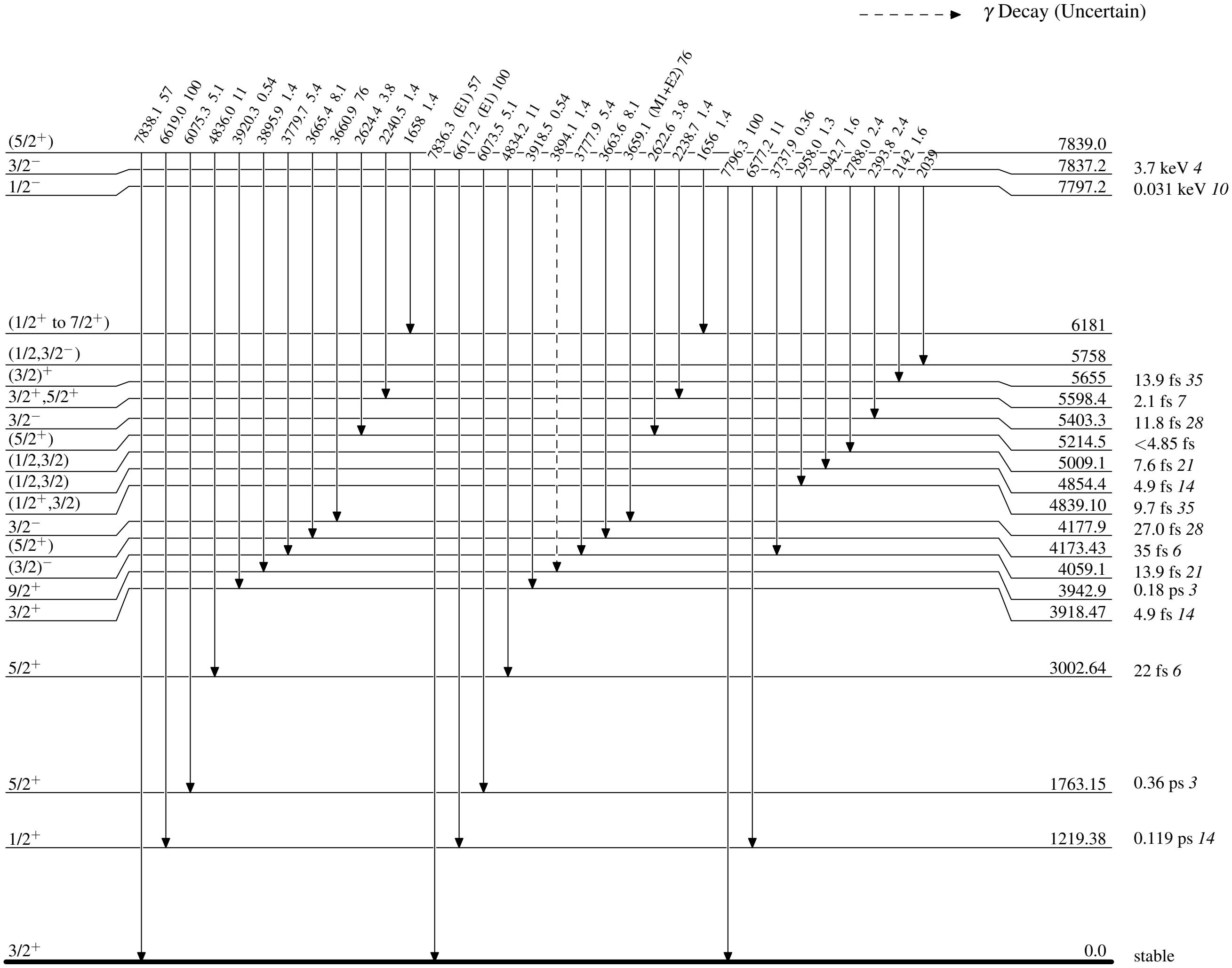
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

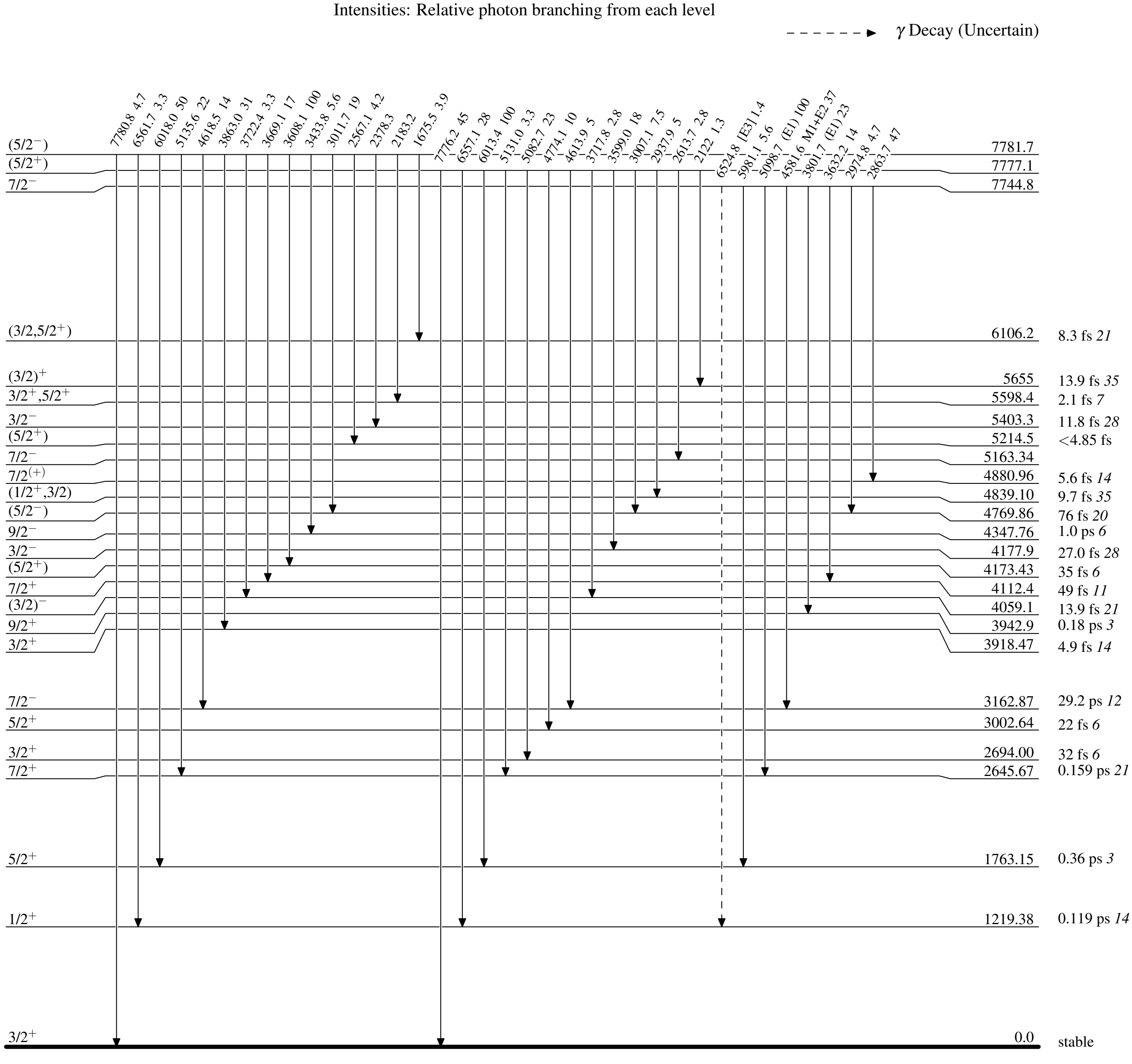


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

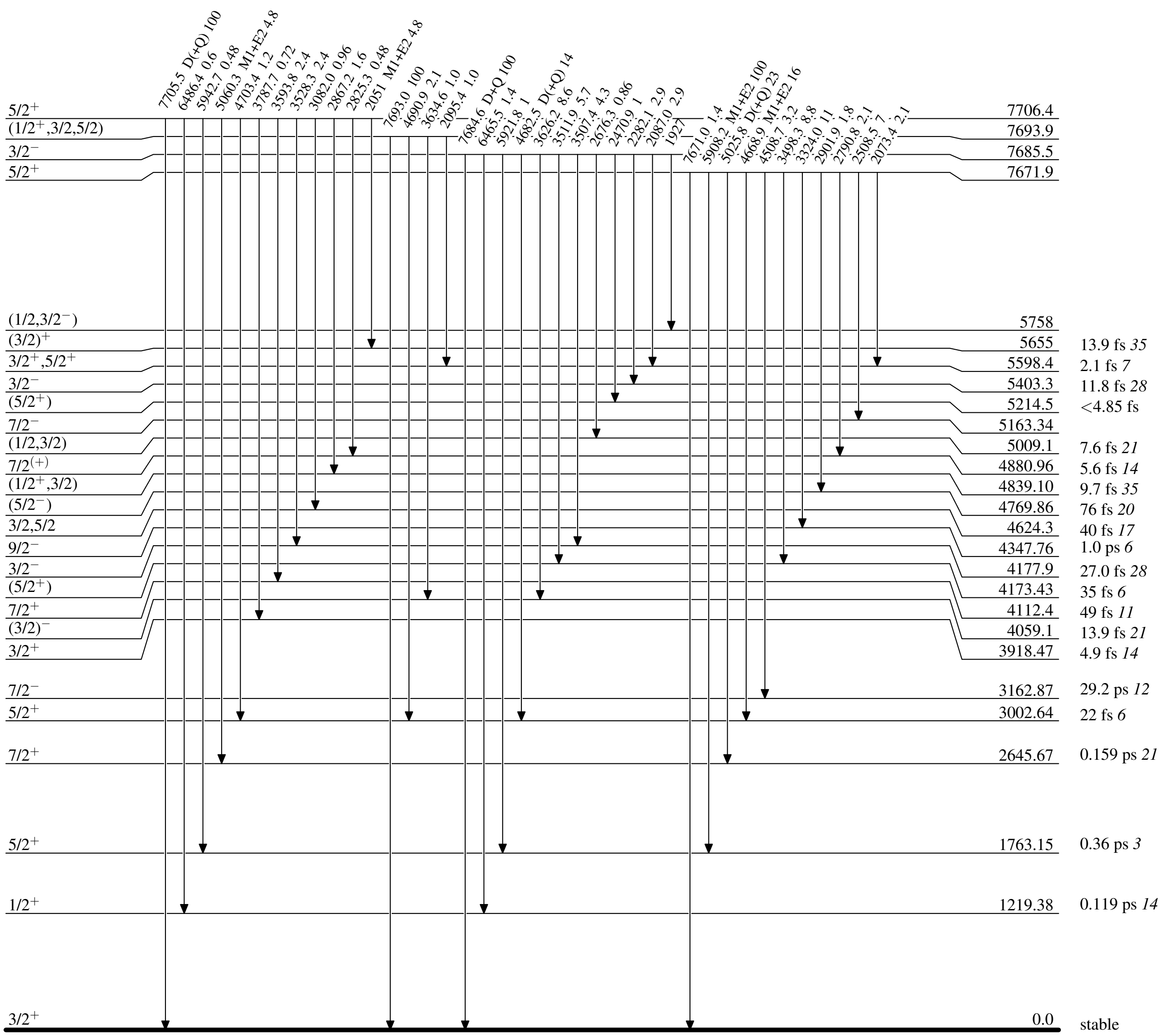


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

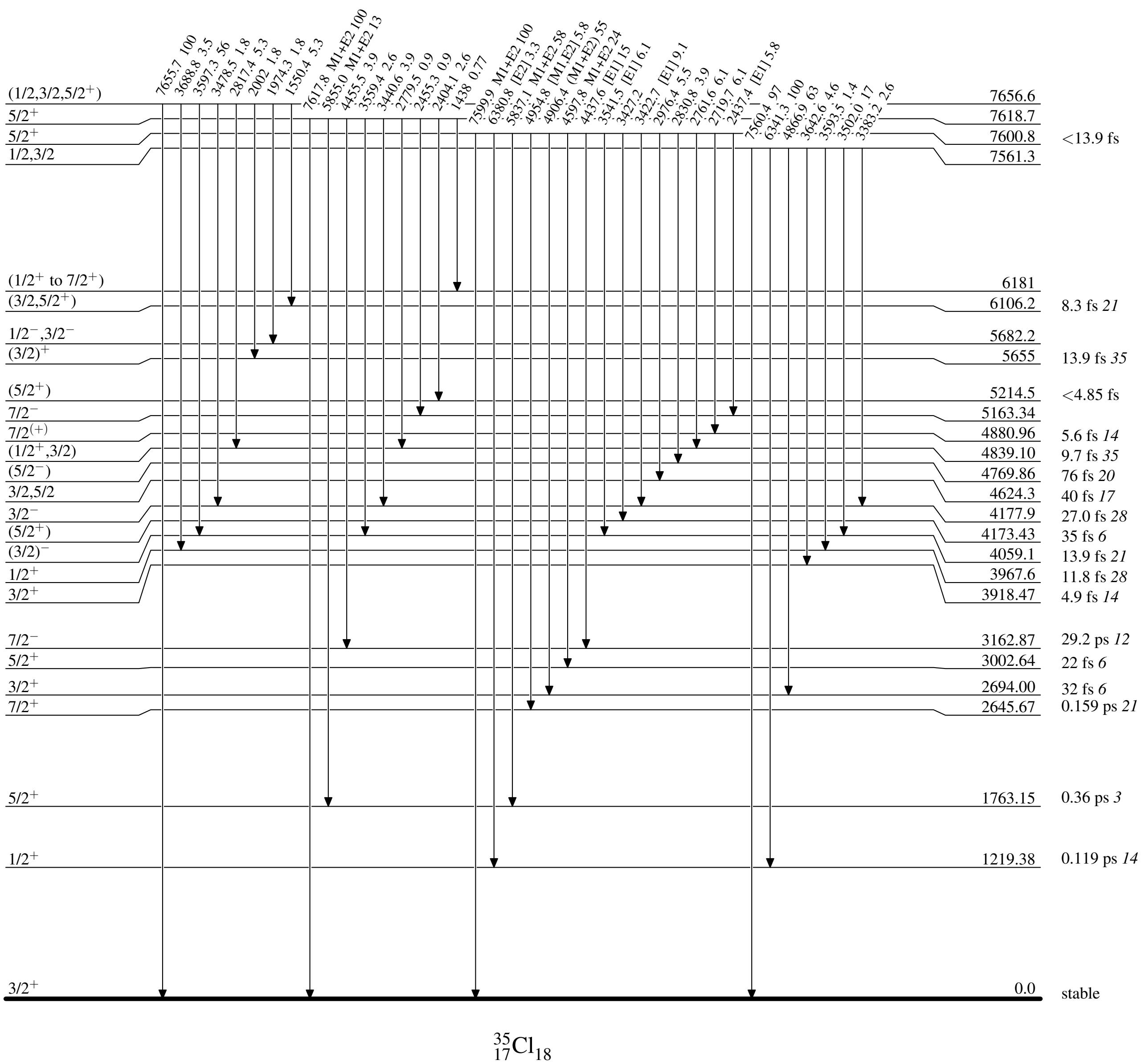


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

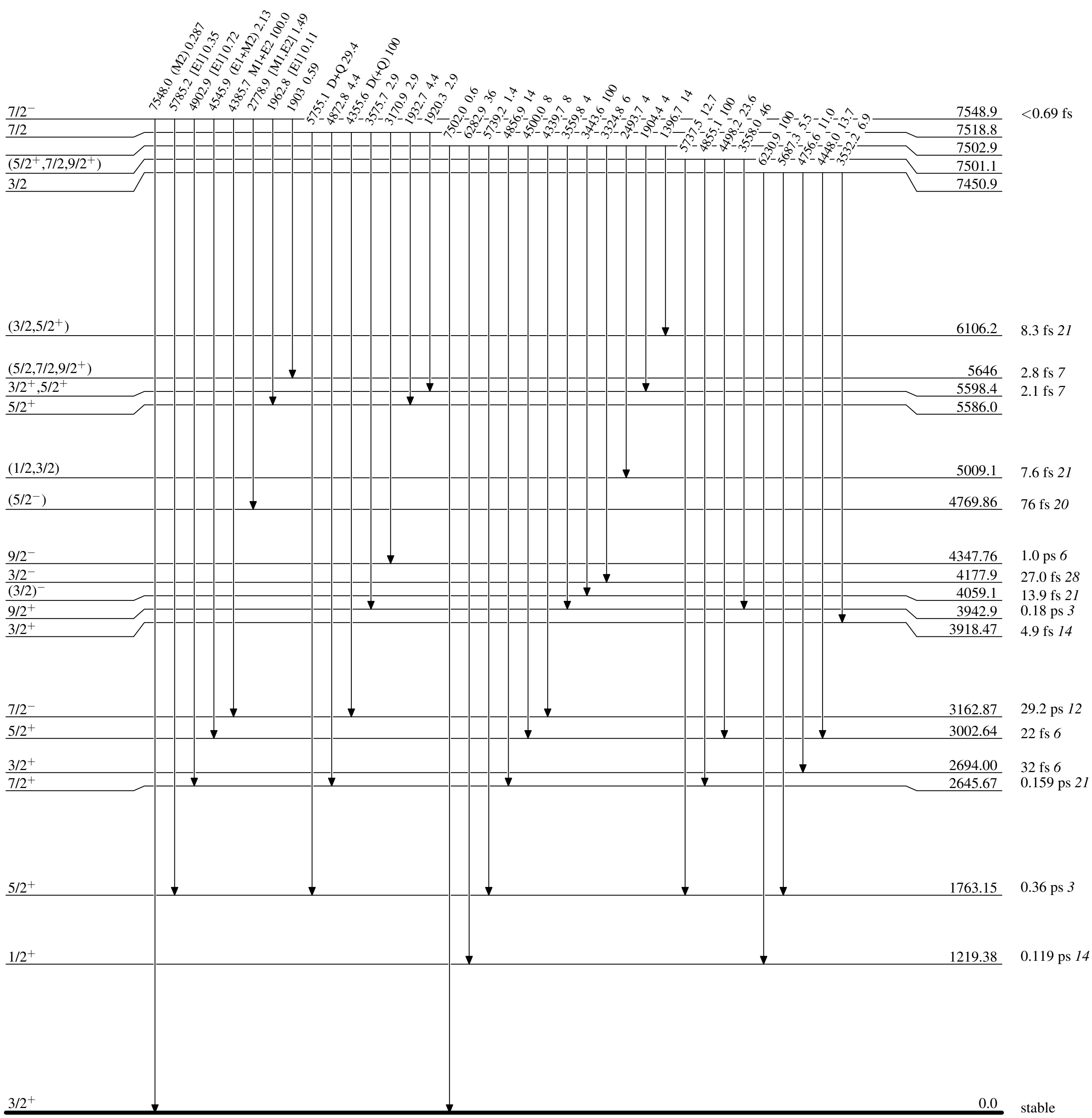
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

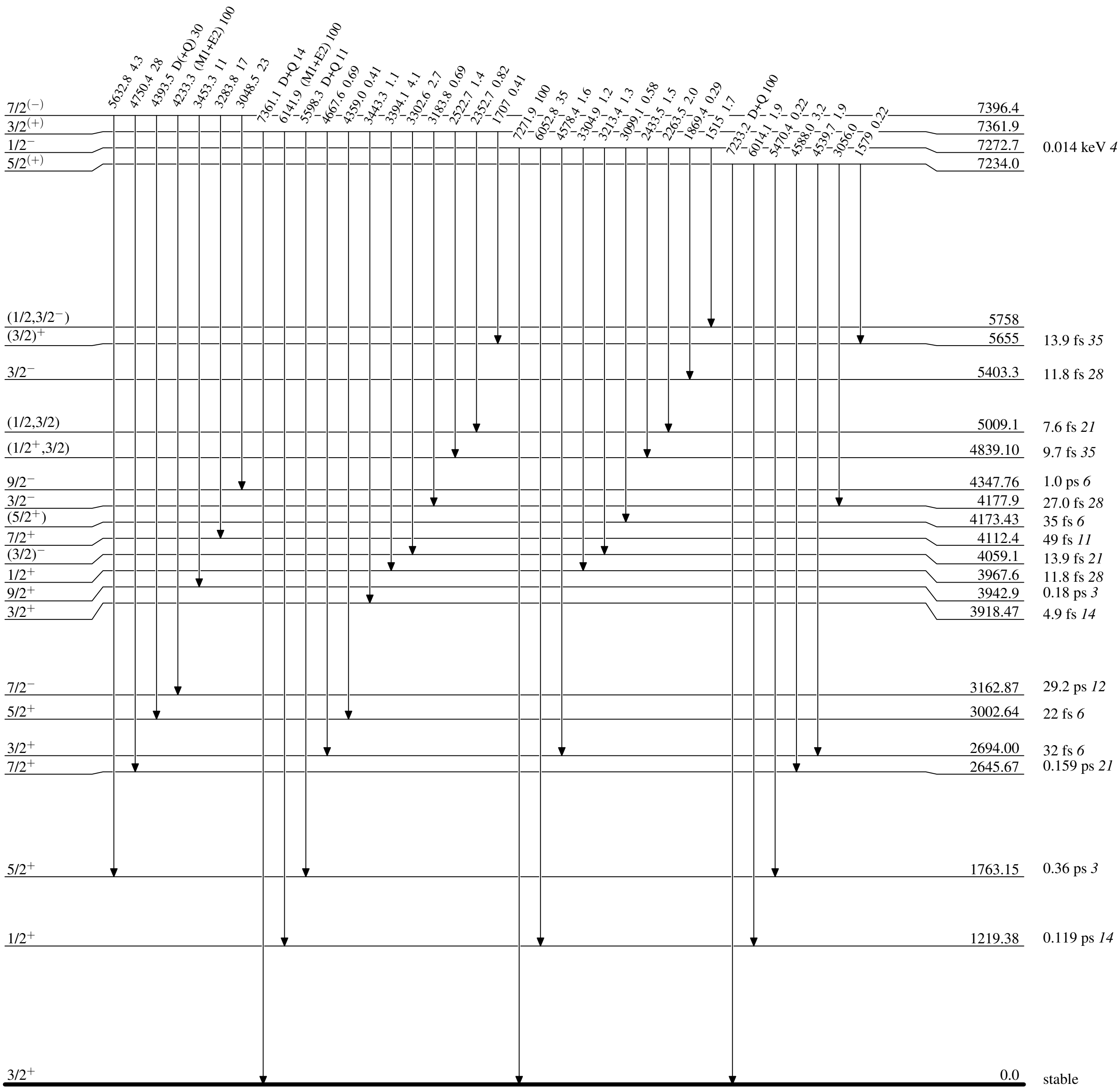


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level

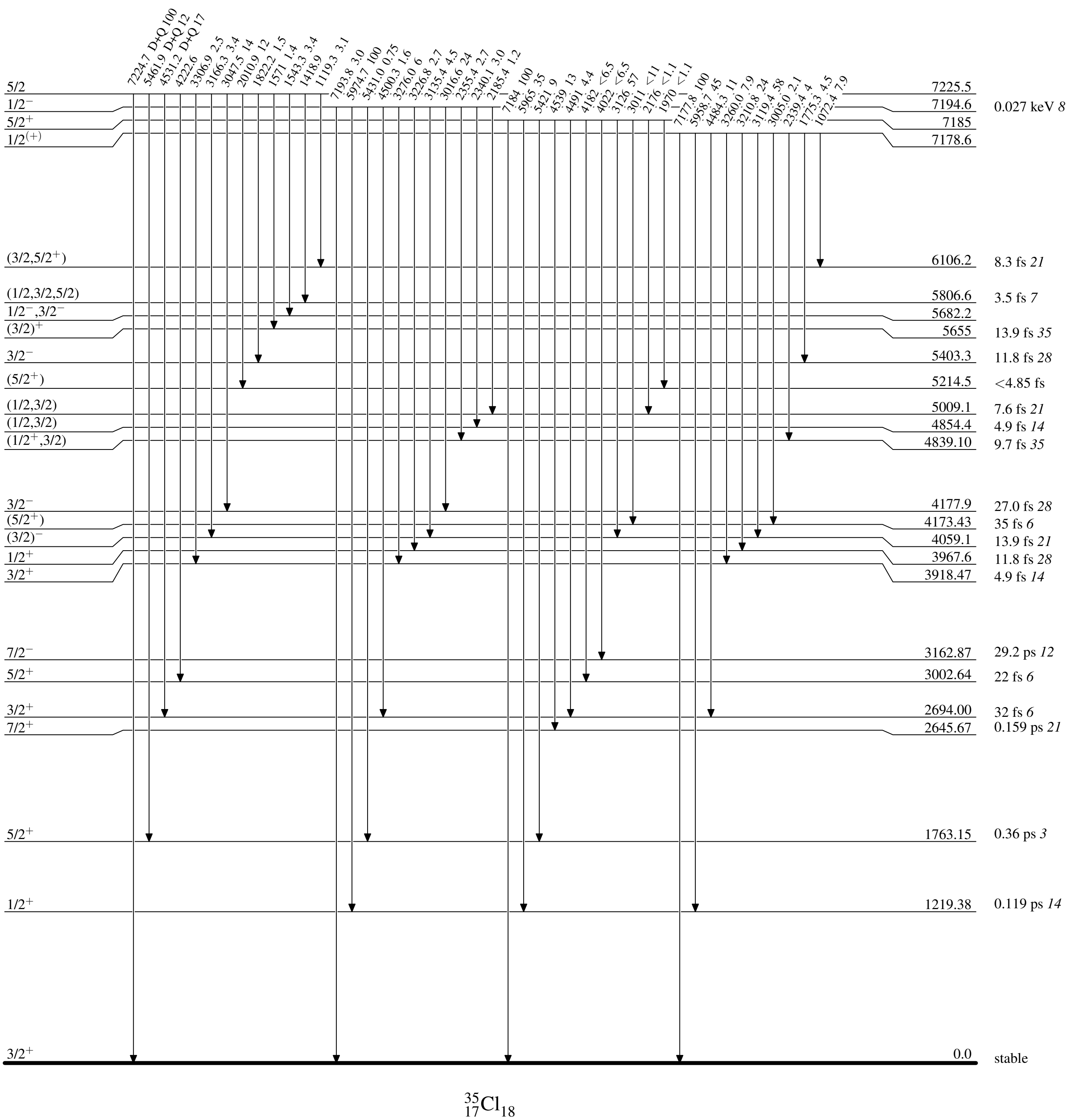


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

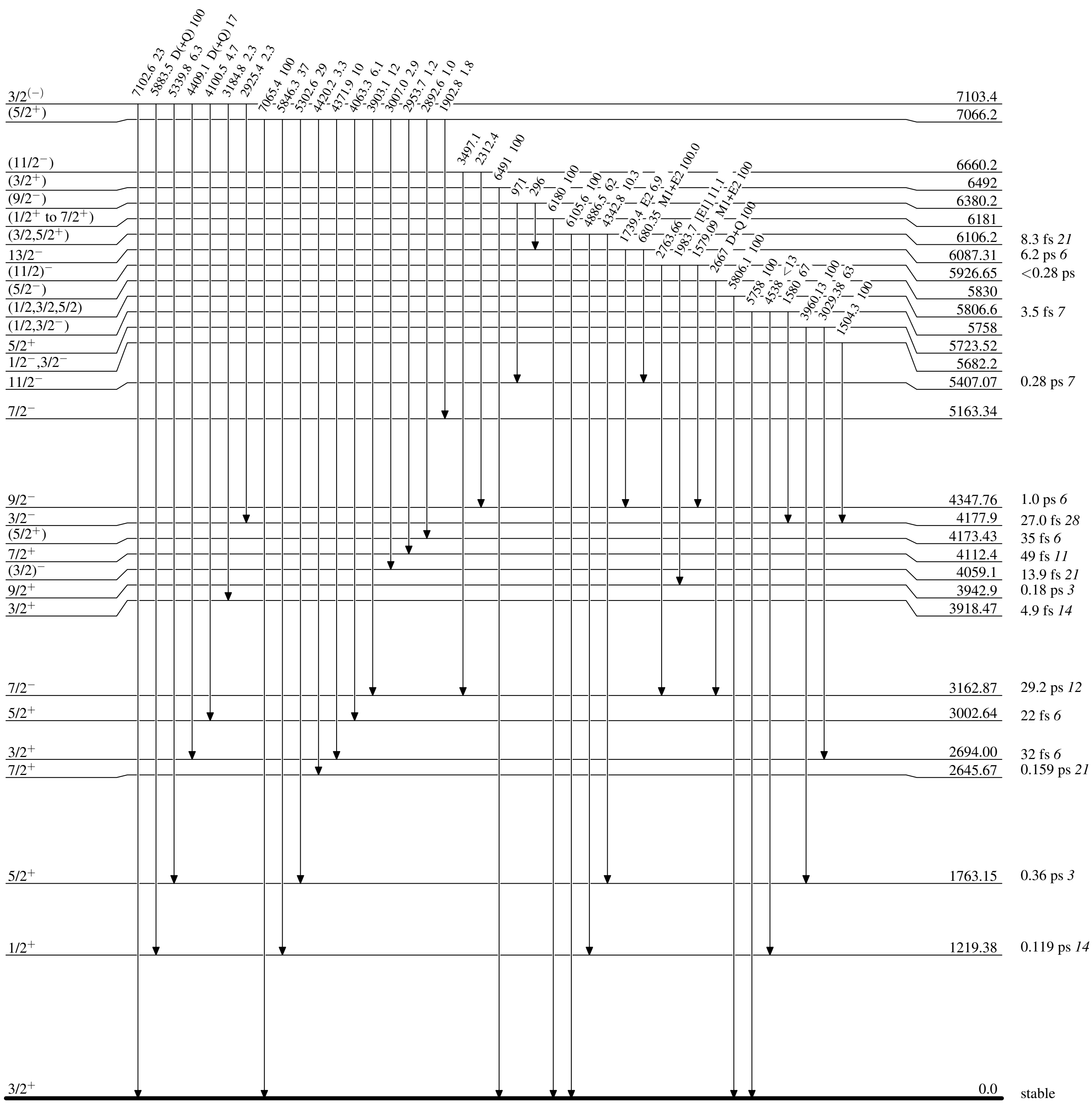
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



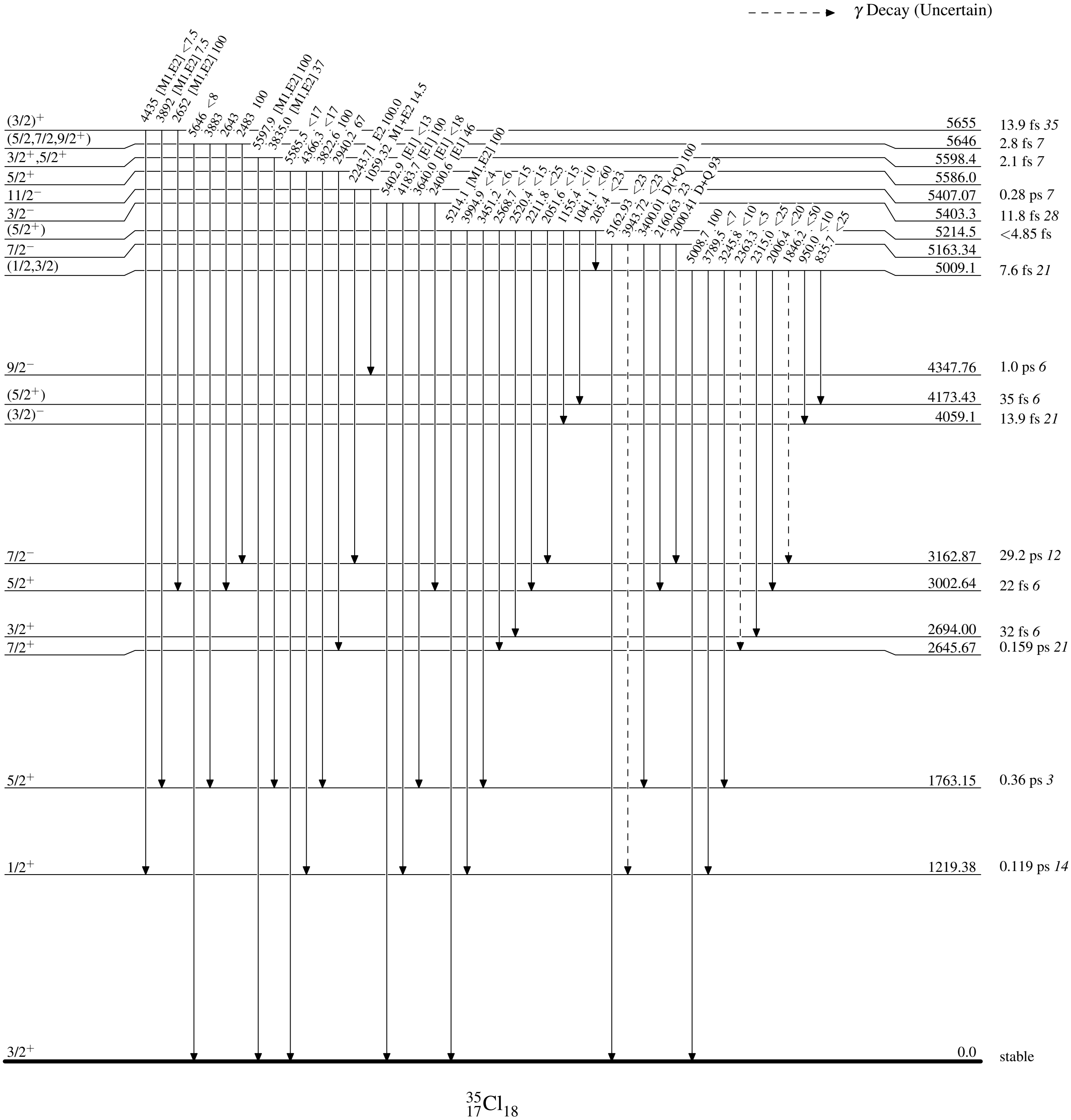
$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level



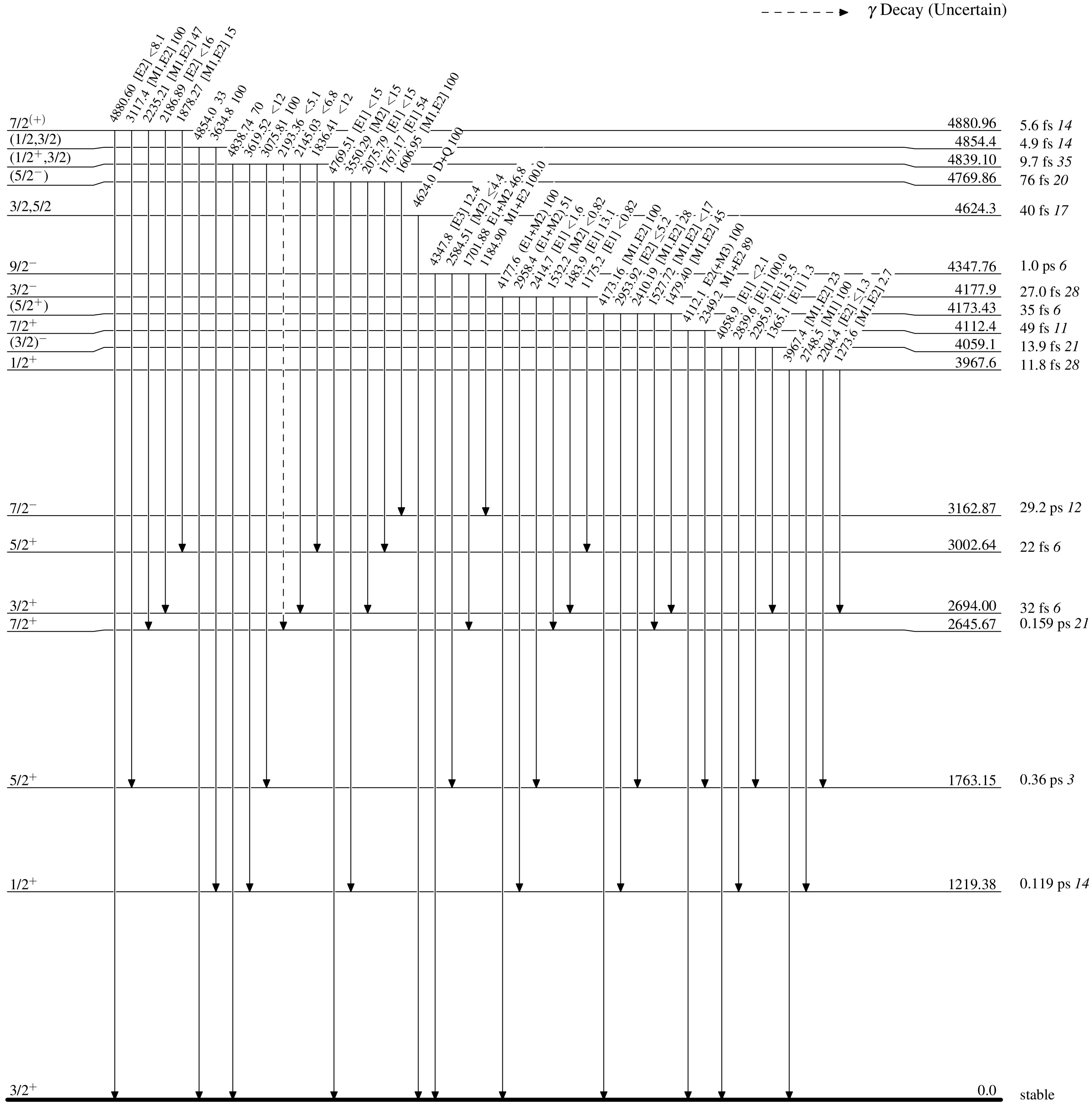
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

γ Decay (Uncertain)

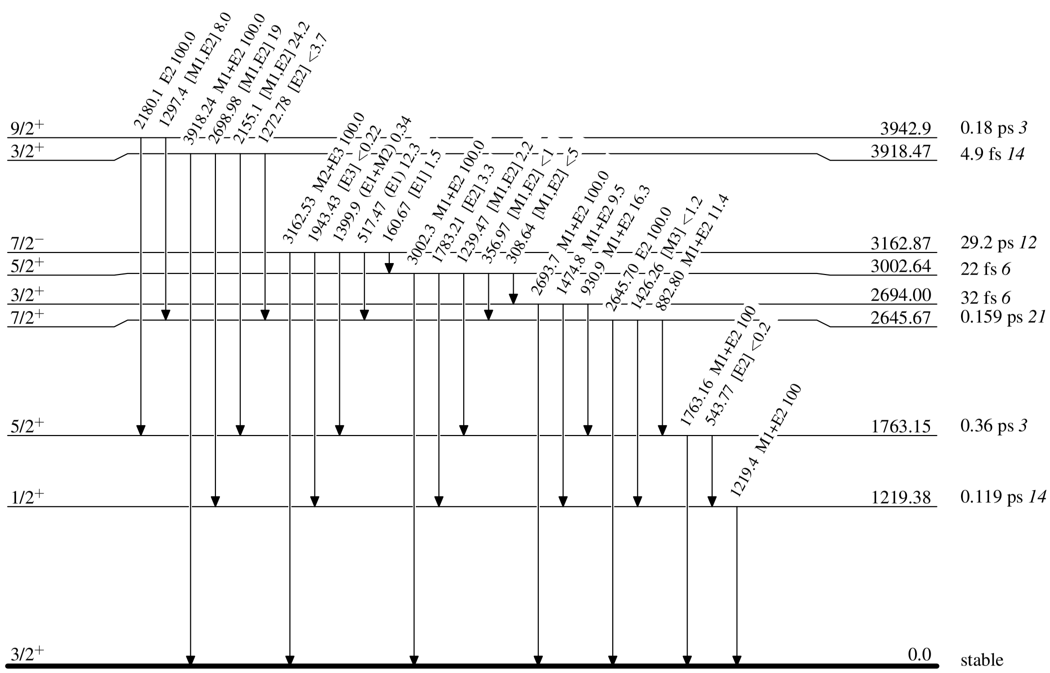


$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas

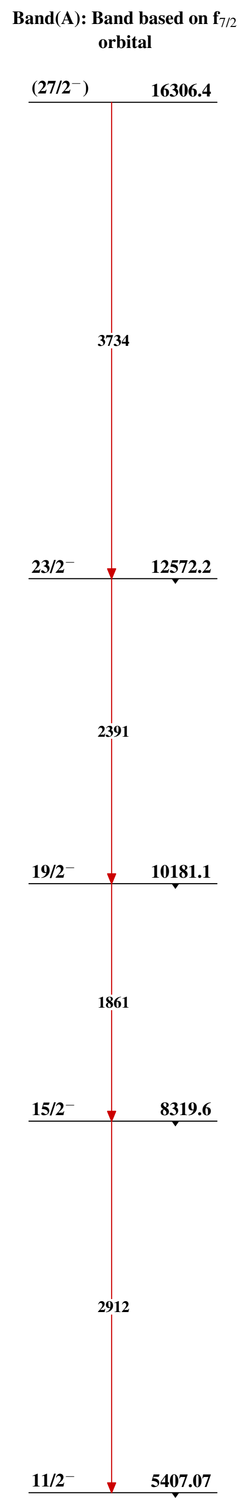
Level Scheme (continued)

Intensities: Relative photon branching from each level



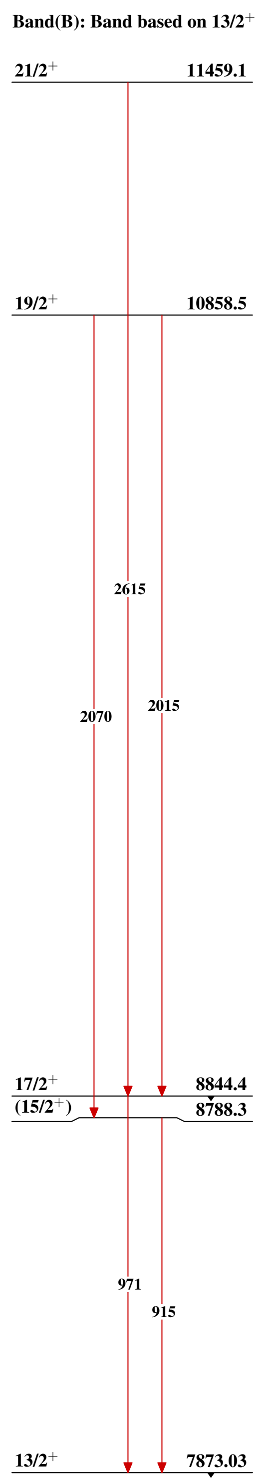
$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas



$^{35}_{17}Cl_{18}$

Adopted Levels, Gammas (continued)



$^{35}_{17}Cl_{18}$